FINAL TRIM AND STABILITY BOOKLET
OF
KB-25
(YARD NO. CHISTI/012)

AS AMENDED ON PAGE NO.23,25,65-67

APPROVED
FOR COMPLIANCE WITH AS MENTIONED (SEE LETTER) REQUIREMENTS
ON BEHALF OF GOVT. OF INDIA
INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 07 Jul 2020
SEE LETTER E-105859-140713

Rev.	Description	Date	Prepared	Review

Issued for : Approval

Client:	Project:
K.B.SHIPPING & CO.,JAMNAGAR	**KB-25**
Builder:	**(YARD NO. CHISTI/012)**
CHISTI MARINE ENGINEERING WORKS	

Doc #				Document Title:	Prepared	JD
CM20-V-105-0203				**FINAL TRIM AND STABILITY BOOKLET**	Checked	ND
Scale	Size	Sheet	Date		Review 1	
NTS	A4	1 of 171	18-03-2020		Review 2	

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GENERAL PARTICULARS

Name of the Vessel	KB-25
Client	K.B. SHIPPING & CO., JAMNAGAR
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/012
Type of Vessel	PONTOON
Class Notation	卐SU-PONTOON, "INDIAN COASTAL SERVICE" DESCRIPTIVE NOTE "FOR CRANE OPERATION" "DECK STRENGHTENED FOR {10T/M^2}

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	15.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	02.200 M
Draft (Ext.)	02.212 M
Displacement(Ext.)	1569.15 T
Lightship	368.27 T
Deadweight	1200.88 T
Gross Tonnage	564
Net Tonnage	170

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.

2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.

3) The minimum range of stability should be:
For L <= 100m: 20^0
For L >= 150m: 15^0
For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta C \leq 15°$
- $GZC \leq 0,6\ GZMAX$
- $A1 \geq 0.4\ ATOT$

where:

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC, GZMAX: Defined in Fig 1

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle equal to the lesser of:

- heeling angle θR of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θF, corresponding to flooding of unprotected openings

ATOT : Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b=(Pd-Zz)/\Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z: Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM.The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

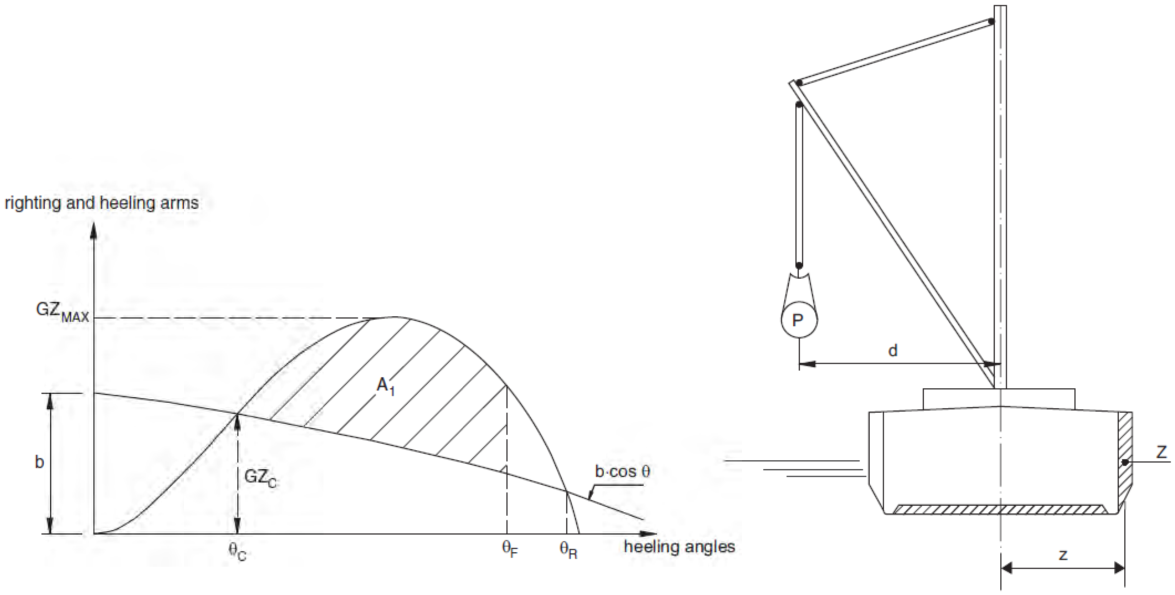


FIGURE 1: CARGO LIFTING

<u>Intact stability criteria in the event of sudden loss of cargo during lifting</u>

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A2 / A1 \geq 1$
- $\theta 2 - \theta 3 \geq 20°$

where:

A1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ1 to the heeling angle θC (see Fig 2)

A2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle θ2 (see Fig 2)

A3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θC to the heeling angle θ3 (see Fig 2)

θ1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ2 : Heeling angle corresponding to the lesser of θR and θF

θC : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ3 : Maximum heeling angle due to roll, at which A3 = A1, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θR : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θF : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

b = Zz cosθ/Δ

HEELING AND RIGHTING ARMS

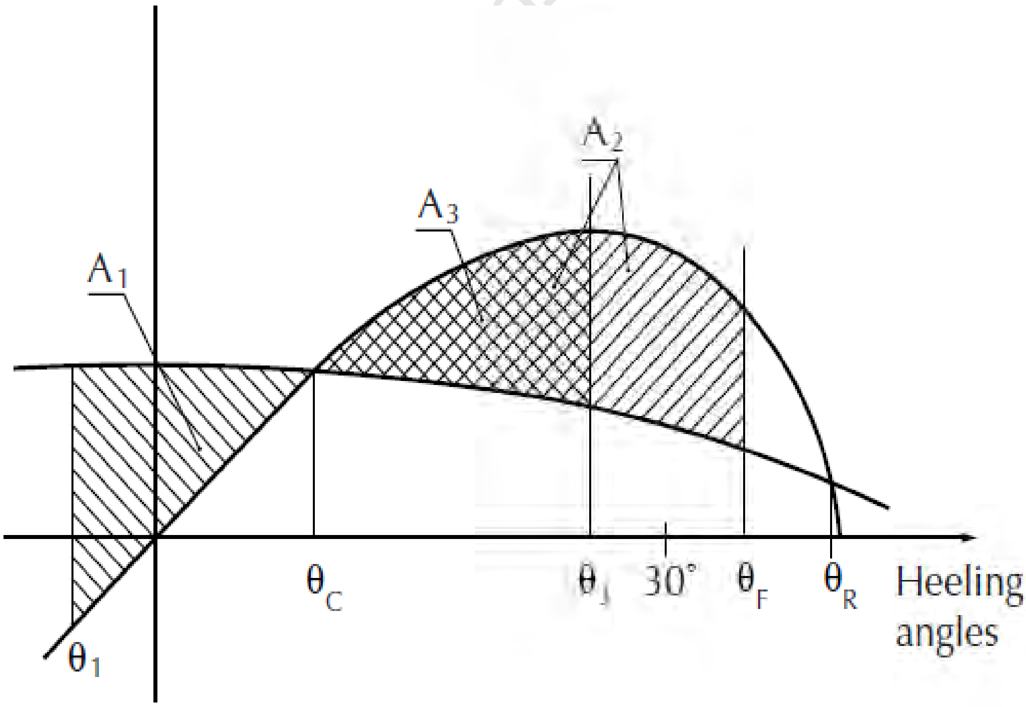


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.

2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.

3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.

4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.

5. In this booklet hydrostatic particulars and cross curves are given at following trims.

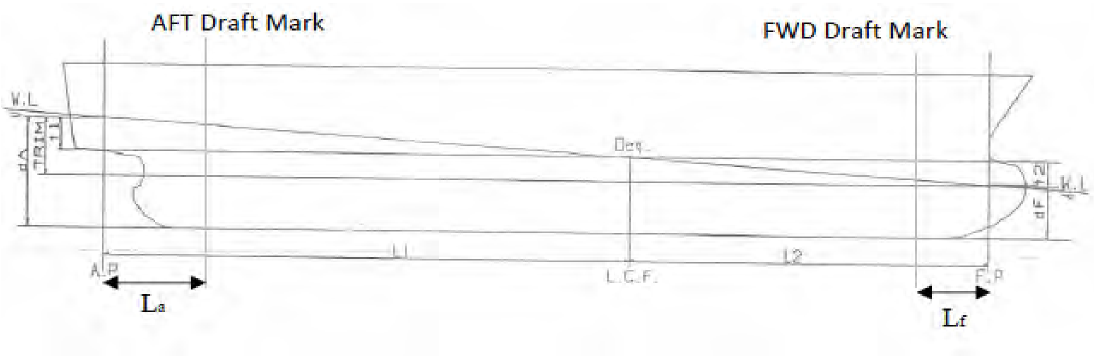
   -0.50m,-0.25m, 0.00m,0.25m, 0.50m

   These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

b) Longitudinal centre of buoyancy, LCB

c) Displacement, Δ

d) Moment to change trim, MCT1cm

e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$Trim = \frac{\Delta * (LCB-LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * Trim}{LBP.} + T\ (extreme)$$

$$T_{FP} = T_{AP} - Trim$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{Trim * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{Trim * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

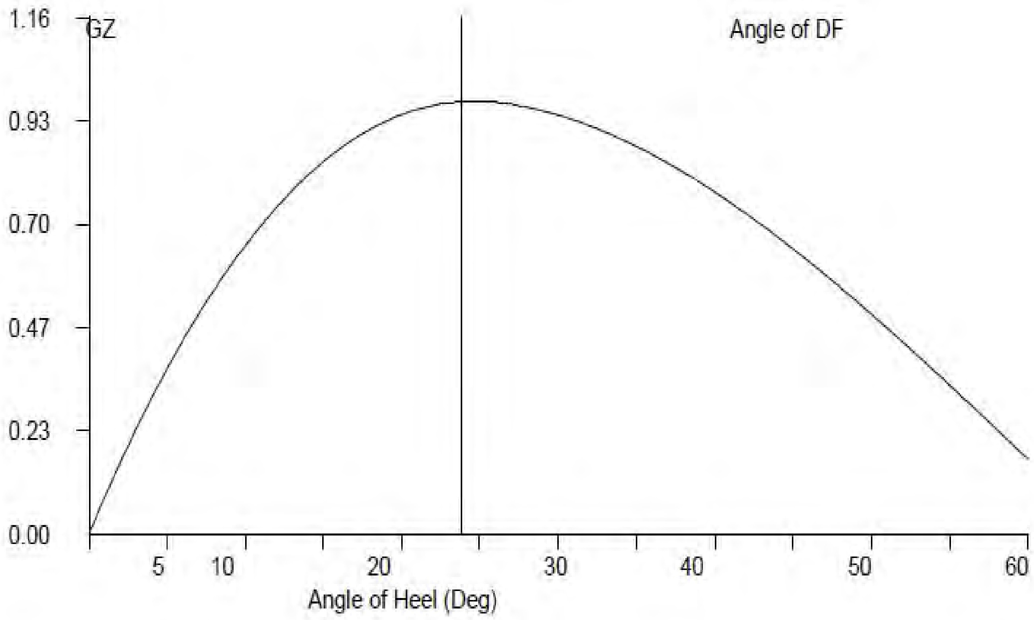
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

1) GZ = KN - KGo Sin (theta)
2) Area under 0-10 = 10*0.01745*[GZ0 + GZ10] / 2
3) Area under 10-20 = 10*0.01745*[GZ10 + GZ20] / 2
4) Area under 20-30 = 10*0.01745*[GZ20 + GZ30] / 2
5) Area under 30-40 = 10*0.01745*[GZ30 + GZ40] / 2
6) Area under 40-50 = 10*0.01745*[GZ40 + GZ50] / 2
7) Area under 50-60 = 10*0.01745*[GZ50 + GZ60] / 2
8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples

1 Loading the Vessel

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Ballast

Displacement	638.84	Tonnes
LCG	24.464	M Fwd of AP
VCG (KG)	3.527	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

Free Surface moment for a Tank = Moment of Inertia (M^4) for a Tank x Density of Liquid in Tank

KGo (Fluid) = KG (Solid) + Free Surface Correction.

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

Trim = (LCB-LCG) X displacement / (MCT x 100) + Trim*

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

CONDITION 1 - LIGHTSHIP + CRANE + BALLAST

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
WBTK6.P	39.01	1.025	39.990	47.660	1905.923	-4.967	-198.630	1.915	76.581	41.850
WBTK6.C	39.61	1.025	40.600	47.664	1935.158	0.000	0.000	1.904	77.302	41.850
WBTK6.S	39.01	1.025	39.990	47.660	1905.923	4.967	198.630	1.915	76.581	41.850
TOTAL TANK			120.580	47.661	5747.005	0.000	0.000	1.911	230.464	125.550

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			368.270	25.000	9,206.750	0.000	0.000	3.000	1104.810
CRANE			150.000	4.500	675.000	0.000	0.000	6.120	918.000
DECK CARGO			0.000	0.000	0.000	0.000	0.000	0.000	0.000
LOAD LIFTED			0.000	0.000	0.000	0.000	0.000	0.000	0.000
TOTAL FIXED WEIGHT			518.270	19.067	9881.750	0.000	0.000	3.903	2022.81

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			120.580	47.661	5747.005	0.000	0.000	1.911	230.464
TOTAL FIXED WEIGHT			518.270	19.067	9881.750	0.000	0.000	3.903	2022.810
TOTAL WEIGHT			638.850	24.464	15628.755	0.000	0.000	3.527	2253.274

CONDITION 1 - LIGHTSHIP + CRANE + BALLAST

Free Surface Correction for VCG = FSM/Displacement

	0.197	m
VCGc	3.724	m

From the MaxVCG table,

Displacement	638.85	T
Allowable Max VCG	13.528	m

VCGc < Maxvcg PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	638.850	T
Draft ,Tm @ 25.00f =	0.960	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	24.824	T-m/cm
KMT =	21.260	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.138 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.000 (MS),Tmid 0.960 m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2
0.891 m

Draft@0(AP),Taft =Tm + Trim/2
1.028 m

CONDITION 1 - LIGHTSHIP + CRANE + BALLAST

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	24.824	T-m	Draft (Mean)	0.960 m
LCB	25.000	m	Draft AP (USK)	1.028 m
			Draft FP (USK)	0.891 m
Metacentric Height				
KMt	21.260	m		
KG	3.527	m		
KG0	3.724	m		
GMt	17.536	m	**Trim**	0.138 m

The condition shown above is first worked
using Even Keel hydrostatic perticulars
Program then interpolates the hydrostatic perticulars
from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	24.370	T-m	Draft (Mean)	0.959 m
LCB	24.455	m	Draft AP (USK)	1.030 m
			Draft FP (USK)	0.889 m
Metacentric Height				
KMt	21.204	m		
KG	3.527	m		
KG0	3.724	m		
GMt	17.480	m	**Trim**	0.141 m

Trim is calculaed as follows:
Trim = (LCB - LCG) X displacement / (MCT X 100) + Trim*
Where Trim* is trim at which hydrostatics were calculated.

CONDITION 1 - LIGHTSHIP + CRANE + BALLAST

Intact Stability (Primary Criterion)

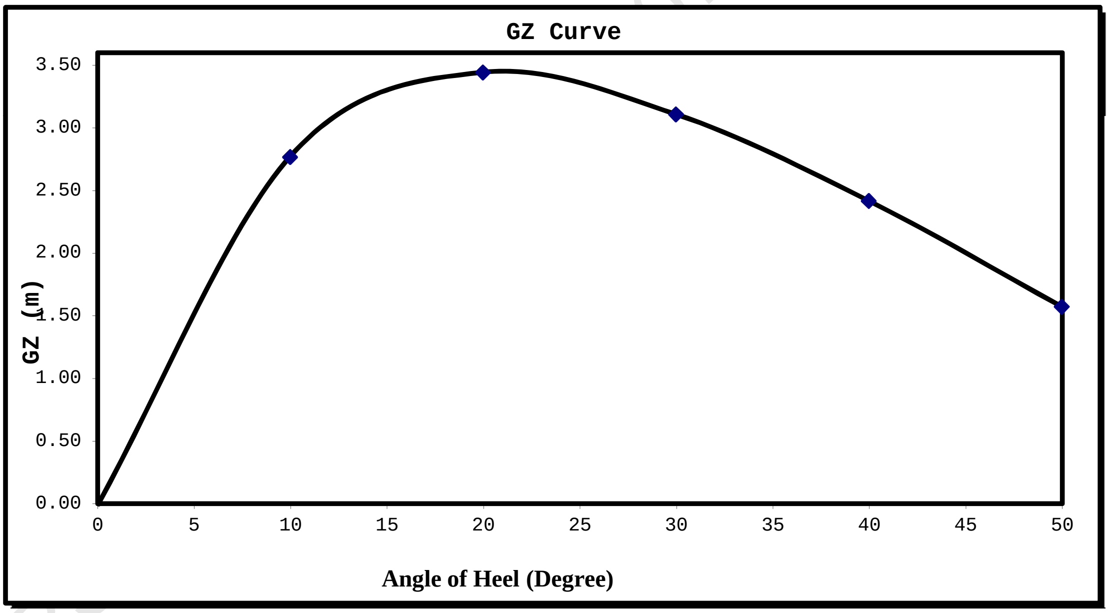
			Minimum Required
Displacement	638.85	T	
KG0	3.724	M	
GMt	17.480	M	0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	-0.012	3.413	4.715	4.970	4.810	4.425
GZ (M)	-0.012	2.769	3.444	3.110	2.418	1.574

	Computed values		Minimum Required	
Residual Area from abs 0 deg to MAxRA	0.8416	m-rad	0.080	m-rad
Angle from Equilibrium to RAZero or flood	Large	m-rad	20.00	Deg.
Angle from Equilibrium to 50% DK Imm Angle	8.59	m-rad	0.00	Deg.

Plot GZ values against the angles of heel as above.
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates.Join the point to the origin.
This is the Tangent to the GZ Curve at the Origin.Find the Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding
If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	0.959	M
LCF from AP	24.874	M
LCB from AP	24.455	M
MCT	24.370	T-M/Cm
Trim	0.141	M
Draft (AP)	1.030	M
Draft (FP)	0.889	M
KG	3.527	M
GoMt	17.480	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (MLD) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
TOTAL TANK										

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			368.270	25.000	9,206.750	0.000	0.000	3.000	1104.810
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

Free Surface Correction for VCG = FSM/Displacement

m

VCGc m

From the MaxVCG table,

Displacement T

Allowable Max VCG m

$\boxed{\text{VCGc < Maxvcg}}$ PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact

Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = T

Draft ,Tm @ 25.00f = m

LCB = m

LCF = m

MCT = T-m/cm

KMT = m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)

m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.000 (MS),Tmid m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2

m

Draft@0(AP),Taft =Tm + Trim/2

m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

MULTIPLY BY	**TO CONVERT FROM**	**TO OBTAIN**	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	TO OBTAIN	**TO CONVERT FROM**	**MULTIPLY**

Relation between Weight and volume

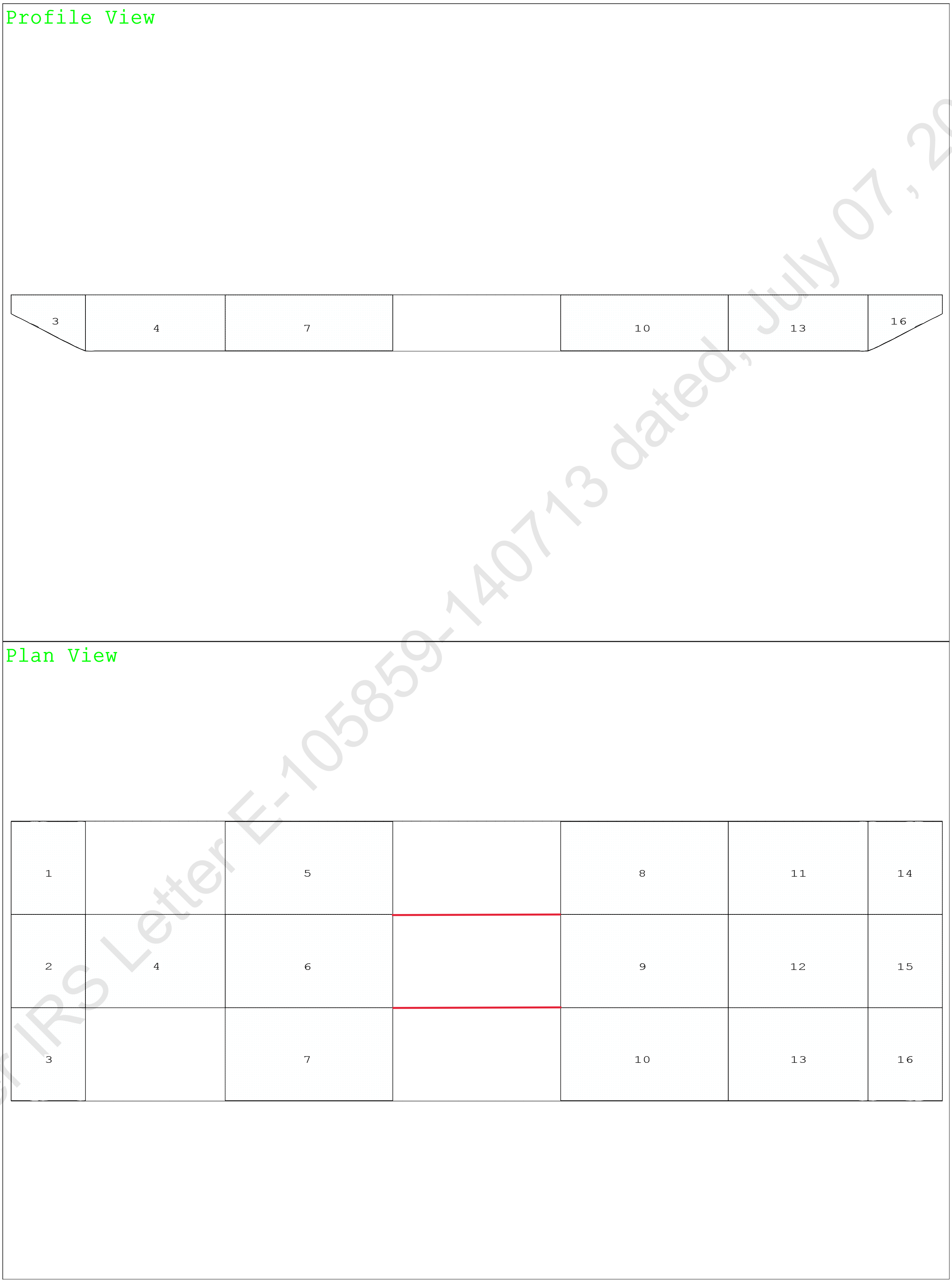
1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill off this opening is a \`potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
   If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

   As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

Condition Graphic



Tanks

1 WBTK1.P.100% SALT WATER
2 WBTK1.C.100% SALT WATER
3 WBTK1.S.100% SALT WATER
4 WBTK2.C.100% SALT WATER
5 WBTK3.P.100% SALT WATER
6 WBTK3.C.100% SALT WATER
7 WBTK3.S.100% SALT WATER
8 WBTK4.P.100% SALT WATER
9 WBTK4.C.100% SALT WATER
10 WBTK4.S.100% SALT WATER
11 WBTK5.P.100% SALT WATER
12 WBTK5.C.100% SALT WATER
13 WBTK5.S.100% SALT WATER
14 WBTK6.P.100% SALT WATER
15 WBTK6.C.100% SALT WATER
16 WBTK6.S.100% SALT WATER

11-09-19 11:58:46 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

TANK STATUS
Trim: zero, Heel: zero

Part	Cu.M.	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
WBTK1.P	39.0	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTK1.C	39.6	1.025	40.60	2.336f	0.000	1.904	41.85*
WBTK1.S	39.0	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTK2.C	110.3	1.025	113.02	7.750f	0.000	1.500	78.48*
WBTK3.P	131.2	1.025	134.48	16.000f	4.980p	1.511	94.17*
WBTK3.C	132.3	1.025	135.61	16.000f	0.000	1.500	94.17*
WBTK3.S	131.2	1.025	134.48	16.000f	4.980s	1.511	94.17*
WBTK4.P	131.2	1.025	134.48	34.000f	4.980p	1.511	94.17*
WBTK4.C	132.3	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK4.S	131.2	1.025	134.48	34.000f	4.980s	1.511	94.17*
WBTK5.P	109.3	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	110.3	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	109.3	1.025	112.08	42.251f	4.980s	1.511	78.49*
WBTK6.P	39.0	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	39.6	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTK6.S	39.0	1.025	39.99	47.660f	4.967s	1.915	41.84*
Total Tanks--------->			1,500.47	27.577f	0.000	1.572	1130.05

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

PAGE NO.25-39, NOT A PART OF THIS APPROVAL

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK1.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.001	0.054	0.05	3.920f	4.536p	0.056	0.033	0.502
200	.005	0.201	0.21	3.847f	4.720p	0.133	0.208	3.954
300	.012	0.469	0.48	3.766f	4.784p	0.202	0.244	5.469
400	.021	0.829	0.85	3.698f	4.808p	0.266	0.318	7.650
500	.033	1.280	1.31	3.635f	4.825p	0.332	0.358	9.400
600	.047	1.830	1.88	3.571f	4.845p	0.398	0.413	11.320
700	.063	2.475	2.54	3.507f	4.861p	0.464	0.496	13.933
800	.082	3.215	3.30	3.444f	4.876p	0.530	0.553	15.913
900	.104	4.050	4.15	3.381f	4.887p	0.596	0.615	17.878
1000	.128	4.980	5.10	3.318f	4.896p	0.663	0.677	19.939
1100	.154	6.005	6.16	3.254f	4.904p	0.729	0.738	21.934
1200	.183	7.126	7.30	3.190f	4.911p	0.795	0.801	24.042
1300	.214	8.332	8.54	3.128f	4.917p	0.861	0.775	25.381
1400	.246	9.593	9.83	3.072f	4.920p	0.926	0.825	27.288
1500	.281	10.979	11.25	3.010f	4.925p	0.992	0.995	30.123
1600	.320	12.486	12.80	2.943f	4.929p	1.059	1.055	31.968
1700	.361	14.087	14.44	2.876f	4.934p	1.126	1.010	33.008
1800	.405	15.780	16.17	2.811f	4.935p	1.194	1.166	35.784
1900	.451	17.576	18.02	2.744f	4.938p	1.261	1.262	38.398
2000	.499	19.466	19.95	2.678f	4.941p	1.328	1.307	40.474
2100	.549	21.406	21.94	2.618f	4.944p	1.393	1.198	40.879
2200	.599	23.352	23.94	2.568f	4.948p	1.456	1.106	41.198
2300	.649	25.304	25.94	2.524f	4.951p	1.517	1.026	41.482
2400	.699	27.259	27.94	2.487f	4.954p	1.577	0.955	41.627
2500	.749	29.216	29.95	2.454f	4.957p	1.636	0.893	41.737
2600	.799	31.175	31.95	2.426f	4.959p	1.693	0.837	41.760
2700	.849	33.133	33.96	2.401f	4.962p	1.750	0.788	41.783
2800	.900	35.092	35.97	2.378f	4.964p	1.805	0.744	41.807
2900	.950	37.052	37.98	2.358f	4.966p	1.861	0.705	41.831
3000	1.000	39.012	39.99	2.340f	4.967p	1.915		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK1.P Reference Point: Long.= 2.000f Trans.= 5.005p Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK1.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.002	0.068	0.07	3.908f	0.000	0.056	0.042	0.739
200	.006	0.245	0.25	3.836f	0.000	0.130	0.188	5.060
300	.013	0.532	0.55	3.765f	0.000	0.197	0.248	7.182
400	.023	0.916	0.94	3.699f	0.000	0.261	0.305	9.219
500	.035	1.393	1.43	3.634f	0.000	0.327	0.362	11.227
600	.050	1.967	2.02	3.570f	0.000	0.392	0.425	13.283
700	.067	2.635	2.70	3.506f	0.000	0.458	0.484	15.298
800	.086	3.399	3.48	3.441f	0.000	0.524	0.546	17.330
900	.108	4.258	4.36	3.377f	0.000	0.590	0.609	19.374
1000	.132	5.213	5.34	3.312f	0.000	0.656	0.672	21.417
1100	.158	6.264	6.42	3.248f	0.000	0.722	0.735	23.461
1200	.187	7.411	7.60	3.183f	0.000	0.788	0.798	25.505
1300	.218	8.646	8.86	3.120f	0.000	0.855	0.833	26.842
1400	.252	9.974	10.22	3.056f	0.000	0.921	0.926	29.592
1500	.288	11.400	11.69	2.992f	0.000	0.987	0.991	31.646
1600	.326	12.931	13.25	2.927f	0.000	1.054	1.058	33.731
1700	.368	14.557	14.92	2.861f	0.000	1.120	1.119	35.750
1800	.411	16.278	16.69	2.796f	0.000	1.187	1.186	37.839
1900	.457	18.097	18.55	2.730f	0.000	1.254	1.251	39.897
2000	.505	20.010	20.51	2.665f	0.000	1.320	1.306	41.854
2100	.555	21.970	22.52	2.606f	0.000	1.385	1.190	41.854
2200	.604	23.930	24.53	2.556f	0.000	1.448	1.092	41.854
2300	.654	25.890	26.54	2.514f	0.000	1.509	1.009	41.854
2400	.703	27.850	28.55	2.478f	0.000	1.568	0.938	41.854
2500	.753	29.810	30.56	2.446f	0.000	1.626	0.877	41.854
2600	.802	31.770	32.56	2.419f	0.000	1.683	0.823	41.854
2700	.852	33.730	34.57	2.394f	0.000	1.739	0.775	41.854
2800	.901	35.690	36.58	2.373f	0.000	1.795	0.732	41.854
2900	.951	37.650	38.59	2.353f	0.000	1.850	0.694	41.854
3000	1.000	39.610	40.60	2.336f	0.000	1.904		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK1.C Reference Point: Long.= 2.000f Trans.= 0.000 Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK1.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.001	0.054	0.05	3.920f	4.536s	0.056	0.033	0.502
200	.005	0.201	0.21	3.847f	4.720s	0.133	0.208	3.954
300	.012	0.469	0.48	3.766f	4.784s	0.202	0.244	5.469
400	.021	0.829	0.85	3.698f	4.808s	0.266	0.318	7.650
500	.033	1.280	1.31	3.635f	4.825s	0.332	0.358	9.400
600	.047	1.830	1.88	3.571f	4.845s	0.398	0.413	11.320
700	.063	2.475	2.54	3.507f	4.861s	0.464	0.496	13.933
800	.082	3.215	3.30	3.444f	4.876s	0.530	0.553	15.913
900	.104	4.050	4.15	3.381f	4.887s	0.596	0.615	17.878
1000	.128	4.980	5.10	3.318f	4.896s	0.663	0.677	19.939
1100	.154	6.005	6.16	3.254f	4.904s	0.729	0.738	21.934
1200	.183	7.126	7.30	3.190f	4.911s	0.795	0.801	24.042
1300	.214	8.332	8.54	3.128f	4.917s	0.861	0.775	25.381
1400	.246	9.593	9.83	3.072f	4.920s	0.926	0.825	27.288
1500	.281	10.979	11.25	3.010f	4.925s	0.992	0.995	30.123
1600	.320	12.486	12.80	2.943f	4.929s	1.059	1.055	31.968
1700	.361	14.087	14.44	2.876f	4.934s	1.126	1.010	33.008
1800	.405	15.780	16.17	2.811f	4.935s	1.194	1.166	35.784
1900	.451	17.576	18.02	2.744f	4.938s	1.261	1.262	38.398
2000	.499	19.466	19.95	2.678f	4.941s	1.328	1.307	40.474
2100	.549	21.406	21.94	2.618f	4.944s	1.393	1.198	40.879
2200	.599	23.352	23.94	2.568f	4.948s	1.456	1.106	41.198
2300	.649	25.304	25.94	2.524f	4.951s	1.517	1.026	41.482
2400	.699	27.259	27.94	2.487f	4.954s	1.577	0.955	41.627
2500	.749	29.216	29.95	2.454f	4.957s	1.636	0.893	41.737
2600	.799	31.175	31.95	2.426f	4.959s	1.693	0.837	41.760
2700	.849	33.133	33.96	2.401f	4.962s	1.750	0.788	41.783
2800	.900	35.092	35.97	2.378f	4.964s	1.805	0.744	41.807
2900	.950	37.052	37.98	2.358f	4.966s	1.861	0.705	41.831
3000	1.000	39.012	39.99	2.340f	4.967s	1.915		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK1.S Reference Point: Long.= 2.000f Trans.= 5.005s Vert.= 0.000
(Zero Sounding is at the Reference Point.)

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TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK3.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.031	4.013	4.11	16.000f	4.775p	0.050	68.243	73.332
200	.062	8.114	8.32	16.000f	4.800p	0.101	34.484	78.219
300	.094	12.304	12.61	16.000f	4.826p	0.152	23.226	83.319
400	.126	16.582	17.00	16.000f	4.851p	0.203	17.593	88.636
500	.160	20.948	21.47	16.000f	4.877p	0.254	14.210	94.165
600	.193	25.358	25.99	16.000f	4.899p	0.306	11.739	94.173
700	.227	29.768	30.51	16.000f	4.914p	0.357	10.000	94.173
800	.261	34.178	35.03	16.000f	4.925p	0.408	8.710	94.173
900	.294	38.588	39.55	16.000f	4.933p	0.458	7.714	94.173
1000	.328	42.998	44.07	16.000f	4.940p	0.509	6.923	94.173
1100	.361	47.408	48.59	16.000f	4.946p	0.559	6.279	94.173
1200	.395	51.818	53.11	16.000f	4.950p	0.609	5.745	94.173
1300	.429	56.228	57.63	16.000f	4.954p	0.659	5.294	94.173
1400	.462	60.638	62.15	16.000f	4.958p	0.710	4.909	94.173
1500	.496	65.048	66.67	16.000f	4.960p	0.760	4.576	94.173
1600	.529	69.458	71.19	16.000f	4.963p	0.810	4.286	94.173
1700	.563	73.868	75.71	16.000f	4.965p	0.860	4.030	94.173
1800	.597	78.278	80.23	16.000f	4.967p	0.910	3.803	94.173
1900	.630	82.688	84.75	16.000f	4.969p	0.960	3.600	94.173
2000	.664	87.098	89.28	16.000f	4.970p	1.011	3.418	94.173
2100	.697	91.508	93.80	16.000f	4.972p	1.061	3.253	94.173
2200	.731	95.918	98.32	16.000f	4.973p	1.111	3.103	94.173
2300	.765	100.328	102.84	16.000f	4.974p	1.161	2.967	94.173
2400	.798	104.738	107.36	16.000f	4.975p	1.211	2.842	94.173
2500	.832	109.148	111.88	16.000f	4.976p	1.261	2.727	94.173
2600	.866	113.558	116.40	16.000f	4.977p	1.311	2.621	94.173
2700	.899	117.968	120.92	16.000f	4.978p	1.361	2.523	94.173
2800	.933	122.378	125.44	16.000f	4.979p	1.411	2.432	94.173
2900	.966	126.788	129.96	16.000f	4.980p	1.461	2.348	94.173
3000	1.000	131.197	134.48	16.000f	4.980p	1.511		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK3.P Reference Point: Long.= 16.000f Trans.= 5.000p Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK3.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.033	4.410	4.52	16.000f	0.000	0.050	67.499	94.171
200	.067	8.820	9.04	16.000f	0.000	0.100	33.749	94.171
300	.100	13.230	13.56	16.000f	0.000	0.150	22.500	94.171
400	.133	17.640	18.08	16.000f	0.000	0.200	16.875	94.171
500	.167	22.050	22.60	16.000f	0.000	0.250	13.500	94.171
600	.200	26.460	27.12	16.000f	0.000	0.300	11.250	94.171
700	.233	30.870	31.64	16.000f	0.000	0.350	9.643	94.171
800	.267	35.280	36.16	16.000f	0.000	0.400	8.437	94.171
900	.300	39.690	40.68	16.000f	0.000	0.450	7.500	94.171
1000	.333	44.100	45.20	16.000f	0.000	0.500	6.750	94.171
1100	.367	48.510	49.72	16.000f	0.000	0.550	6.136	94.171
1200	.400	52.920	54.24	16.000f	0.000	0.600	5.625	94.171
1300	.433	57.330	58.76	16.000f	0.000	0.650	5.192	94.171
1400	.467	61.740	63.28	16.000f	0.000	0.700	4.821	94.171
1500	.500	66.150	67.80	16.000f	0.000	0.750	4.500	94.171
1600	.533	70.560	72.32	16.000f	0.000	0.800	4.219	94.171
1700	.567	74.970	76.84	16.000f	0.000	0.850	3.971	94.171
1800	.600	79.380	81.36	16.000f	0.000	0.900	3.750	94.171
1900	.633	83.790	85.88	16.000f	0.000	0.950	3.553	94.171
2000	.667	88.200	90.40	16.000f	0.000	1.000	3.375	94.171
2100	.700	92.610	94.92	16.000f	0.000	1.050	3.214	94.171
2200	.733	97.020	99.45	16.000f	0.000	1.100	3.068	94.171
2300	.767	101.430	103.97	16.000f	0.000	1.150	2.935	94.171
2400	.800	105.840	108.49	16.000f	0.000	1.200	2.812	94.171
2500	.833	110.250	113.01	16.000f	0.000	1.250	2.700	94.171
2600	.867	114.660	117.53	16.000f	0.000	1.300	2.596	94.171
2700	.900	119.070	122.05	16.000f	0.000	1.350	2.500	94.171
2800	.933	123.479	126.57	16.000f	0.000	1.400	2.411	94.171
2900	.967	127.889	131.09	16.000f	0.000	1.450	2.328	94.171
3000	1.000	132.299	135.61	16.000f	0.000	1.500		

Soundings in mm.---------Other distances in METERS.---------------------------
WBTK3.C Reference Point: Long.= 16.000f Trans.= 0.000 Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK3.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.031	4.013	4.11	16.000f	4.775s	0.050	68.243	73.332
200	.062	8.114	8.32	16.000f	4.800s	0.101	34.484	78.219
300	.094	12.304	12.61	16.000f	4.826s	0.152	23.226	83.319
400	.126	16.582	17.00	16.000f	4.851s	0.203	17.593	88.636
500	.160	20.948	21.47	16.000f	4.877s	0.254	14.210	94.165
600	.193	25.358	25.99	16.000f	4.899s	0.306	11.739	94.173
700	.227	29.768	30.51	16.000f	4.914s	0.357	10.000	94.173
800	.261	34.178	35.03	16.000f	4.925s	0.408	8.710	94.173
900	.294	38.588	39.55	16.000f	4.933s	0.458	7.714	94.173
1000	.328	42.998	44.07	16.000f	4.940s	0.509	6.923	94.173
1100	.361	47.408	48.59	16.000f	4.946s	0.559	6.279	94.173
1200	.395	51.818	53.11	16.000f	4.950s	0.609	5.745	94.173
1300	.429	56.228	57.63	16.000f	4.954s	0.659	5.294	94.173
1400	.462	60.638	62.15	16.000f	4.958s	0.710	4.909	94.173
1500	.496	65.048	66.67	16.000f	4.960s	0.760	4.576	94.173
1600	.529	69.458	71.19	16.000f	4.963s	0.810	4.286	94.173
1700	.563	73.868	75.71	16.000f	4.965s	0.860	4.030	94.173
1800	.597	78.278	80.23	16.000f	4.967s	0.910	3.803	94.173
1900	.630	82.688	84.75	16.000f	4.969s	0.960	3.600	94.173
2000	.664	87.098	89.28	16.000f	4.970s	1.011	3.418	94.173
2100	.697	91.508	93.80	16.000f	4.972s	1.061	3.253	94.173
2200	.731	95.918	98.32	16.000f	4.973s	1.111	3.103	94.173
2300	.765	100.328	102.84	16.000f	4.974s	1.161	2.967	94.173
2400	.798	104.738	107.36	16.000f	4.975s	1.211	2.842	94.173
2500	.832	109.148	111.88	16.000f	4.976s	1.261	2.727	94.173
2600	.866	113.558	116.40	16.000f	4.977s	1.311	2.621	94.173
2700	.899	117.968	120.92	16.000f	4.978s	1.361	2.523	94.173
2800	.933	122.378	125.44	16.000f	4.979s	1.411	2.432	94.173
2900	.966	126.788	129.96	16.000f	4.980s	1.461	2.348	94.173
3000	1.000	131.197	134.48	16.000f	4.980s	1.511		

Soundings in mm.---------Other distances in METERS.---------------------------
WBTK3.S Reference Point: Long.= 16.000f Trans.= 5.000s Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK4.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.031	4.013	4.11	34.000f	4.775p	0.050	68.237	73.332
200	.062	8.114	8.32	34.000f	4.800p	0.101	34.480	78.219
300	.094	12.304	12.61	34.000f	4.826p	0.152	23.227	83.318
400	.126	16.582	17.00	34.000f	4.851p	0.203	17.594	88.635
500	.160	20.948	21.47	34.000f	4.877p	0.254	14.209	94.165
600	.193	25.358	25.99	34.000f	4.899p	0.306	11.741	94.172
700	.227	29.768	30.51	34.000f	4.914p	0.357	10.001	94.172
800	.261	34.178	35.03	34.000f	4.925p	0.408	8.711	94.172
900	.294	38.588	39.55	34.000f	4.933p	0.458	7.715	94.172
1000	.328	42.998	44.07	34.000f	4.940p	0.509	6.924	94.172
1100	.361	47.408	48.59	34.000f	4.946p	0.559	6.280	94.172
1200	.395	51.818	53.11	34.000f	4.950p	0.609	5.745	94.172
1300	.429	56.228	57.63	34.000f	4.954p	0.659	5.295	94.172
1400	.462	60.638	62.15	34.000f	4.958p	0.710	4.910	94.172
1500	.496	65.048	66.67	34.000f	4.960p	0.760	4.577	94.172
1600	.529	69.458	71.19	34.000f	4.963p	0.810	4.286	94.172
1700	.563	73.868	75.71	34.000f	4.965p	0.860	4.030	94.172
1800	.597	78.278	80.23	34.000f	4.967p	0.910	3.803	94.172
1900	.630	82.688	84.75	34.000f	4.969p	0.960	3.600	94.172
2000	.664	87.098	89.27	34.000f	4.970p	1.011	3.418	94.172
2100	.697	91.508	93.80	34.000f	4.972p	1.061	3.253	94.172
2200	.731	95.918	98.32	34.000f	4.973p	1.111	3.104	94.172
2300	.765	100.328	102.84	34.000f	4.974p	1.161	2.967	94.172
2400	.798	104.738	107.36	34.000f	4.975p	1.211	2.842	94.172
2500	.832	109.148	111.88	34.000f	4.976p	1.261	2.728	94.172
2600	.866	113.558	116.40	34.000f	4.977p	1.311	2.622	94.172
2700	.899	117.968	120.92	34.000f	4.978p	1.361	2.524	94.172
2800	.933	122.378	125.44	34.000f	4.979p	1.411	2.433	94.172
2900	.966	126.788	129.96	34.000f	4.980p	1.461	2.348	94.172
3000	1.000	131.197	134.48	34.000f	4.980p	1.511		

Soundings in mm.---------Other distances in METERS.---------------------------
WBTK4.P Reference Point: Long.= 34.000f Trans.= 5.000p Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK4.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.033	4.410	4.52	34.000f	0.000	0.050	67.502	94.171
200	.067	8.820	9.04	34.000f	0.000	0.100	33.751	94.171
300	.100	13.230	13.56	34.000f	0.000	0.150	22.501	94.171
400	.133	17.640	18.08	34.000f	0.000	0.200	16.876	94.171
500	.167	22.050	22.60	34.000f	0.000	0.250	13.500	94.171
600	.200	26.460	27.12	34.000f	0.000	0.300	11.250	94.171
700	.233	30.870	31.64	34.000f	0.000	0.350	9.643	94.171
800	.267	35.280	36.16	34.000f	0.000	0.400	8.438	94.171
900	.300	39.690	40.68	34.000f	0.000	0.450	7.500	94.171
1000	.333	44.100	45.20	34.000f	0.000	0.500	6.750	94.171
1100	.367	48.510	49.72	34.000f	0.000	0.550	6.137	94.171
1200	.400	52.920	54.24	34.000f	0.000	0.600	5.625	94.171
1300	.433	57.330	58.76	34.000f	0.000	0.650	5.192	94.171
1400	.467	61.740	63.28	34.000f	0.000	0.700	4.822	94.171
1500	.500	66.150	67.80	34.000f	0.000	0.750	4.500	94.171
1600	.533	70.560	72.32	34.000f	0.000	0.800	4.219	94.171
1700	.567	74.970	76.84	34.000f	0.000	0.850	3.971	94.171
1800	.600	79.380	81.36	34.000f	0.000	0.900	3.750	94.171
1900	.633	83.790	85.88	34.000f	0.000	0.950	3.553	94.171
2000	.667	88.200	90.40	34.000f	0.000	1.000	3.375	94.171
2100	.700	92.610	94.92	34.000f	0.000	1.050	3.214	94.171
2200	.733	97.020	99.45	34.000f	0.000	1.100	3.068	94.171
2300	.767	101.430	103.97	34.000f	0.000	1.150	2.935	94.171
2400	.800	105.840	108.49	34.000f	0.000	1.200	2.813	94.171
2500	.833	110.249	113.01	34.000f	0.000	1.250	2.700	94.171
2600	.867	114.659	117.53	34.000f	0.000	1.300	2.596	94.171
2700	.900	119.069	122.05	34.000f	0.000	1.350	2.500	94.171
2800	.933	123.479	126.57	34.000f	0.000	1.400	2.411	94.171
2900	.967	127.889	131.09	34.000f	0.000	1.450	2.328	94.171
3000	1.000	132.299	135.61	34.000f	0.000	1.500		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK4.C Reference Point: Long.= 34.000f Trans.= 0.000 Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK4.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.031	4.013	4.11	34.000f	4.775s	0.050	68.237	73.332
200	.062	8.114	8.32	34.000f	4.800s	0.101	34.480	78.219
300	.094	12.304	12.61	34.000f	4.826s	0.152	23.227	83.318
400	.126	16.582	17.00	34.000f	4.851s	0.203	17.594	88.635
500	.160	20.948	21.47	34.000f	4.877s	0.254	14.209	94.165
600	.193	25.358	25.99	34.000f	4.899s	0.306	11.741	94.172
700	.227	29.768	30.51	34.000f	4.914s	0.357	10.001	94.172
800	.261	34.178	35.03	34.000f	4.925s	0.408	8.711	94.172
900	.294	38.588	39.55	34.000f	4.933s	0.458	7.715	94.172
1000	.328	42.998	44.07	34.000f	4.940s	0.509	6.924	94.172
1100	.361	47.408	48.59	34.000f	4.946s	0.559	6.280	94.172
1200	.395	51.818	53.11	34.000f	4.950s	0.609	5.745	94.172
1300	.429	56.228	57.63	34.000f	4.954s	0.659	5.295	94.172
1400	.462	60.638	62.15	34.000f	4.958s	0.710	4.910	94.172
1500	.496	65.048	66.67	34.000f	4.960s	0.760	4.577	94.172
1600	.529	69.458	71.19	34.000f	4.963s	0.810	4.286	94.172
1700	.563	73.868	75.71	34.000f	4.965s	0.860	4.030	94.172
1800	.597	78.278	80.23	34.000f	4.967s	0.910	3.803	94.172
1900	.630	82.688	84.75	34.000f	4.969s	0.960	3.600	94.172
2000	.664	87.098	89.27	34.000f	4.970s	1.011	3.418	94.172
2100	.697	91.508	93.80	34.000f	4.972s	1.061	3.253	94.172
2200	.731	95.918	98.32	34.000f	4.973s	1.111	3.104	94.172
2300	.765	100.328	102.84	34.000f	4.974s	1.161	2.967	94.172
2400	.798	104.738	107.36	34.000f	4.975s	1.211	2.842	94.172
2500	.832	109.148	111.88	34.000f	4.976s	1.261	2.728	94.172
2600	.866	113.558	116.40	34.000f	4.977s	1.311	2.622	94.172
2700	.899	117.968	120.92	34.000f	4.978s	1.361	2.524	94.172
2800	.933	122.378	125.44	34.000f	4.979s	1.411	2.433	94.172
2900	.966	126.788	129.96	34.000f	4.980s	1.461	2.348	94.172
3000	1.000	131.197	134.48	34.000f	4.980s	1.511		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK4.S Reference Point: Long.= 34.000f Trans.= 5.000s Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK5.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.027	2.964	3.04	42.266f	4.772p	0.044	53.336	60.667
200	.058	6.374	6.53	42.257f	4.798p	0.095	25.348	64.715
300	.090	9.857	10.10	42.255f	4.823p	0.145	16.739	68.945
400	.123	13.413	13.75	42.254f	4.849p	0.197	12.559	73.355
500	.156	17.043	17.47	42.253f	4.874p	0.248	10.087	77.949
600	.189	20.718	21.24	42.252f	4.897p	0.300	8.316	78.486
700	.223	24.393	25.00	42.252f	4.912p	0.351	7.062	78.486
800	.257	28.068	28.77	42.252f	4.924p	0.401	6.138	78.485
900	.290	31.743	32.54	42.252f	4.933p	0.452	5.428	78.485
1000	.324	35.419	36.30	42.251f	4.940p	0.502	4.865	78.485
1100	.358	39.094	40.07	42.251f	4.945p	0.553	4.407	78.484
1200	.391	42.769	43.84	42.251f	4.950p	0.603	4.029	78.484
1300	.425	46.444	47.61	42.251f	4.954p	0.653	3.710	78.484
1400	.458	50.119	51.37	42.251f	4.957p	0.704	3.437	78.483
1500	.492	53.794	55.14	42.251f	4.960p	0.754	3.203	78.483
1600	.526	57.469	58.91	42.251f	4.963p	0.804	2.998	78.482
1700	.559	61.144	62.67	42.251f	4.965p	0.854	2.818	78.482
1800	.593	64.819	66.44	42.251f	4.967p	0.904	2.658	78.482
1900	.626	68.494	70.21	42.251f	4.969p	0.954	2.515	78.481
2000	.660	72.170	73.97	42.251f	4.970p	1.005	2.387	78.481
2100	.694	75.845	77.74	42.251f	4.972p	1.055	2.271	78.480
2200	.727	79.520	81.51	42.251f	4.973p	1.105	2.167	78.480
2300	.761	83.195	85.27	42.251f	4.974p	1.155	2.071	78.480
2400	.794	86.870	89.04	42.251f	4.975p	1.205	1.983	78.479
2500	.828	90.545	92.81	42.251f	4.976p	1.255	1.903	78.479
2600	.862	94.220	96.58	42.251f	4.977p	1.305	1.828	78.479
2700	.895	97.895	100.34	42.251f	4.978p	1.355	1.760	78.478
2800	.929	101.570	104.11	42.251f	4.979p	1.405	1.696	78.478
2900	.962	105.245	107.88	42.251f	4.980p	1.455	1.637	78.477
3000	.996	108.920	111.64	42.251f	4.980p	1.505	1.582	78.477
3012	1.000	109.348	112.08	42.251f	4.980p	1.511		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK5.P Reference Point: Long.= 42.250f Trans.= 5.003p Vert.= -0.012
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK5.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.030	3.261	3.34	42.265f	0.000	0.044	52.821	78.476
200	.063	6.936	7.11	42.257f	0.000	0.094	24.836	78.476
300	.096	10.611	10.88	42.255f	0.000	0.144	16.234	78.476
400	.130	14.286	14.64	42.253f	0.000	0.194	12.058	78.476
500	.163	17.961	18.41	42.253f	0.000	0.244	9.591	78.476
600	.196	21.636	22.18	42.252f	0.000	0.294	7.962	78.476
700	.230	25.311	25.94	42.252f	0.000	0.344	6.806	78.476
800	.263	28.986	29.71	42.252f	0.000	0.394	5.943	78.476
900	.296	32.661	33.48	42.252f	0.000	0.444	5.274	78.476
1000	.330	36.336	37.24	42.251f	0.000	0.494	4.741	78.476
1100	.363	40.011	41.01	42.251f	0.000	0.544	4.306	78.476
1200	.396	43.686	44.78	42.251f	0.000	0.594	3.943	78.476
1300	.430	47.361	48.55	42.251f	0.000	0.644	3.637	78.476
1400	.463	51.036	52.31	42.251f	0.000	0.694	3.375	78.476
1500	.496	54.711	56.08	42.251f	0.000	0.744	3.149	78.476
1600	.530	58.386	59.85	42.251f	0.000	0.794	2.951	78.476
1700	.563	62.061	63.61	42.251f	0.000	0.844	2.776	78.476
1800	.596	65.736	67.38	42.251f	0.000	0.894	2.621	78.476
1900	.630	69.411	71.15	42.251f	0.000	0.944	2.482	78.476
2000	.663	73.086	74.91	42.251f	0.000	0.994	2.357	78.476
2100	.696	76.761	78.68	42.251f	0.000	1.044	2.244	78.476
2200	.729	80.436	82.45	42.251f	0.000	1.094	2.142	78.476
2300	.763	84.111	86.21	42.251f	0.000	1.144	2.048	78.476
2400	.796	87.786	89.98	42.251f	0.000	1.194	1.962	78.476
2500	.829	91.461	93.75	42.251f	0.000	1.244	1.884	78.476
2600	.863	95.136	97.51	42.251f	0.000	1.294	1.811	78.476
2700	.896	98.811	101.28	42.250f	0.000	1.344	1.743	78.476
2800	.929	102.486	105.05	42.250f	0.000	1.394	1.681	78.476
2900	.963	106.161	108.82	42.250f	0.000	1.444	1.623	78.476
3000	.996	109.836	112.58	42.250f	0.000	1.494	1.568	78.476
3012	1.000	110.264	113.02	42.250f	0.000	1.500		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK5.C Reference Point: Long.= 42.250f Trans.= 0.000 Vert.= -0.012
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK5.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.027	2.964	3.04	42.266f	4.772s	0.044	53.336	60.667
200	.058	6.374	6.53	42.257f	4.798s	0.095	25.348	64.715
300	.090	9.857	10.10	42.255f	4.823s	0.145	16.739	68.945
400	.123	13.413	13.75	42.254f	4.849s	0.197	12.559	73.355
500	.156	17.043	17.47	42.253f	4.874s	0.248	10.087	77.949
600	.189	20.718	21.24	42.252f	4.897s	0.300	8.316	78.486
700	.223	24.393	25.00	42.252f	4.912s	0.351	7.062	78.486
800	.257	28.068	28.77	42.252f	4.924s	0.401	6.138	78.485
900	.290	31.743	32.54	42.252f	4.933s	0.452	5.428	78.485
1000	.324	35.419	36.30	42.251f	4.940s	0.502	4.865	78.485
1100	.358	39.094	40.07	42.251f	4.945s	0.553	4.407	78.484
1200	.391	42.769	43.84	42.251f	4.950s	0.603	4.029	78.484
1300	.425	46.444	47.61	42.251f	4.954s	0.653	3.710	78.484
1400	.458	50.119	51.37	42.251f	4.957s	0.704	3.437	78.483
1500	.492	53.794	55.14	42.251f	4.960s	0.754	3.203	78.483
1600	.526	57.469	58.91	42.251f	4.963s	0.804	2.998	78.482
1700	.559	61.144	62.67	42.251f	4.965s	0.854	2.818	78.482
1800	.593	64.819	66.44	42.251f	4.967s	0.904	2.658	78.482
1900	.626	68.494	70.21	42.251f	4.969s	0.954	2.515	78.481
2000	.660	72.170	73.97	42.251f	4.970s	1.005	2.387	78.481
2100	.694	75.845	77.74	42.251f	4.972s	1.055	2.271	78.480
2200	.727	79.520	81.51	42.251f	4.973s	1.105	2.167	78.480
2300	.761	83.195	85.27	42.251f	4.974s	1.155	2.071	78.480
2400	.794	86.870	89.04	42.251f	4.975s	1.205	1.983	78.479
2500	.828	90.545	92.81	42.251f	4.976s	1.255	1.903	78.479
2600	.862	94.220	96.58	42.251f	4.977s	1.305	1.828	78.479
2700	.895	97.895	100.34	42.251f	4.978s	1.355	1.760	78.478
2800	.929	101.570	104.11	42.251f	4.979s	1.405	1.696	78.478
2900	.962	105.245	107.88	42.251f	4.980s	1.455	1.637	78.477
3000	.996	108.920	111.64	42.251f	4.980s	1.505	1.582	78.477
3012	1.000	109.348	112.08	42.251f	4.980s	1.511		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK5.S Reference Point: Long.= 42.250f Trans.= 5.003s Vert.= -0.012
(Zero Sounding is at the Reference Point.)

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GHS 16.50

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK6.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.001	0.054	0.05	46.080f	4.536p	0.056	0.033	0.486
200	.005	0.201	0.21	46.153f	4.720p	0.133	0.203	3.952
300	.012	0.469	0.48	46.234f	4.784p	0.202	0.245	5.459
400	.021	0.829	0.85	46.302f	4.808p	0.266	0.316	7.643
500	.033	1.280	1.31	46.365f	4.825p	0.332	0.358	9.389
600	.047	1.830	1.88	46.429f	4.845p	0.398	0.414	11.307
700	.063	2.475	2.54	46.493f	4.861p	0.464	0.497	13.927
800	.082	3.215	3.30	46.556f	4.876p	0.530	0.553	15.907
900	.104	4.050	4.15	46.619f	4.887p	0.596	0.614	17.872
1000	.128	4.980	5.10	46.682f	4.896p	0.663	0.676	19.932
1100	.154	6.005	6.16	46.746f	4.904p	0.729	0.736	21.927
1200	.183	7.126	7.30	46.810f	4.911p	0.795	0.801	24.036
1300	.214	8.332	8.54	46.872f	4.917p	0.861	0.776	25.371
1400	.246	9.593	9.83	46.928f	4.920p	0.926	0.825	27.268
1500	.281	10.979	11.25	46.990f	4.925p	0.992	0.995	30.116
1600	.320	12.486	12.80	47.057f	4.929p	1.059	1.054	31.960
1700	.361	14.087	14.44	47.124f	4.934p	1.126	1.009	32.987
1800	.405	15.780	16.17	47.189f	4.935p	1.194	1.166	35.771
1900	.451	17.576	18.02	47.256f	4.938p	1.261	1.262	38.392
2000	.499	19.466	19.95	47.322f	4.941p	1.328	1.306	40.468
2100	.549	21.406	21.94	47.382f	4.944p	1.393	1.198	40.873
2200	.599	23.352	23.94	47.432f	4.948p	1.456	1.105	41.193
2300	.649	25.304	25.94	47.476f	4.951p	1.517	1.027	41.478
2400	.699	27.259	27.94	47.513f	4.954p	1.577	0.955	41.624
2500	.749	29.216	29.95	47.546f	4.957p	1.636	0.894	41.736
2600	.799	31.175	31.95	47.574f	4.959p	1.693	0.838	41.760
2700	.849	33.133	33.96	47.599f	4.962p	1.750	0.789	41.783
2800	.900	35.092	35.97	47.622f	4.964p	1.805	0.744	41.807
2900	.950	37.052	37.98	47.642f	4.966p	1.861	0.706	41.831
3000	1.000	39.012	39.99	47.660f	4.967p	1.915		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK6.P Reference Point: Long.= 48.000f Trans.= 5.005p Vert.= 0.000
(Zero Sounding is at the Reference Point.)

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK6.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.002	0.068	0.07	46.092f	0.000	0.056	0.039	0.715
200	.006	0.245	0.25	46.164f	0.000	0.130	0.184	5.060
300	.013	0.532	0.55	46.235f	0.000	0.197	0.248	7.182
400	.023	0.916	0.94	46.301f	0.000	0.261	0.306	9.219
500	.035	1.393	1.43	46.366f	0.000	0.327	0.362	11.227
600	.050	1.967	2.02	46.430f	0.000	0.392	0.424	13.283
700	.067	2.635	2.70	46.494f	0.000	0.458	0.486	15.298
800	.086	3.399	3.48	46.559f	0.000	0.524	0.546	17.330
900	.108	4.258	4.36	46.623f	0.000	0.590	0.609	19.374
1000	.132	5.213	5.34	46.688f	0.000	0.656	0.673	21.417
1100	.158	6.264	6.42	46.752f	0.000	0.722	0.736	23.461
1200	.187	7.411	7.60	46.817f	0.000	0.788	0.798	25.505
1300	.218	8.646	8.86	46.881f	0.000	0.855	0.833	26.837
1400	.252	9.974	10.22	46.944f	0.000	0.921	0.927	29.593
1500	.288	11.400	11.69	47.008f	0.000	0.987	0.992	31.646
1600	.326	12.931	13.25	47.073f	0.000	1.054	1.059	33.731
1700	.368	14.557	14.92	47.139f	0.000	1.120	1.120	35.750
1800	.411	16.278	16.69	47.204f	0.000	1.187	1.187	37.839
1900	.457	18.097	18.55	47.270f	0.000	1.254	1.251	39.897
2000	.505	20.010	20.51	47.335f	0.000	1.320	1.306	41.854
2100	.555	21.970	22.52	47.394f	0.000	1.385	1.190	41.854
2200	.604	23.930	24.53	47.444f	0.000	1.448	1.093	41.854
2300	.654	25.890	26.54	47.486f	0.000	1.509	1.010	41.854
2400	.703	27.850	28.55	47.522f	0.000	1.568	0.939	41.854
2500	.753	29.810	30.56	47.554f	0.000	1.626	0.877	41.854
2600	.802	31.770	32.56	47.581f	0.000	1.683	0.823	41.854
2700	.852	33.730	34.57	47.606f	0.000	1.739	0.775	41.854
2800	.901	35.690	36.58	47.627f	0.000	1.795	0.733	41.854
2900	.951	37.650	38.59	47.647f	0.000	1.850	0.694	41.854
3000	1.000	39.610	40.60	47.664f	0.000	1.904		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK6.C Reference Point: Long.= 48.000f Trans.= 0.000 Vert.= 0.000
(Zero Sounding is at the Reference Point.)

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GHS 16.50

TANK CHARACTERISTICS
No Trim, No Heel
Tank: WBTK6.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity LCG	TCG	VCG	GML	FSM m.-MT
0	.000	0.000	0.00					
100	.001	0.054	0.05	46.080f	4.536s	0.056	0.033	0.486
200	.005	0.201	0.21	46.153f	4.720s	0.133	0.203	3.952
300	.012	0.469	0.48	46.234f	4.784s	0.202	0.241	5.459
400	.021	0.829	0.85	46.302f	4.808s	0.266	0.316	7.643
500	.033	1.280	1.31	46.365f	4.825s	0.332	0.358	9.389
600	.047	1.830	1.88	46.429f	4.845s	0.398	0.414	11.307
700	.063	2.475	2.54	46.493f	4.861s	0.464	0.497	13.927
800	.082	3.215	3.30	46.556f	4.876s	0.530	0.553	15.907
900	.104	4.050	4.15	46.619f	4.887s	0.596	0.614	17.872
1000	.128	4.980	5.10	46.682f	4.896s	0.663	0.677	19.932
1100	.154	6.005	6.16	46.746f	4.904s	0.729	0.736	21.927
1200	.183	7.126	7.30	46.810f	4.911s	0.795	0.801	24.036
1300	.214	8.332	8.54	46.872f	4.917s	0.861	0.776	25.371
1400	.246	9.593	9.83	46.928f	4.920s	0.926	0.825	27.268
1500	.281	10.979	11.25	46.990f	4.925s	0.992	0.995	30.116
1600	.320	12.486	12.80	47.057f	4.929s	1.059	1.054	31.960
1700	.361	14.087	14.44	47.124f	4.934s	1.126	1.009	32.987
1800	.405	15.780	16.17	47.189f	4.935s	1.194	1.166	35.771
1900	.451	17.576	18.02	47.256f	4.938s	1.261	1.262	38.392
2000	.499	19.466	19.95	47.322f	4.941s	1.328	1.306	40.468
2100	.549	21.406	21.94	47.382f	4.944s	1.393	1.198	40.873
2200	.599	23.352	23.94	47.432f	4.948s	1.456	1.105	41.193
2300	.649	25.304	25.94	47.476f	4.951s	1.517	1.027	41.478
2400	.699	27.259	27.94	47.513f	4.954s	1.577	0.955	41.624
2500	.749	29.216	29.95	47.546f	4.957s	1.636	0.894	41.736
2600	.799	31.175	31.95	47.574f	4.959s	1.693	0.838	41.760
2700	.849	33.133	33.96	47.599f	4.962s	1.750	0.789	41.783
2800	.900	35.092	35.97	47.622f	4.964s	1.805	0.744	41.807
2900	.950	37.052	37.98	47.642f	4.966s	1.861	0.706	41.831
3000	1.000	39.012	39.99	47.660f	4.967s	1.915		

Soundings in mm.---------Other distances in METERS.--------------------------
WBTK6.S Reference Point: Long.= 48.000f Trans.= 5.005s Vert.= 0.000
(Zero Sounding is at the Reference Point.)

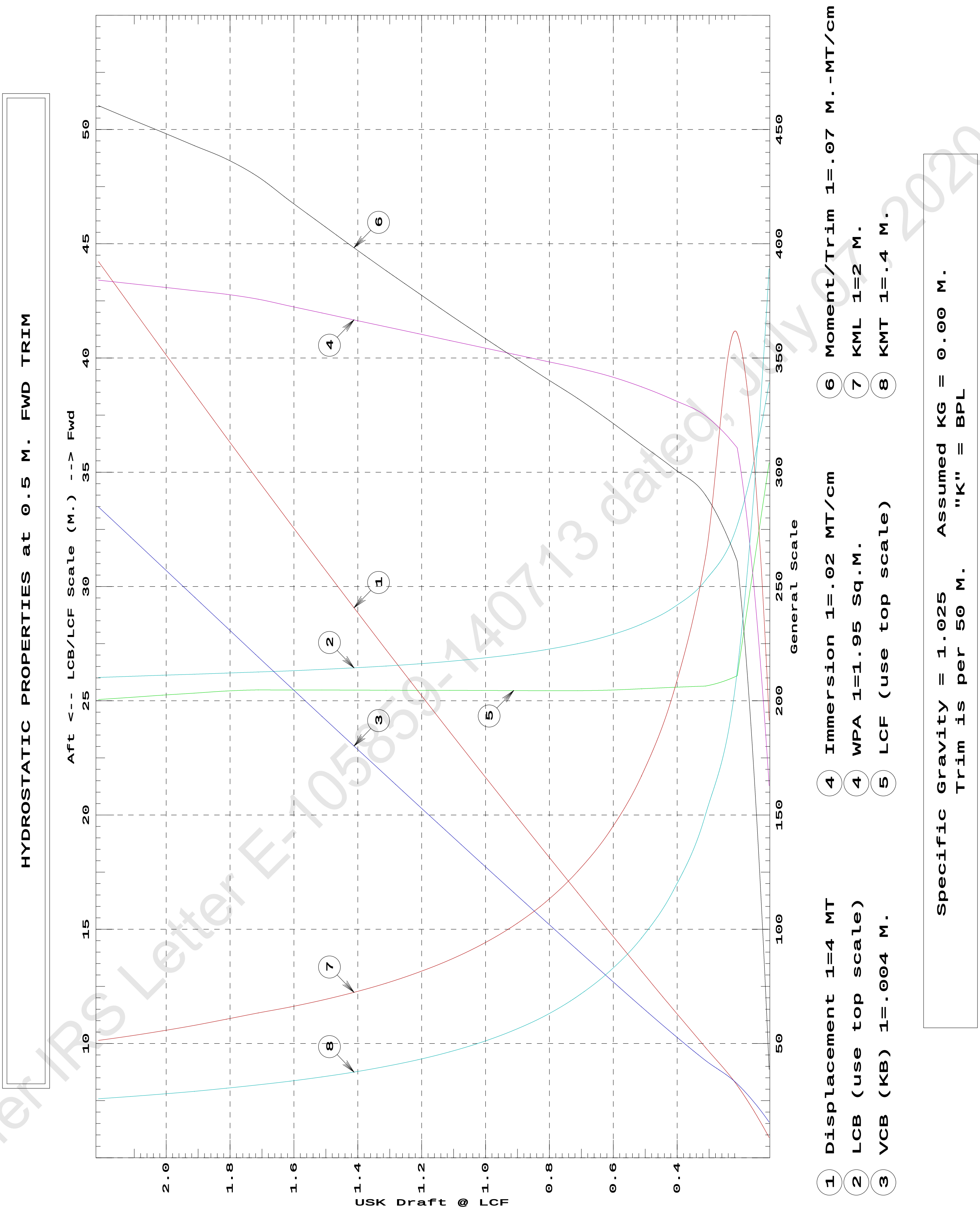
HYDROSTATIC PARTICULARS

HYDROSTATIC PROPERTIES
Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	35.31	38.933f	0.062	3.26	35.422f	2.70	382.72	155.640
0.212	126.67	32.680f	0.129	6.21	26.091f	18.28	721.37	85.218
0.312	193.60	30.258f	0.171	6.49	25.644f	20.31	524.43	59.944
0.412	259.63	29.079f	0.216	6.63	25.587f	21.12	406.76	46.782
0.512	326.98	28.354f	0.264	6.75	25.525f	21.85	334.10	38.458
0.612	395.34	27.859f	0.314	6.84	25.465f	22.58	285.53	32.636
0.712	464.24	27.501f	0.364	6.91	25.439f	23.27	250.56	28.275
0.812	533.64	27.233f	0.414	6.97	25.443f	23.89	223.80	24.906
0.912	603.65	27.026f	0.465	7.03	25.446f	24.52	203.11	22.305
1.012	674.25	26.861f	0.515	7.09	25.451f	25.17	186.64	20.238
1.112	745.46	26.726f	0.566	7.15	25.454f	25.83	173.21	18.554
1.212	817.28	26.614f	0.618	7.21	25.457f	26.50	162.12	17.166
1.312	889.68	26.520f	0.669	7.27	25.462f	27.18	152.75	15.996
1.412	962.69	26.440f	0.721	7.33	25.465f	27.88	144.78	15.003
1.512	1,036.38	26.371f	0.773	7.39	25.470f	28.59	137.93	14.149
1.612	1,110.58	26.311f	0.825	7.45	25.473f	29.31	131.94	13.408
1.712	1,185.40	26.258f	0.877	7.51	25.476f	30.05	126.74	12.763
1.812	1,261.19	26.210f	0.929	7.56	25.427f	30.60	121.30	12.168
1.912	1,337.62	26.163f	0.982	7.59	25.335f	31.00	115.89	11.627
2.012	1,414.37	26.115f	1.034	7.62	25.243f	31.42	111.05	11.147
2.112	1,491.44	26.068f	1.087	7.65	25.148f	31.82	106.66	10.716
2.212	1,568.83	26.020f	1.139	7.68	25.053f	32.23	102.71	10.332

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ LCF
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=2 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

HYDROSTATIC PROPERTIES

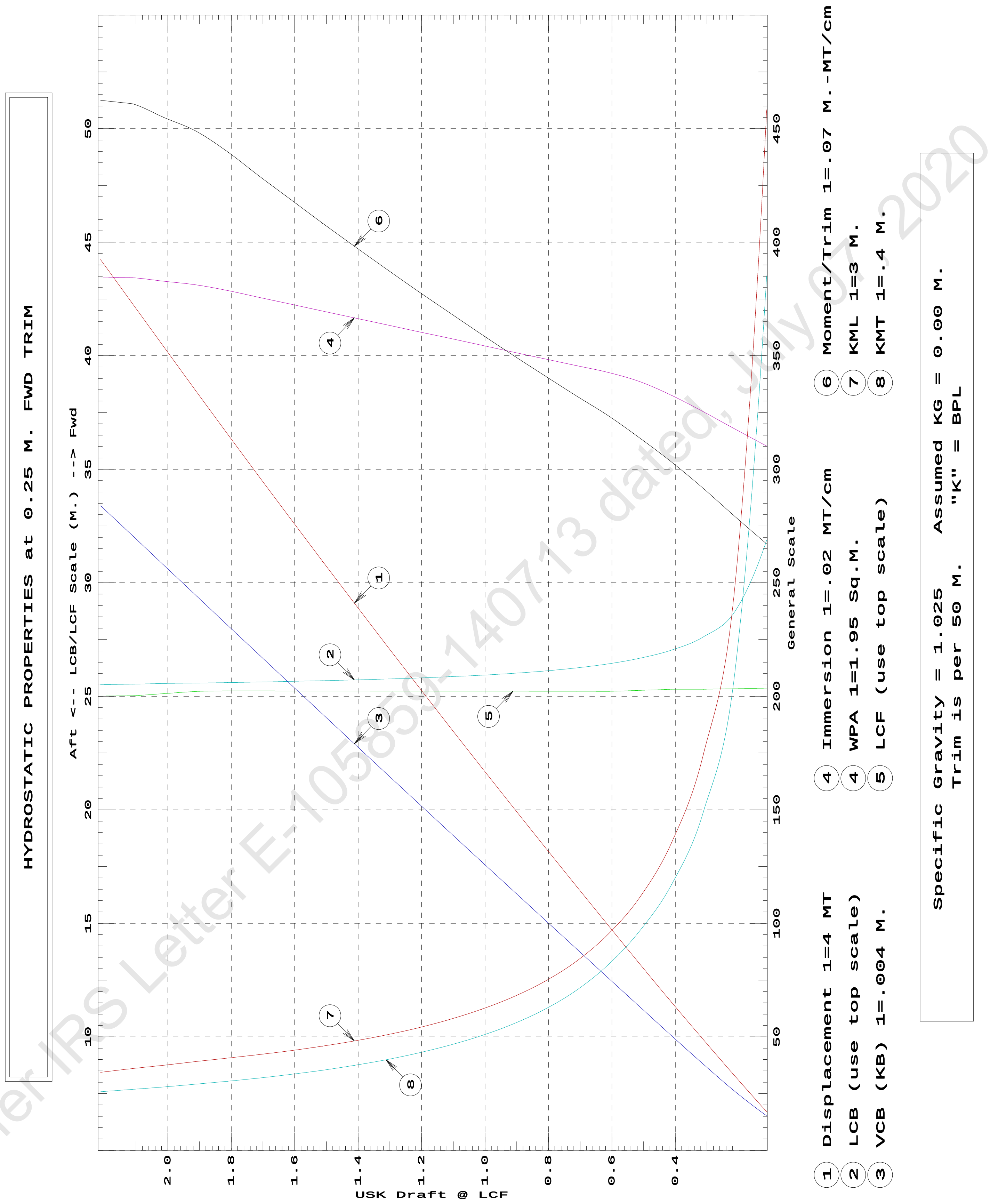
Trim: Fwd 0.250/50.000, No Heel, Fixed VCG = 0.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	67.99	31.878f	0.061	6.20	25.355f	18.70	1375.02	154.077
0.212	130.85	28.739f	0.104	6.35	25.333f	19.54	746.69	84.364
0.312	195.25	27.612f	0.152	6.51	25.309f	20.41	522.70	59.562
0.412	261.01	27.032f	0.202	6.65	25.309f	21.22	406.53	46.823
0.512	328.33	26.675f	0.253	6.77	25.258f	21.96	334.41	38.687
0.612	396.62	26.426f	0.304	6.85	25.218f	22.65	285.59	32.693
0.712	465.44	26.248f	0.355	6.91	25.219f	23.26	249.88	28.193
0.812	534.86	26.114f	0.406	6.97	25.221f	23.89	223.30	24.847
0.912	604.89	26.011f	0.458	7.03	25.223f	24.52	202.66	22.253
1.012	675.51	25.929f	0.509	7.09	25.225f	25.16	186.26	20.193
1.112	746.74	25.862f	0.561	7.15	25.227f	25.83	172.93	18.521
1.212	818.57	25.806f	0.613	7.21	25.229f	26.49	161.82	17.132
1.312	891.00	25.759f	0.664	7.27	25.230f	27.18	152.52	15.970
1.412	964.03	25.719f	0.716	7.33	25.233f	27.88	144.57	14.979
1.512	1,037.66	25.685f	0.768	7.39	25.234f	28.58	137.73	14.127
1.612	1,111.90	25.655f	0.821	7.45	25.237f	29.31	131.80	13.391
1.712	1,186.73	25.629f	0.873	7.51	25.238f	30.04	126.56	12.744
1.812	1,262.18	25.605f	0.925	7.57	25.240f	30.79	121.97	12.177
1.912	1,338.33	25.584f	0.978	7.62	25.209f	31.43	117.41	11.662
2.012	1,415.08	25.561f	1.031	7.66	25.117f	31.84	112.52	11.180
2.112	1,492.21	25.536f	1.083	7.69	25.024f	32.27	108.13	10.753
2.212	1,569.14	25.510f	1.136	7.69	25.009f	32.37	103.16	10.342

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at 0.25 M. FWD TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ LCF
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	68.53	25.000f	0.044	6.21	25.000f	18.72	1365.62	152.943
0.212	131.38	25.000f	0.095	6.36	25.000f	19.61	746.26	84.110
0.312	195.72	25.000f	0.146	6.51	25.000f	20.41	521.46	59.400
0.412	261.52	25.000f	0.197	6.65	25.000f	21.24	406.00	46.750
0.512	328.77	25.000f	0.249	6.79	25.000f	22.05	335.40	38.975
0.612	396.99	25.000f	0.301	6.85	25.000f	22.65	285.27	32.654
0.712	465.83	25.000f	0.352	6.91	25.000f	23.27	249.73	28.175
0.812	535.26	25.000f	0.404	6.97	25.000f	23.88	223.09	24.822
0.912	605.29	25.000f	0.456	7.03	25.000f	24.52	202.53	22.237
1.012	675.92	25.000f	0.507	7.09	25.000f	25.16	186.15	20.181
1.112	747.15	25.000f	0.559	7.15	25.000f	25.82	172.79	18.506
1.212	818.99	25.000f	0.611	7.21	25.000f	26.50	161.77	17.125
1.312	891.43	25.000f	0.663	7.27	25.000f	27.18	152.43	15.959
1.412	964.46	25.000f	0.715	7.33	25.000f	27.87	144.50	14.970
1.512	1,038.11	25.000f	0.767	7.39	25.000f	28.59	137.68	14.122
1.612	1,112.35	25.000f	0.819	7.45	25.000f	29.30	131.72	13.382
1.712	1,187.19	25.000f	0.872	7.51	25.000f	30.04	126.53	12.740
1.812	1,262.64	25.000f	0.924	7.57	25.000f	30.79	121.92	12.171
1.912	1,338.68	25.000f	0.977	7.63	25.000f	31.55	117.84	11.669
2.012	1,415.33	25.000f	1.029	7.69	25.000f	32.24	113.91	11.217
2.112	1,492.22	25.000f	1.082	7.69	25.000f	32.31	108.27	10.755
2.212	1,569.15	25.000f	1.134	7.69	25.000f	32.38	103.19	10.344

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

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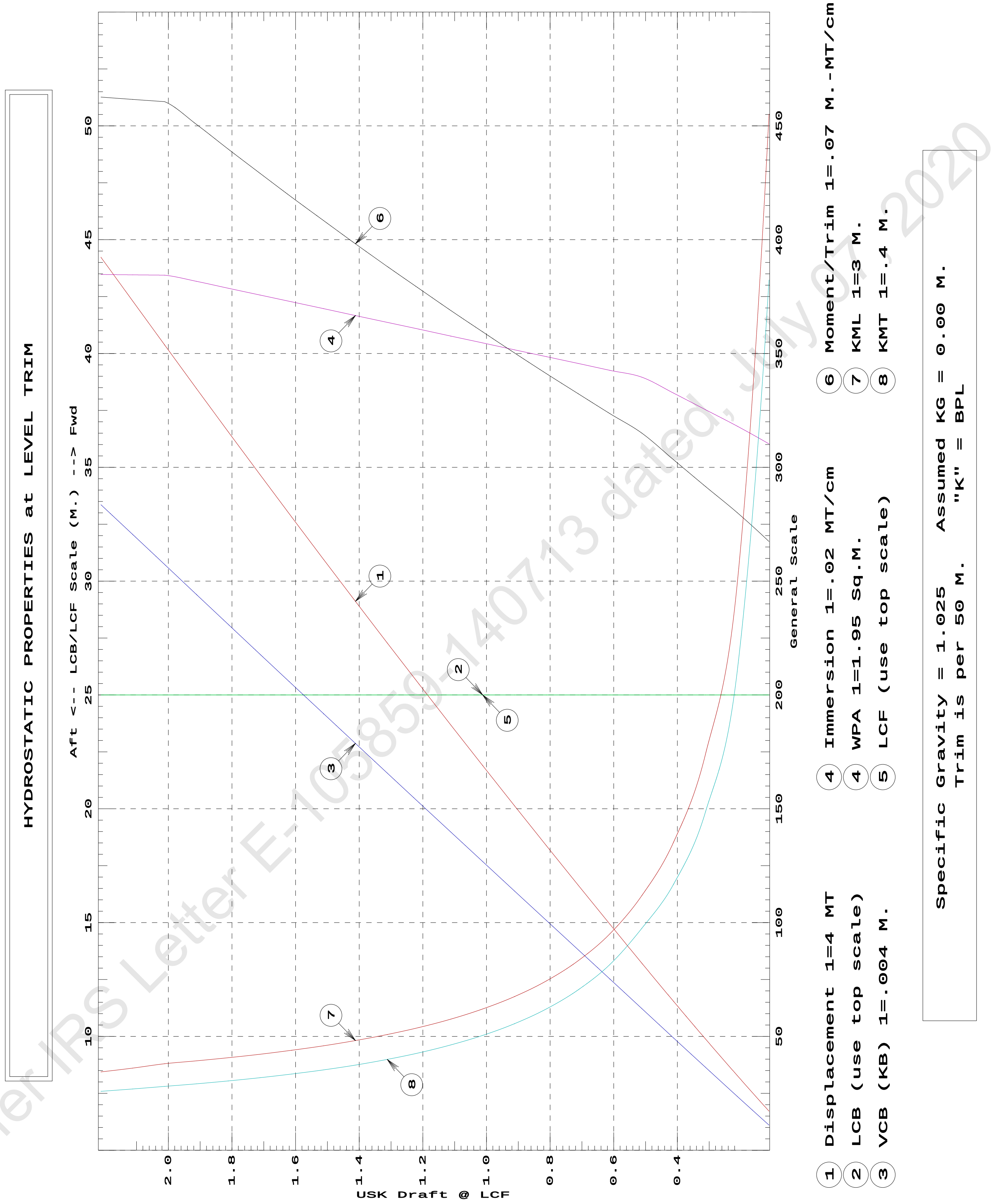


HYDROSTATIC PROPERTIES

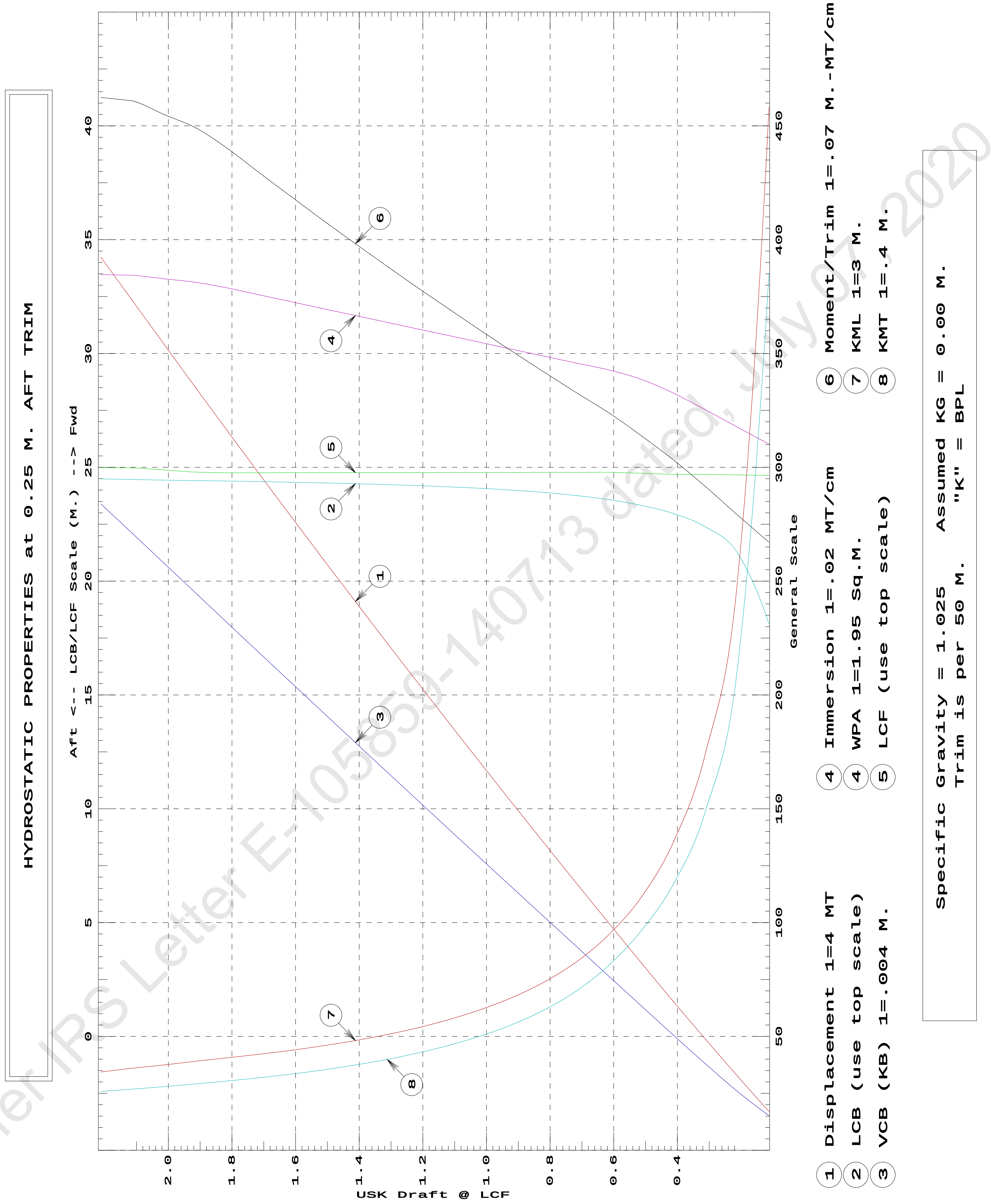
Trim: Aft 0.250/50.000, No Heel, Fixed VCG = 0.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	67.99	18.121f	0.061	6.20	24.645f	18.70	1375.05	154.081
0.212	130.84	21.260f	0.104	6.35	24.667f	19.54	746.72	84.368
0.312	195.22	22.387f	0.152	6.51	24.691f	20.41	522.75	59.569
0.412	260.99	22.967f	0.202	6.65	24.691f	21.22	406.56	46.826
0.512	328.32	23.325f	0.253	6.77	24.742f	21.96	334.41	38.687
0.612	396.63	23.574f	0.304	6.85	24.782f	22.65	285.59	32.692
0.712	465.44	23.752f	0.355	6.91	24.781f	23.26	249.88	28.193
0.812	534.88	23.886f	0.406	6.97	24.779f	23.89	223.29	24.846
0.912	604.91	23.989f	0.458	7.03	24.777f	24.52	202.66	22.253
1.012	675.53	24.071f	0.509	7.09	24.775f	25.16	186.25	20.192
1.112	746.76	24.138f	0.561	7.15	24.773f	25.83	172.92	18.521
1.212	818.59	24.194f	0.613	7.21	24.771f	26.49	161.82	17.132
1.312	891.02	24.241f	0.664	7.27	24.770f	27.18	152.52	15.969
1.412	964.05	24.281f	0.716	7.33	24.767f	27.88	144.57	14.978
1.512	1,037.68	24.315f	0.768	7.39	24.766f	28.58	137.72	14.127
1.612	1,111.92	24.345f	0.821	7.45	24.764f	29.31	131.80	13.391
1.712	1,186.75	24.371f	0.873	7.51	24.762f	30.04	126.56	12.744
1.812	1,262.19	24.395f	0.925	7.57	24.760f	30.79	121.97	12.177
1.912	1,338.33	24.416f	0.978	7.62	24.791f	31.43	117.42	11.662
2.012	1,415.08	24.439f	1.031	7.66	24.883f	31.84	112.52	11.180
2.112	1,492.15	24.464f	1.083	7.69	24.976f	32.27	108.13	10.753
2.212	1,569.13	24.490f	1.136	7.69	24.991f	32.37	103.16	10.342

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at 0.25 M. AFT TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ LCF
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025
Trim is per 50 M.
Assumed KG = 0.00 M.
"K" = BPL

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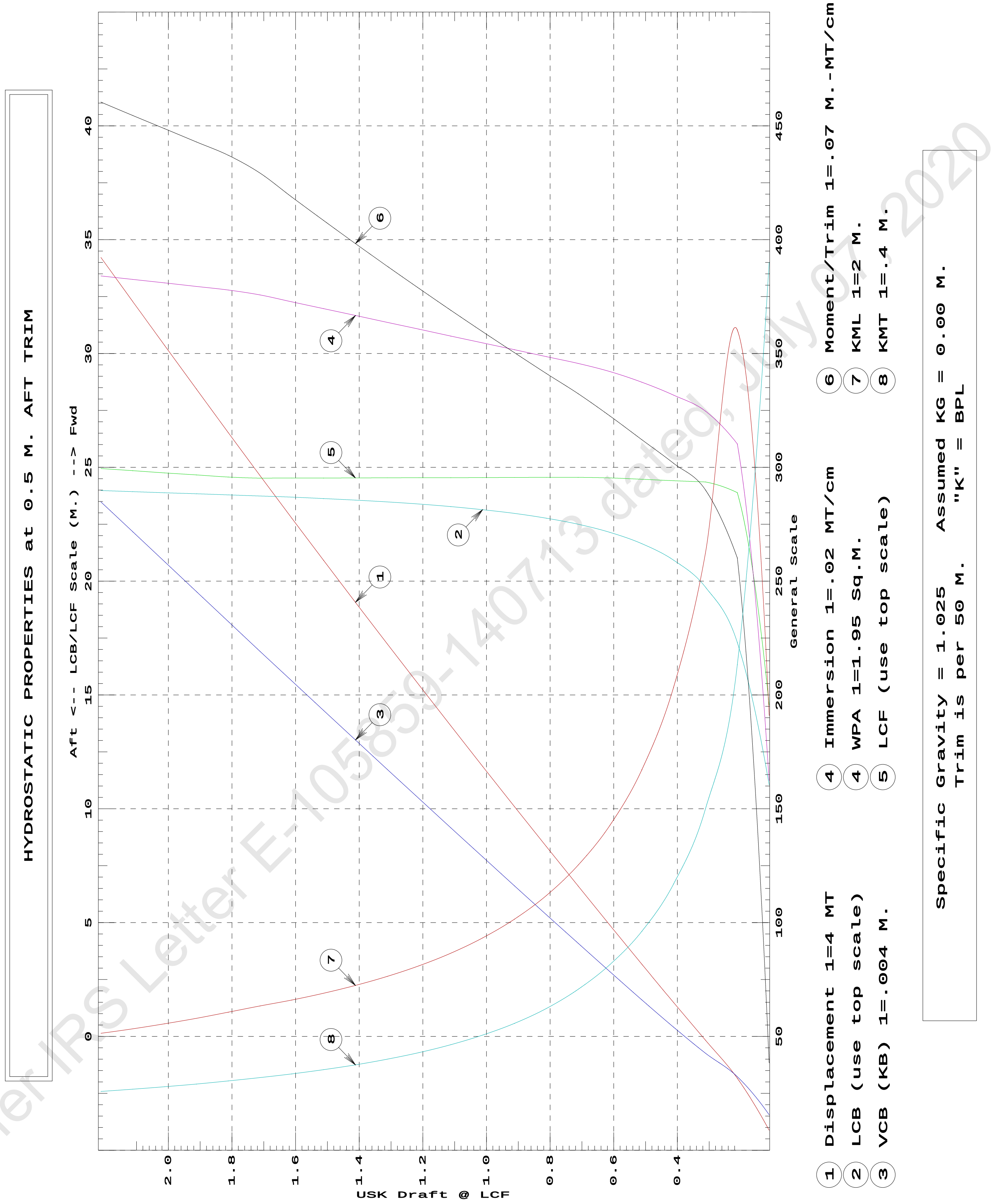
GHS 16.50

HYDROSTATIC PROPERTIES
Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

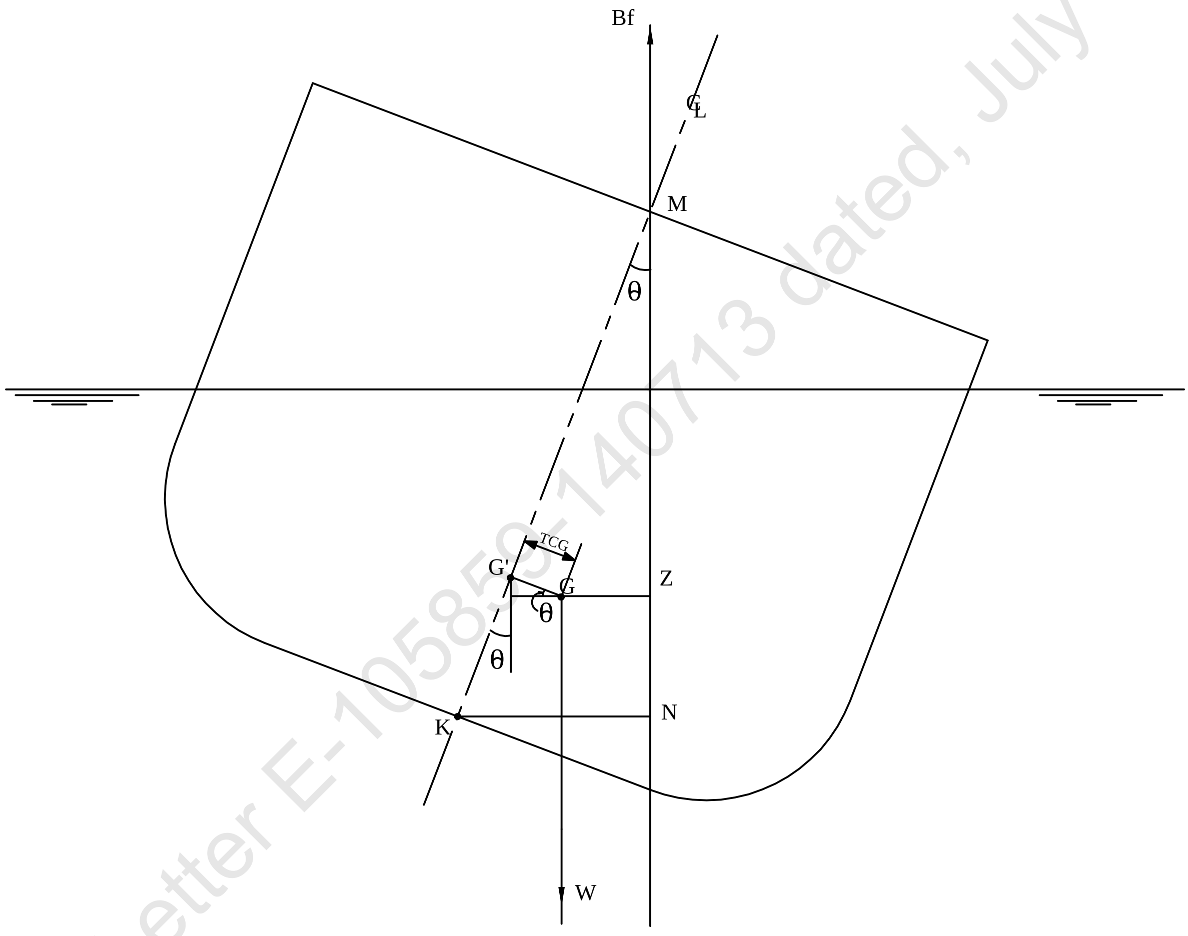
LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	35.14	11.050f	0.062	3.25	14.553f	2.68	381.82	155.997
0.212	126.38	17.305f	0.129	6.21	23.886f	18.21	720.54	85.309
0.312	193.59	19.742f	0.171	6.49	24.356f	20.31	524.44	59.946
0.412	259.63	20.921f	0.216	6.63	24.413f	21.12	406.77	46.784
0.512	326.97	21.646f	0.264	6.75	24.475f	21.85	334.11	38.459
0.612	395.34	22.141f	0.314	6.84	24.535f	22.58	285.54	32.637
0.712	464.31	22.499f	0.364	6.91	24.561f	23.27	250.53	28.272
0.812	533.71	22.767f	0.414	6.97	24.557f	23.89	223.77	24.904
0.912	603.73	22.974f	0.465	7.03	24.554f	24.52	203.09	22.303
1.012	674.39	23.140f	0.515	7.09	24.549f	25.17	186.61	20.234
1.112	745.60	23.274f	0.567	7.15	24.546f	25.83	173.19	18.551
1.212	817.41	23.386f	0.618	7.21	24.543f	26.50	162.10	17.163
1.312	889.82	23.480f	0.669	7.27	24.538f	27.18	152.73	15.994
1.412	962.83	23.560f	0.721	7.33	24.535f	27.88	144.77	15.001
1.512	1,036.44	23.629f	0.773	7.39	24.530f	28.59	137.92	14.149
1.612	1,110.65	23.689f	0.825	7.45	24.527f	29.31	131.94	13.407
1.712	1,185.47	23.742f	0.877	7.51	24.524f	30.05	126.73	12.762
1.812	1,261.17	23.790f	0.929	7.56	24.573f	30.60	121.30	12.168
1.912	1,337.60	23.837f	0.982	7.59	24.665f	31.00	115.89	11.627
2.012	1,414.34	23.885f	1.034	7.62	24.757f	31.42	111.06	11.148
2.112	1,491.42	23.932f	1.087	7.65	24.852f	31.82	106.67	10.716
2.212	1,568.81	23.980f	1.139	7.68	24.947f	32.23	102.71	10.332

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

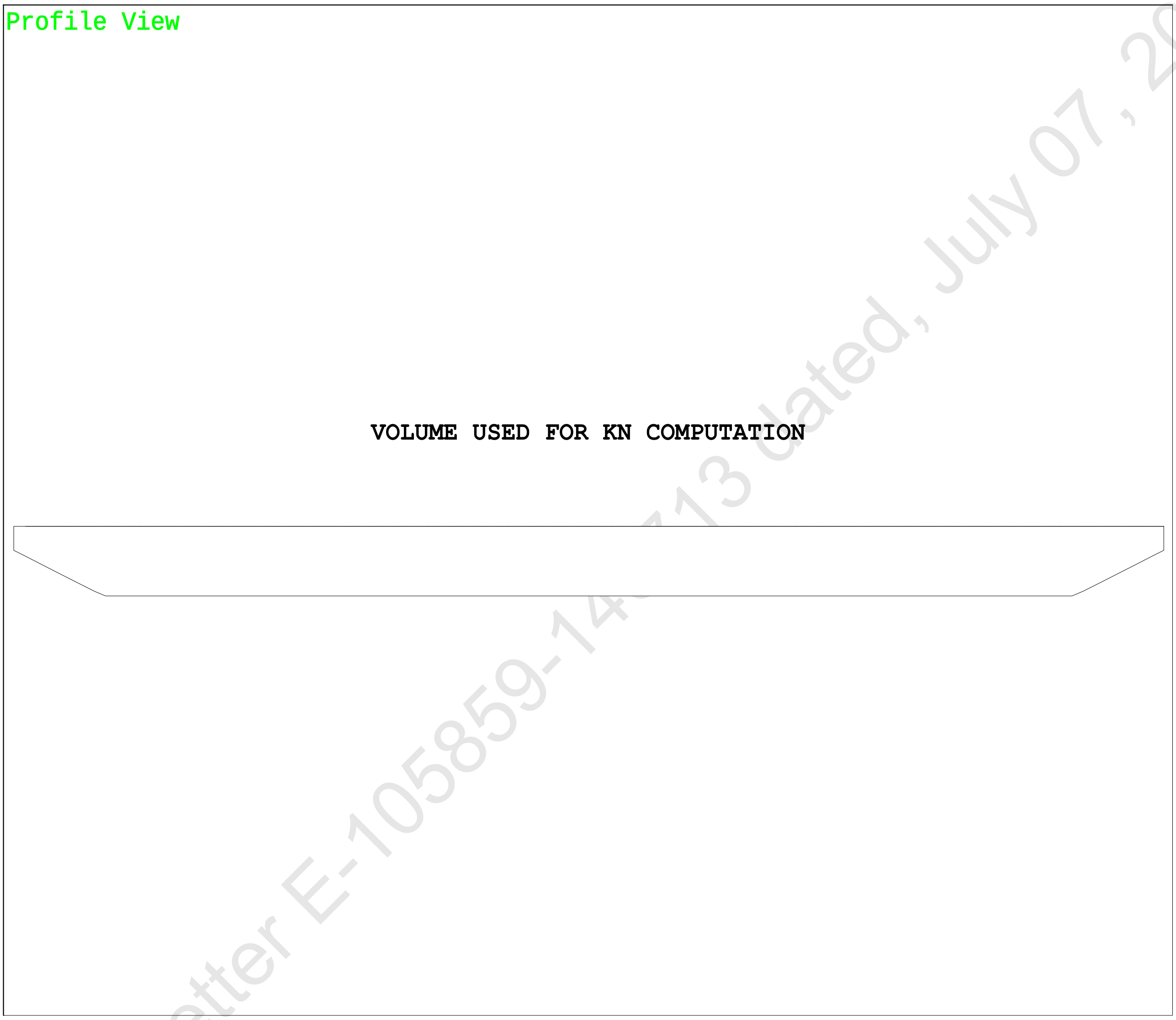
Draft is from USK.


HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
0 5 10 15 20 25 30 35 40
General Scale
50 100 150 200 250 300 350 400 450
USK Draft @ LCF
2.0 1.8 1.6 1.4 1.2 1.0 0.8 0.6 0.4
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=2 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025   Assumed KG = 0.00 M.
Trim is per 50 M.   "K" = BPL

<u>KN CONCEPT</u>



GZ = KN - KG' SIN θ - TCG COS θ

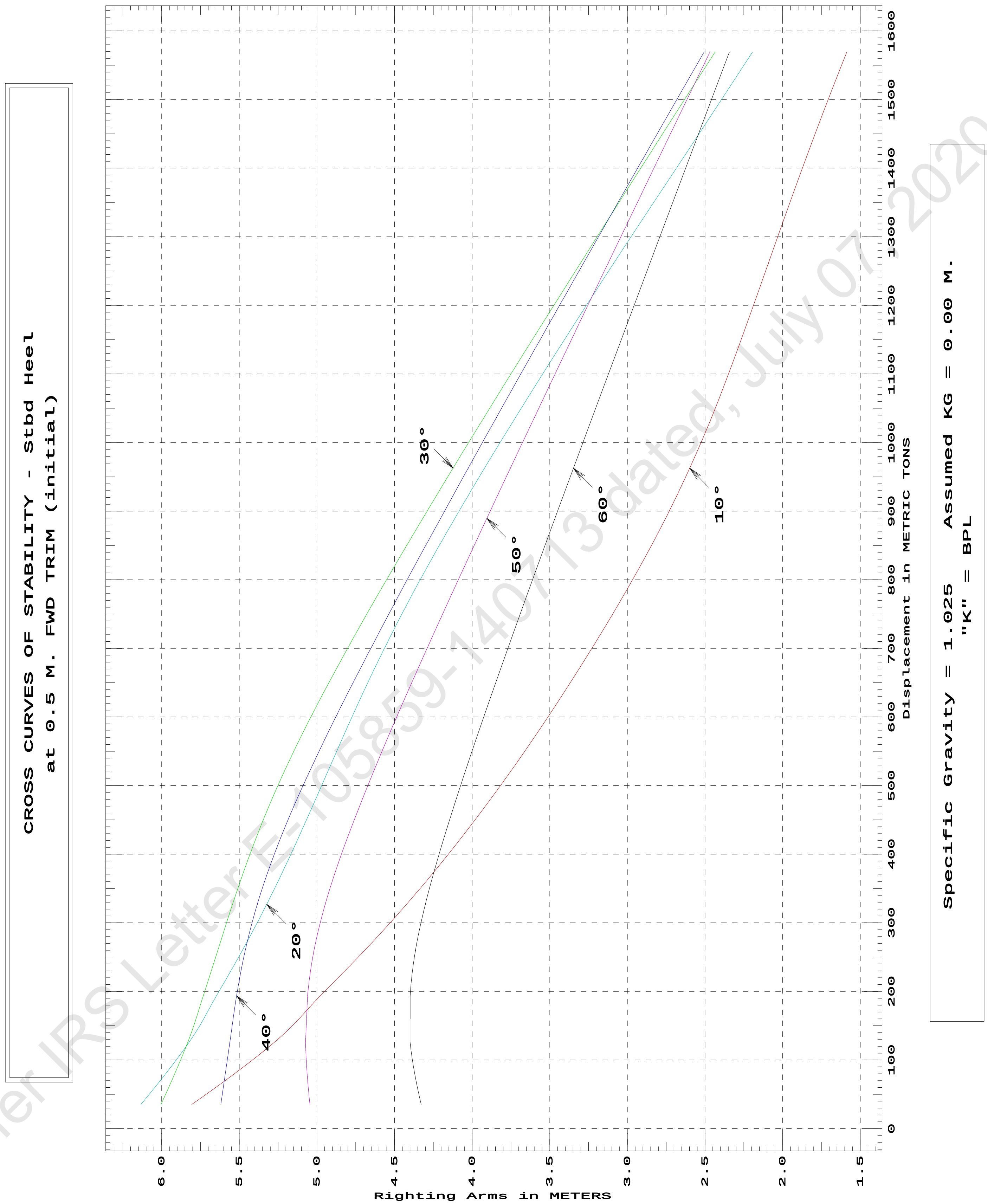

Profile View
VOLUME USED FOR KN COMPUTATION

CROSS CURVE TABLES

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
35.31	5.807s	6.134s	6.004s	5.619s	5.045s	4.329s
126.67	5.270s	5.821s	5.830s	5.560s	5.073s	4.400s
193.60	4.978s	5.647s	5.732s	5.517s	5.060s	4.400s
259.63	4.693s	5.480s	5.639s	5.462s	5.019s	4.366s
326.98	4.423s	5.323s	5.543s	5.384s	4.947s	4.301s
395.34	4.170s	5.176s	5.439s	5.282s	4.849s	4.219s
464.24	3.934s	5.039s	5.319s	5.159s	4.735s	4.126s
533.64	3.712s	4.905s	5.183s	5.020s	4.611s	4.025s
603.65	3.501s	4.767s	5.030s	4.870s	4.479s	3.920s
674.25	3.300s	4.621s	4.865s	4.712s	4.342s	3.811s
745.46	3.107s	4.463s	4.690s	4.547s	4.200s	3.698s
817.28	2.924s	4.293s	4.507s	4.377s	4.054s	3.584s
889.68	2.754s	4.111s	4.318s	4.203s	3.905s	3.467s
962.69	2.599s	3.920s	4.123s	4.026s	3.754s	3.348s
1,036.38	2.458s	3.722s	3.924s	3.844s	3.600s	3.227s
1,110.58	2.331s	3.516s	3.721s	3.661s	3.444s	3.105s
1,185.40	2.211s	3.305s	3.514s	3.474s	3.286s	2.981s
1,261.19	2.093s	3.088s	3.303s	3.284s	3.125s	2.855s
1,337.62	1.973s	2.867s	3.089s	3.091s	2.962s	2.728s
1,414.37	1.850s	2.642s	2.873s	2.898s	2.799s	2.601s
1,491.44	1.721s	2.418s	2.656s	2.703s	2.634s	2.473s
1,568.83	1.588s	2.195s	2.436s	2.507s	2.469s	2.344s

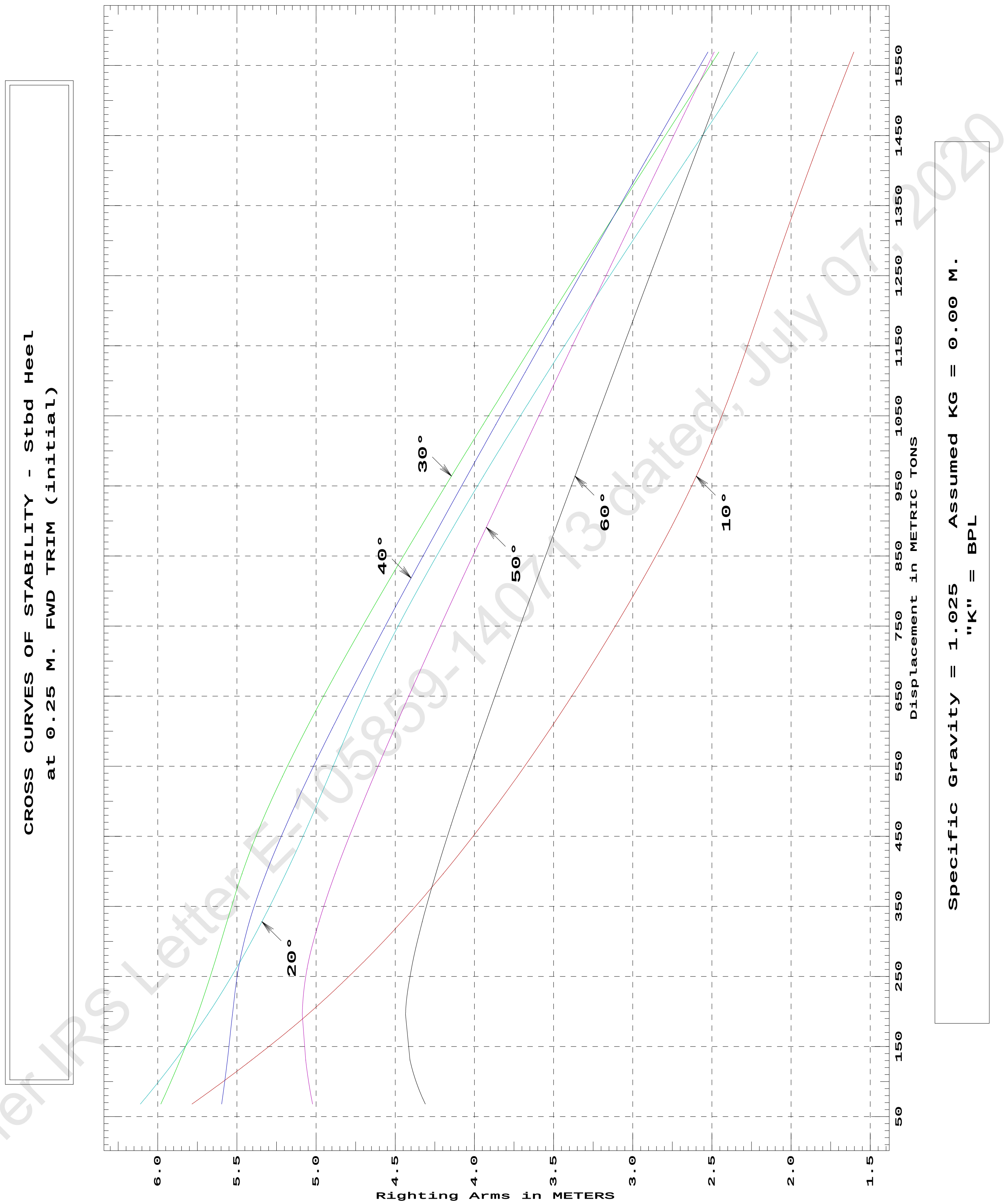
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
0
100
200
300
400
500
600
700
800
900
1000
1100
1200
1300
1400
1500
1600
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement	Heel Angles in Degrees					
METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
67.99	5.785s	6.110s	5.981s	5.596s	5.022s	4.310s
130.85	5.407s	5.889s	5.859s	5.560s	5.066s	4.407s
195.25	5.056s	5.686s	5.750s	5.529s	5.087s	4.435s
261.01	4.744s	5.506s	5.653s	5.490s	5.057s	4.398s
328.33	4.459s	5.342s	5.563s	5.421s	4.981s	4.329s
396.62	4.197s	5.192s	5.468s	5.317s	4.880s	4.244s
465.44	3.956s	5.053s	5.352s	5.190s	4.763s	4.149s
534.86	3.730s	4.921s	5.214s	5.048s	4.637s	4.047s
604.89	3.516s	4.789s	5.058s	4.896s	4.503s	3.940s
675.51	3.313s	4.647s	4.891s	4.737s	4.364s	3.830s
746.74	3.119s	4.489s	4.714s	4.571s	4.221s	3.717s
818.57	2.932s	4.317s	4.530s	4.400s	4.075s	3.601s
891.00	2.757s	4.135s	4.340s	4.225s	3.925s	3.484s
964.03	2.599s	3.943s	4.145s	4.046s	3.773s	3.365s
1,037.66	2.459s	3.744s	3.945s	3.865s	3.619s	3.244s
1,111.90	2.334s	3.538s	3.742s	3.680s	3.462s	3.121s
1,186.73	2.219s	3.326s	3.534s	3.493s	3.303s	2.997s
1,262.18	2.107s	3.110s	3.324s	3.303s	3.143s	2.871s
1,338.33	1.989s	2.888s	3.110s	3.111s	2.980s	2.744s
1,415.08	1.865s	2.663s	2.893s	2.917s	2.816s	2.616s
1,492.21	1.736s	2.435s	2.675s	2.721s	2.651s	2.487s
1,569.14	1.604s	2.210s	2.456s	2.525s	2.486s	2.358s

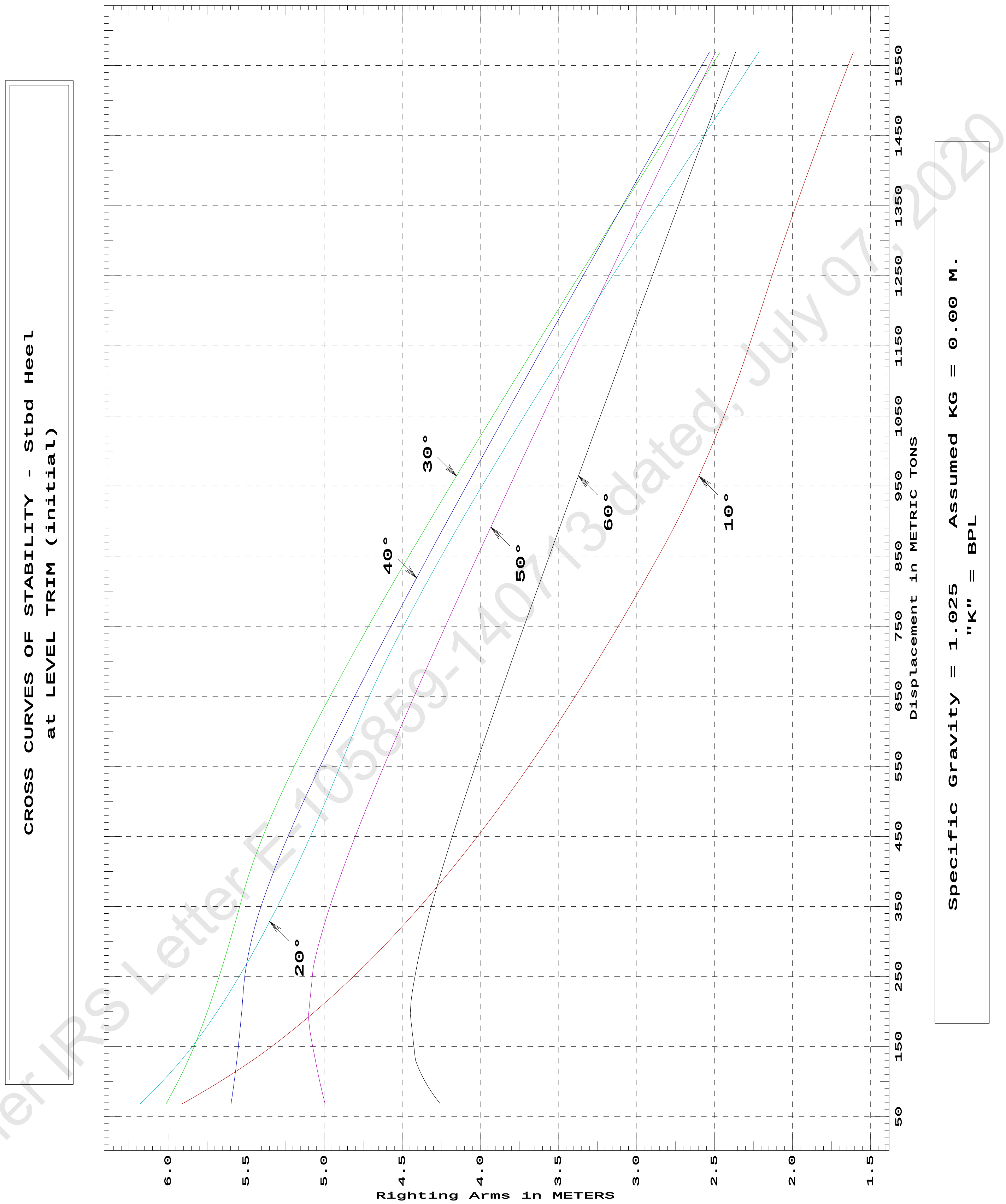
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.25 M. FWD TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
50
150
250
350
450
550
650
750
850
950
1050
1150
1250
1350
1450
1550

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
68.53	5.910s	6.179s	6.013s	5.596s	4.993s	4.256s
131.38	5.453s	5.913s	5.870s	5.558s	5.057s	4.416s
195.72	5.082s	5.699s	5.755s	5.529s	5.100s	4.448s
261.52	4.760s	5.514s	5.657s	5.502s	5.071s	4.409s
328.77	4.471s	5.349s	5.568s	5.434s	4.993s	4.339s
396.99	4.206s	5.197s	5.480s	5.328s	4.890s	4.252s
465.83	3.963s	5.057s	5.363s	5.200s	4.772s	4.156s
535.26	3.735s	4.925s	5.223s	5.058s	4.645s	4.054s
605.29	3.521s	4.798s	5.067s	4.905s	4.511s	3.947s
675.92	3.317s	4.657s	4.899s	4.745s	4.372s	3.836s
747.15	3.123s	4.498s	4.723s	4.579s	4.228s	3.723s
818.99	2.936s	4.326s	4.539s	4.407s	4.082s	3.607s
891.43	2.757s	4.143s	4.347s	4.232s	3.932s	3.490s
964.46	2.599s	3.950s	4.152s	4.053s	3.780s	3.370s
1,038.11	2.459s	3.751s	3.952s	3.871s	3.625s	3.249s
1,112.35	2.334s	3.545s	3.748s	3.687s	3.468s	3.126s
1,187.19	2.222s	3.333s	3.541s	3.499s	3.309s	3.002s
1,262.64	2.112s	3.116s	3.330s	3.310s	3.149s	2.877s
1,338.68	1.994s	2.895s	3.117s	3.117s	2.986s	2.750s
1,415.33	1.871s	2.670s	2.900s	2.923s	2.822s	2.621s
1,492.22	1.742s	2.441s	2.682s	2.728s	2.657s	2.493s
1,569.15	1.609s	2.216s	2.463s	2.531s	2.492s	2.363s

Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
50
150
250
350
450
550
650
750
850
950
1050
1150
1250
1350
1450
1550
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5

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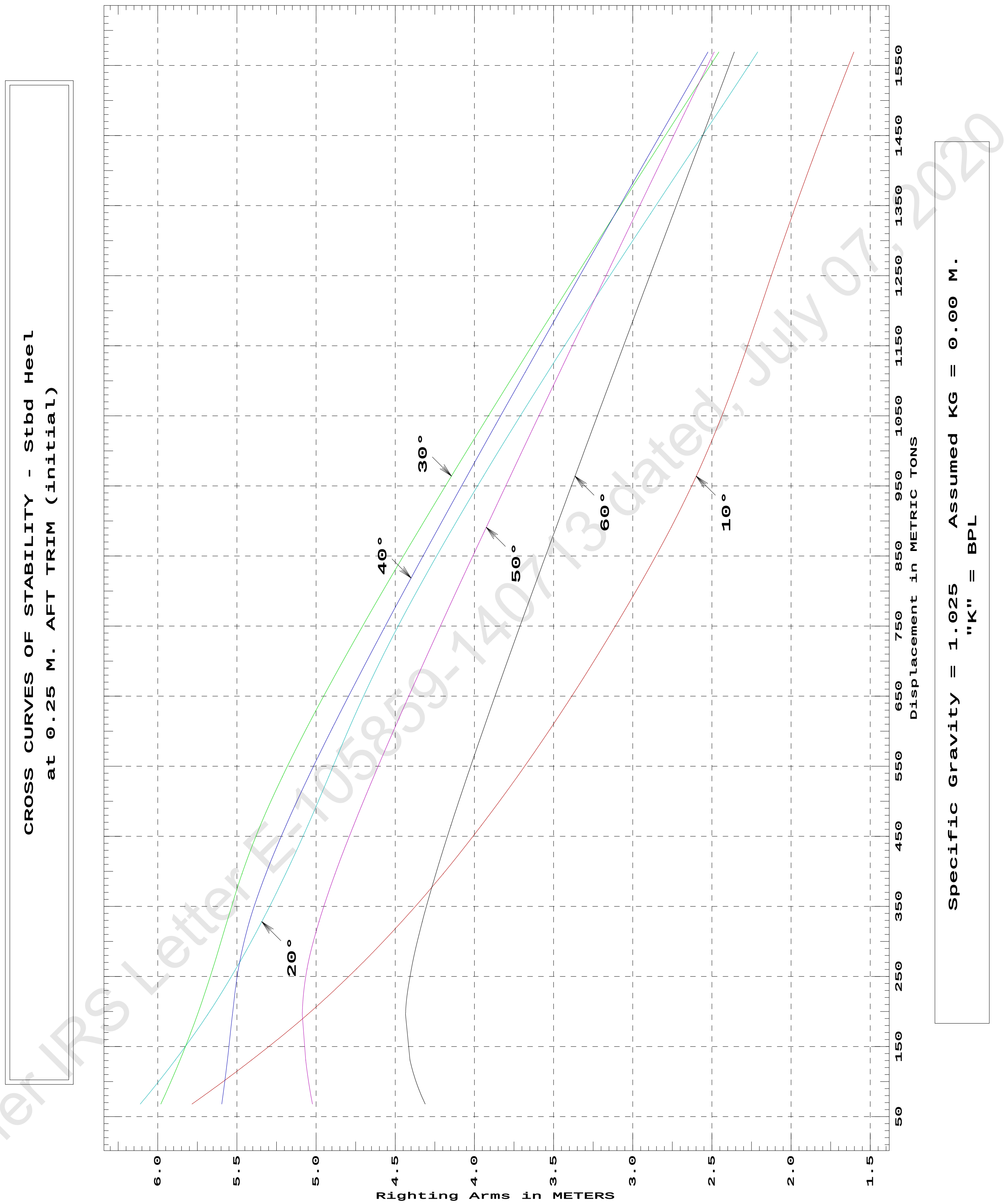
GHS 16.50

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
67.99	5.784s	6.111s	5.981s	5.596s	5.023s	4.311s
130.84	5.407s	5.889s	5.860s	5.560s	5.066s	4.407s
195.22	5.057s	5.686s	5.750s	5.529s	5.087s	4.435s
260.99	4.744s	5.506s	5.653s	5.490s	5.057s	4.398s
328.32	4.459s	5.343s	5.563s	5.421s	4.981s	4.329s
396.63	4.197s	5.192s	5.468s	5.317s	4.880s	4.244s
465.44	3.956s	5.053s	5.352s	5.190s	4.763s	4.149s
534.88	3.730s	4.921s	5.214s	5.048s	4.637s	4.047s
604.91	3.516s	4.790s	5.058s	4.896s	4.503s	3.940s
675.53	3.313s	4.647s	4.891s	4.737s	4.364s	3.830s
746.76	3.119s	4.489s	4.714s	4.571s	4.221s	3.717s
818.59	2.932s	4.317s	4.530s	4.400s	4.075s	3.601s
891.02	2.757s	4.135s	4.340s	4.225s	3.925s	3.484s
964.05	2.599s	3.943s	4.145s	4.046s	3.773s	3.365s
1,037.68	2.459s	3.743s	3.945s	3.865s	3.619s	3.243s
1,111.92	2.334s	3.538s	3.742s	3.680s	3.462s	3.121s
1,186.75	2.219s	3.326s	3.534s	3.493s	3.303s	2.997s
1,262.19	2.107s	3.110s	3.324s	3.303s	3.143s	2.871s
1,338.33	1.989s	2.888s	3.110s	3.111s	2.980s	2.744s
1,415.08	1.865s	2.663s	2.893s	2.917s	2.816s	2.616s
1,492.15	1.736s	2.435s	2.675s	2.721s	2.651s	2.487s
1,569.13	1.604s	2.210s	2.456s	2.525s	2.486s	2.358s

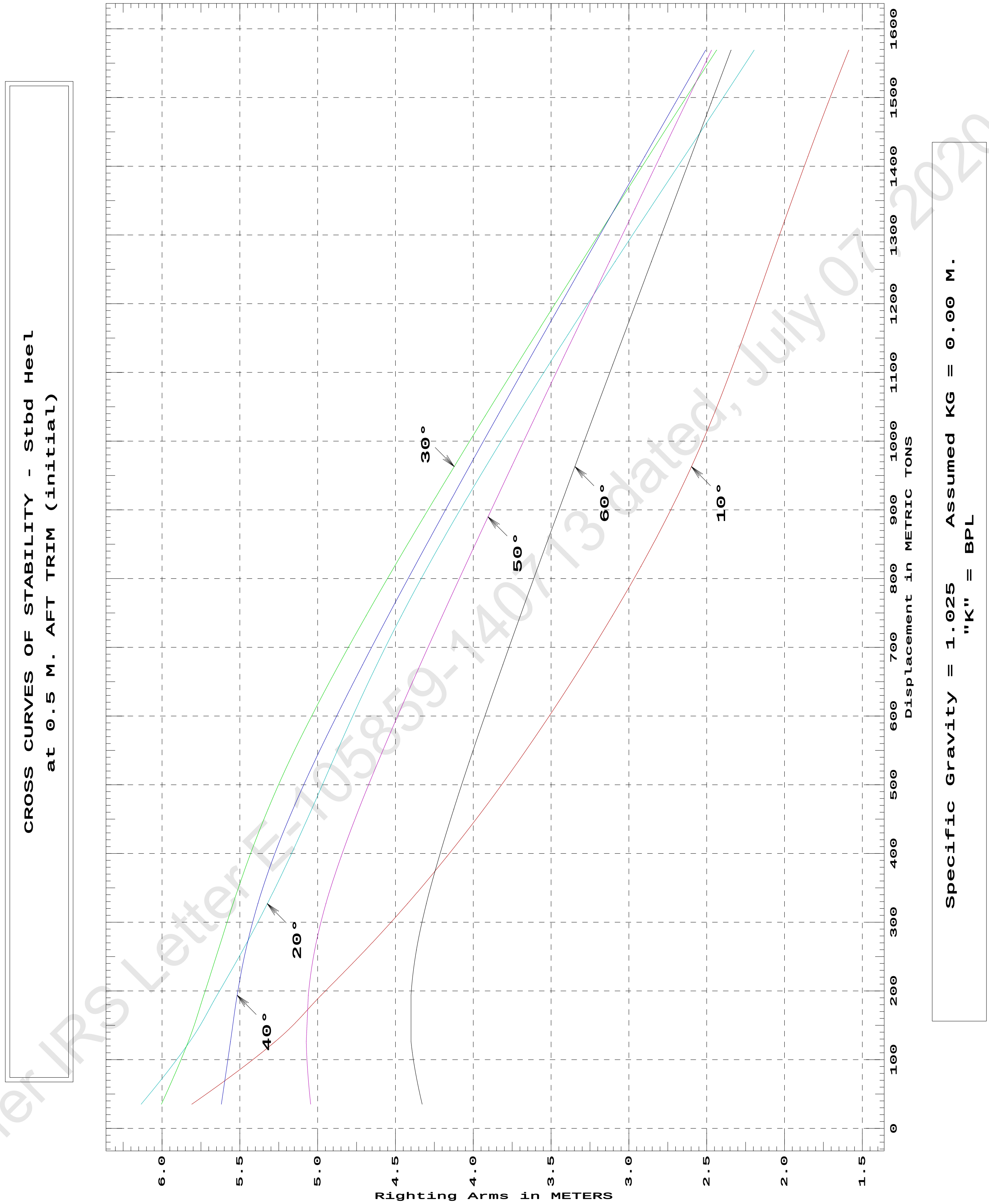
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.25 M. AFT TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
50
150
250
350
450
550
650
750
850
950
1050
1150
1250
1350
1450
1550

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
35.14	5.809s	6.135s	6.004s	5.619s	5.045s	4.328s
126.38	5.272s	5.822s	5.831s	5.560s	5.073s	4.400s
193.59	4.978s	5.647s	5.732s	5.517s	5.060s	4.400s
259.63	4.693s	5.481s	5.639s	5.462s	5.019s	4.366s
326.97	4.423s	5.324s	5.543s	5.384s	4.947s	4.301s
395.34	4.170s	5.177s	5.439s	5.282s	4.849s	4.219s
464.31	3.934s	5.039s	5.319s	5.159s	4.735s	4.125s
533.71	3.712s	4.905s	5.183s	5.020s	4.611s	4.025s
603.73	3.501s	4.767s	5.030s	4.870s	4.479s	3.920s
674.39	3.300s	4.621s	4.864s	4.711s	4.341s	3.810s
745.60	3.107s	4.463s	4.689s	4.547s	4.199s	3.698s
817.41	2.924s	4.292s	4.506s	4.377s	4.054s	3.583s
889.82	2.754s	4.111s	4.317s	4.203s	3.905s	3.466s
962.83	2.598s	3.920s	4.123s	4.025s	3.753s	3.347s
1,036.44	2.458s	3.721s	3.924s	3.844s	3.600s	3.227s
1,110.65	2.331s	3.516s	3.721s	3.660s	3.443s	3.105s
1,185.47	2.211s	3.305s	3.514s	3.474s	3.285s	2.981s
1,261.17	2.093s	3.088s	3.303s	3.284s	3.125s	2.855s
1,337.60	1.973s	2.867s	3.089s	3.091s	2.962s	2.728s
1,414.34	1.850s	2.643s	2.873s	2.898s	2.799s	2.601s
1,491.42	1.721s	2.418s	2.656s	2.703s	2.634s	2.473s
1,568.81	1.588s	2.195s	2.437s	2.507s	2.469s	2.344s

Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
0
100
200
300
400
500
600
700
800
900
1000
1100
1200
1300
1400
1500
1600
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5

DEADWEIGHT TABLE

Draft M	Disp T	Dwt T	
0.612	396.99	28.72	
0.712	465.83	97.56	
0.812	535.26	166.99	
0.912	605.29	237.02	
1.012	675.92	307.65	
1.112	747.15	378.88	
1.212	818.99	450.72	
1.312	891.43	523.16	
1.412	964.46	596.19	
1.512	1038.11	669.84	
1.612	1112.35	744.08	
1.712	1187.19	818.92	
1.812	1262.64	894.37	
1.912	1338.68	970.41	
2.012	1415.33	1047.06	
2.112	1492.22	1123.95	
2.212	1569.15	1200.88	Full Load Displacement

MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind

Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.372	94%	0d	12d
500.00	14.274	47%	0d	9d
750.00	12.752	16%	0d	6d
1,000.00	10.916	0%	0d	5d
1,250.00	8.412	0%	2d	3d
1,500.00	5.807	0%	6d	2d
# 1,750.00	3.468	0%	15d	1d

Heeling moment is present from: wind

Trim = Fwd 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.445	95%	0d	14d
500.00	14.325	51%	0d	10d
750.00	12.852	14%	0d	7d
1,000.00	11.011	0%	0d	5d
1,250.00	8.477	0%	2d	4d
1,500.00	5.833	0%	6d	3d
# 1,750.00	3.494	0%	15d	2d

Heeling moment is present from: wind

Trim = zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.471	95%	0d	17d
500.00	14.340	51%	0d	11d
750.00	12.878	13%	0d	8d
1,000.00	11.044	0%	0d	6d
1,250.00	8.477	0%	2d	4d
1,500.00	5.832	0%	6d	3d
# 1,750.00	3.467	0%	LARGE	2d

Heeling moment is present from: wind

Trim = Aft 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.445	95%	0d	14d
500.00	14.325	51%	0d	10d
750.00	12.852	14%	0d	7d
1,000.00	11.011	0%	0d	5d
1,250.00	8.462	0%	2d	4d
1,500.00	5.832	0%	6d	3d
# 1,750.00	3.493	0%	LARGE	2d

MAX.DISPLACEMENT ALLOWED IS 1569.15 T ONLY

MAX VCG DOES NOT COVER CRANE OPERATION/LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW,WITH CRANE IN STOWED POSITION

Heeling moment is present from: wind
Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

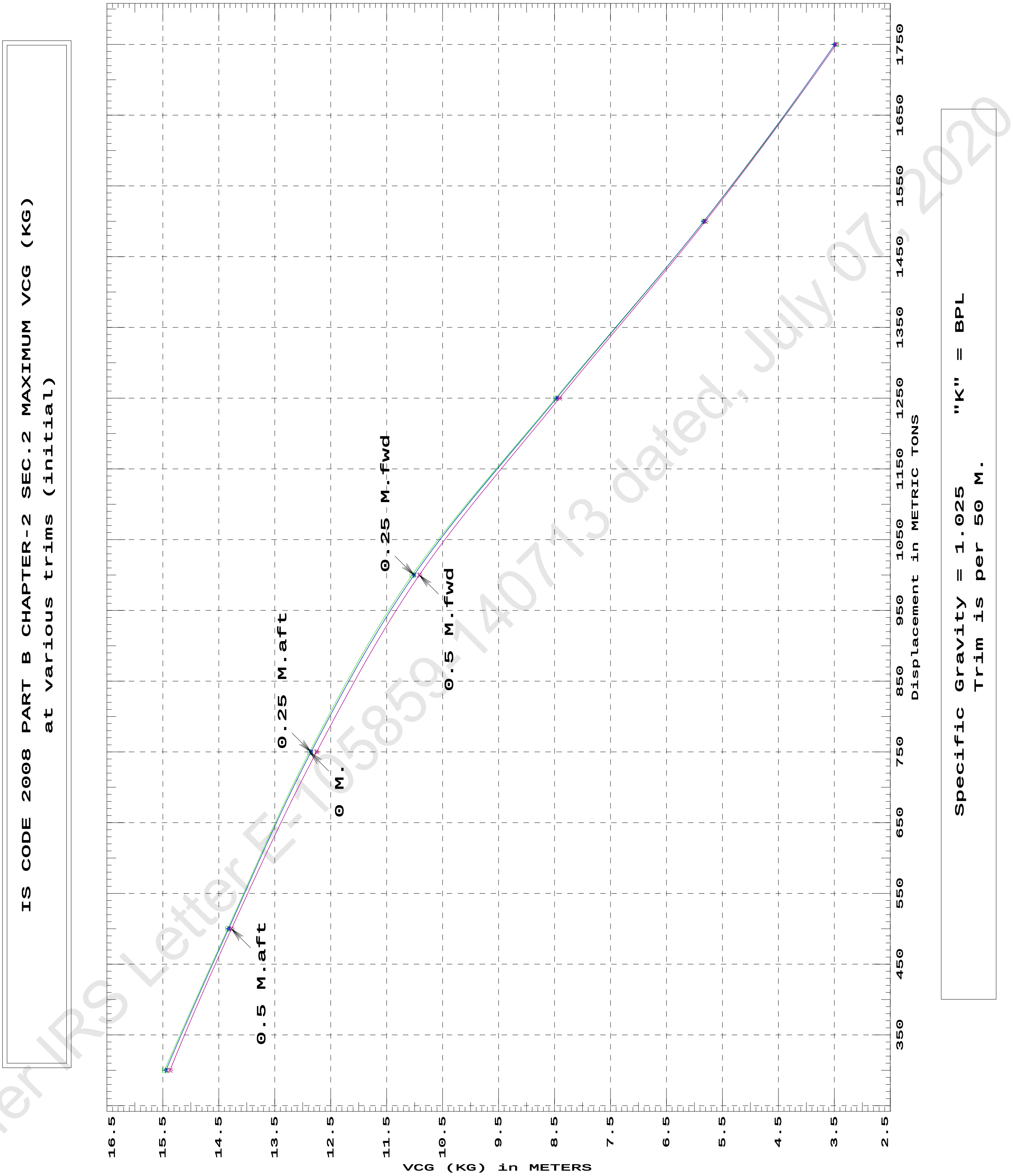
Displacement METRIC TONS	Max VCG	LIM1	LIM2	LIM3
		--- Margins ---		
300.00	15.373	94%	0d	12d
500.00	14.274	47%	0d	9d
750.00	12.752	16%	0d	6d
1,000.00	10.917	0%	0d	5d
1,250.00	8.412	0%	2d	3d
1,500.00	5.805	0%	6d	2d
# 1,750.00	3.469	0%	15d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM--------IS CODE 2008 PART B CHAPTER-2 SEC.2--------		Min/Max
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad
(2) Angle from Equilibrium to RAzero or Flood	>	20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg

MAX.DISPLACEMENT ALLOWED IS 1569.15 T ONLY

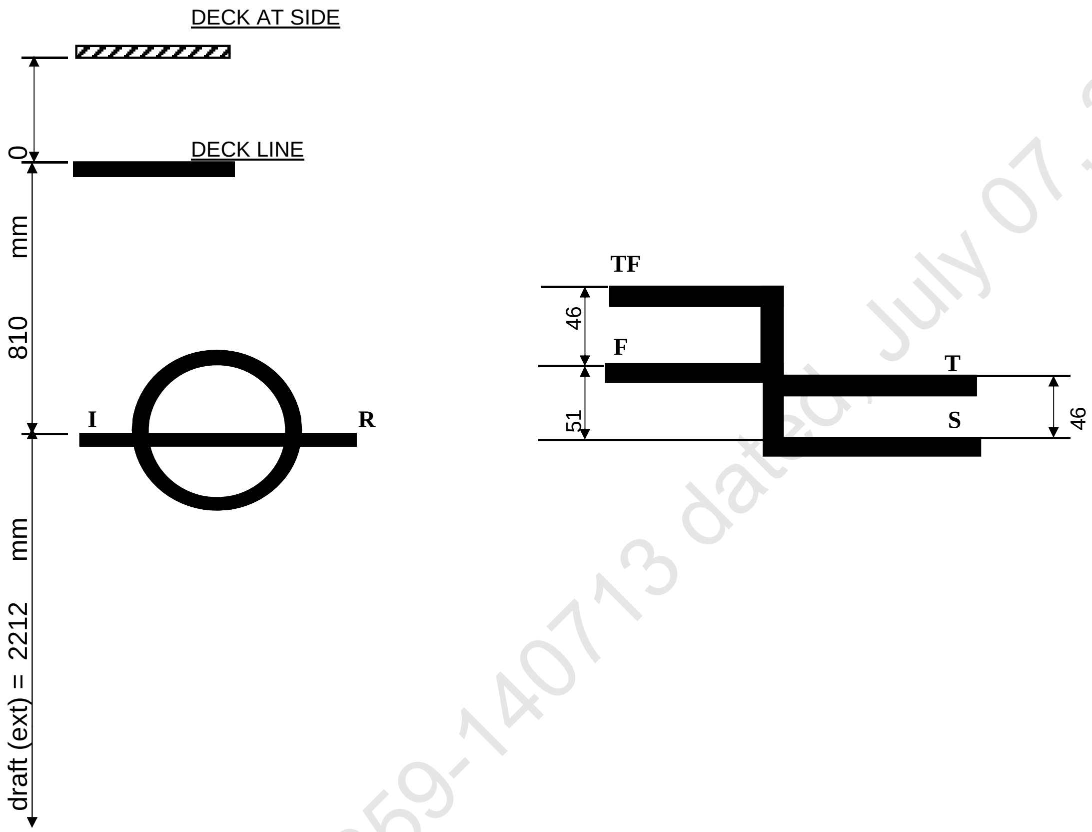
MAX VCG DOES NOT COVER CRANE OPERATION/LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW,WITH CRANE IN STOWED POSITION



MAX.DISPLACEMENT ALLOWED IS 1569.15 T ONLY

MAX VCG DOES NOT COVER CRANE OPERATION/LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW,WITH CRANE IN STOWED POSITION

TABLE OF FREEBOARDS



LOADLINE	FREEBOARD (mm)	DRAFT (ext) KL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1569.15	1200.88
FRESH WATER	759	2.263	1568.79	1200.52
TROPICAL	764	2.258	1604.13	1235.86
TROPICAL FRESH WATER	713	2.309	1603.68	1235.41

Deck Plate Thickness = 10 mm
Keel Plate Thickness = 12 mm

LOADING CONDITIONS

	SUMMARY OF CONDITIONS		
CONDITION NO.	1	2	3
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Aft
TOTAL TANK LOAD (T)	0	120.57	409.19
DWT CONST.(T)	0	150.00	751.71
DEADWEIGHT (T)	0	270.57	1160.9
LIGHT WEIGHT (T)	368.27	368.27	368.27
DISPLACEMENT (T)	368.27	638.84	1,529.17
FWD DRAFT (M)	0.570	0.889	1.940
AFT DRAFT (M)	0.570	1.030	2.379
MEAN DRAFT (M)	0.570	0.959	2.160
TRIM (M)	0.000	0.141a	0.438a
L.C.G. (FROM AP) (M)	25.000f	24.464f	24.113f
L.C.B. (FROM AP) (M)	25.000f	24.455f	24.087f
M.C.T. (T-CM)	22.18	24.37	30.79
KG (M)	3	3.527	4.024
GM_T (M)	32.026	17.48	6.278
MAX. GZ (M)	4.208	3.445	0.842

	SUMMARY OF CONDITIONS		
CONDITION NO.	4	5	6
CONDITION	Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Stbd	Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Fwd	Condition 5 : Lightship + Crane + Ballast + Deck Cargo Towing Condition with Max. draft 2.212m
TOTAL TANK LOAD (T)	409.19	409.19	445.16
DWT CONST.(T)	751.71	751.71	755.72
DEADWEIGHT (T)	1160.9	1160.9	1200.88
LIGHT WEIGHT (T)	368.27	368.27	368.27
DISPLACEMENT (T)	1,529.17	1,529.17	1,569.15
FWD DRAFT (M)	2.028	2.116	2.156
AFT DRAFT (M)	2.291	2.204	2.268
MEAN DRAFT (M)	2.160	2.160	2.212
TRIM (M)	0.263a	0.088a	0.113a
L.C.G. (FROM AP) (M)	24.467f	24.821f	24.774f
L.C.B. (FROM AP) (M)	24.451f	24.816f	24.770f
M.C.T. (T-CM)	30.92	31.08	31.42
KG (M)	4.024	4.024	3.078
GM_T (M)	6.291	6.301	7.059
MAX. GZ (M)	0.765	0.858	1.12

	SUMMARY OF CONDITIONS		
CONDITION NO.	7	8	9
CONDITION	Condition 6 : Lightship + Max.Deck Cargo Max. deck cargo loading condition	Condition 7 : Lightship + Load at Aft + Ballast Crane Lifting 36.6T @ 18m Radius Aft	Condition 8 : Lightship + Load at Stbd + Ballast‼Crane Lifting 36.6T @ 18m Radius Stbd
TOTAL TANK LOAD (T)	0	337.18	337.18
DWT CONST.(T)	1200.88	286.6	286.6
DEADWEIGHT (T)	1200.88	623.78	623.78
LIGHT WEIGHT (T)	368.27	368.27	368.27
DISPLACEMENT (T)	1,569.15	992.05	992.06
FWD DRAFT (M)	2.212	1.308	1.434
AFT DRAFT (M)	2.212	1.589	1.456
MEAN DRAFT (M)	2.212	1.449	1.445
TRIM (M)	0	0.281a	0.022a
L.C.G. (FROM AP) (M)	25.000f	24.237f	24.941f
L.C.B. (FROM AP) (M)	25.000f	24.207f	24.938f
M.C.T. (T-CM)	31.32	26.95	27.03
KG (M)	3.383	6.006	6.006
GM_T (M)	6.961	8.395	8.516
MAX. GZ (M)	1.1	1.86	1.249

	SUMMARY OF CONDITIONS
CONDITION NO.	10
CONDITION	Condition 9 : Lightship + Load at Fwd + Ballast Crane Lifting 36.60T @ 18m Radius Fwd
TOTAL TANK LOAD (T)	337.18
DWT CONST.(T)	286.6
DEADWEIGHT (T)	623.78
LIGHT WEIGHT (T)	368.27
DISPLACEMENT (T)	992.05
FWD DRAFT (M)	1.592
AFT DRAFT (M)	1.305
MEAN DRAFT (M)	1.449
TRIM (M)	0.287f
L.C.G. (FROM AP) (M)	25.780f
L.C.B. (FROM AP) (M)	25.810f
M.C.T. (T-CM)	26.95
KG (M)	6.006
GM_T (M)	8.395
MAX. GZ (M)	1.86

PONTOON (YARD NO. CHISTI/009)

Condition 00 : Lightship

Lightship condition

Condition Graphic - Draft: 0.570 Trim: zero Heel: zero

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 00 : Lightship

Lightship condition

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 0.570 @ 50.00f, 0.570 @ 25.00f, 0.570 @ 0.00
                            Trim: zero,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
WEIGHT                                368.27  25.000f  0.000   3.000
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->                 0.00                                0.00
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                     1.025        368.27  25.000f  0.000   0.279
-------------------------------------------------------------------------------
               Righting Arms:                   0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          666.0  25.000f  0.000    303.81  34.747
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        6.83            22.18    301.09  32.026
Distances in METERS.-----------------------------------------Moments in m.-MT.
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                             SUMMARY OF LOADING
         0.0 Cu.M.   (0%) SALT WATER
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DRAFT AT FP (M) =    0.570
DRAFT AT MS (M) =    0.570
DRAFT AT AP (M) =    0.570
DRAFT AT FWD MARK (M) =    0.570
DRAFT AT AFT MARK (M) =    0.570
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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 00 : Lightship

Lightship condition

HYDROSTATIC PROPERTIES
No Trim, No Heel, VCG = 3.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
0.570	368.27	25.000f	0.279	6.83	25.000f	22.18	301.09	32.026

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.000f TCG = 0.000 VCG = 3.000

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
0.558	0.00	0.00	368.17	0.000	-0.025	0.0000	14.925
0.558	0.00	0.04s	368.27	0.000	0.000	-0.0000	14.880
0.537	0.00	5.04s	368.26	0.000	2.634	0.1149	9.880
0.359	0.00	10.04s	368.18	0.000	3.774	0.4048	4.880
0.067	0.00	14.92s	368.32	0.000	4.122	0.7464	0.000
0.059	0.00	15.04s	368.27	0.000	4.127	0.7550	-0.120
-0.307	0.00	20.04s	368.09	0.000	4.208	1.1207	-5.120
-0.713	0.00	25.04s	368.06	0.000	4.146	1.4862	-10.120
-1.144	0.00	30.04s	368.47	0.000	3.991	1.8419	-15.120
-1.583	0.00	35.04s	368.10	0.000	3.747	2.1802	-20.120
-2.006	0.00	40.04s	368.42	0.000	3.419	2.4935	-25.120
-2.417	0.00	45.04s	368.31	0.000	3.035	2.7755	-30.120
-2.807	0.00	50.04s	368.40	0.000	2.608	3.0221	-35.120
-3.178	0.00	55.04s	368.30	0.000	2.148	3.2298	-40.120
-3.525	0.00	60.04s	368.16	0.000	1.662	3.3962	-45.120

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 9.35 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	1.1207 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	14.88 P

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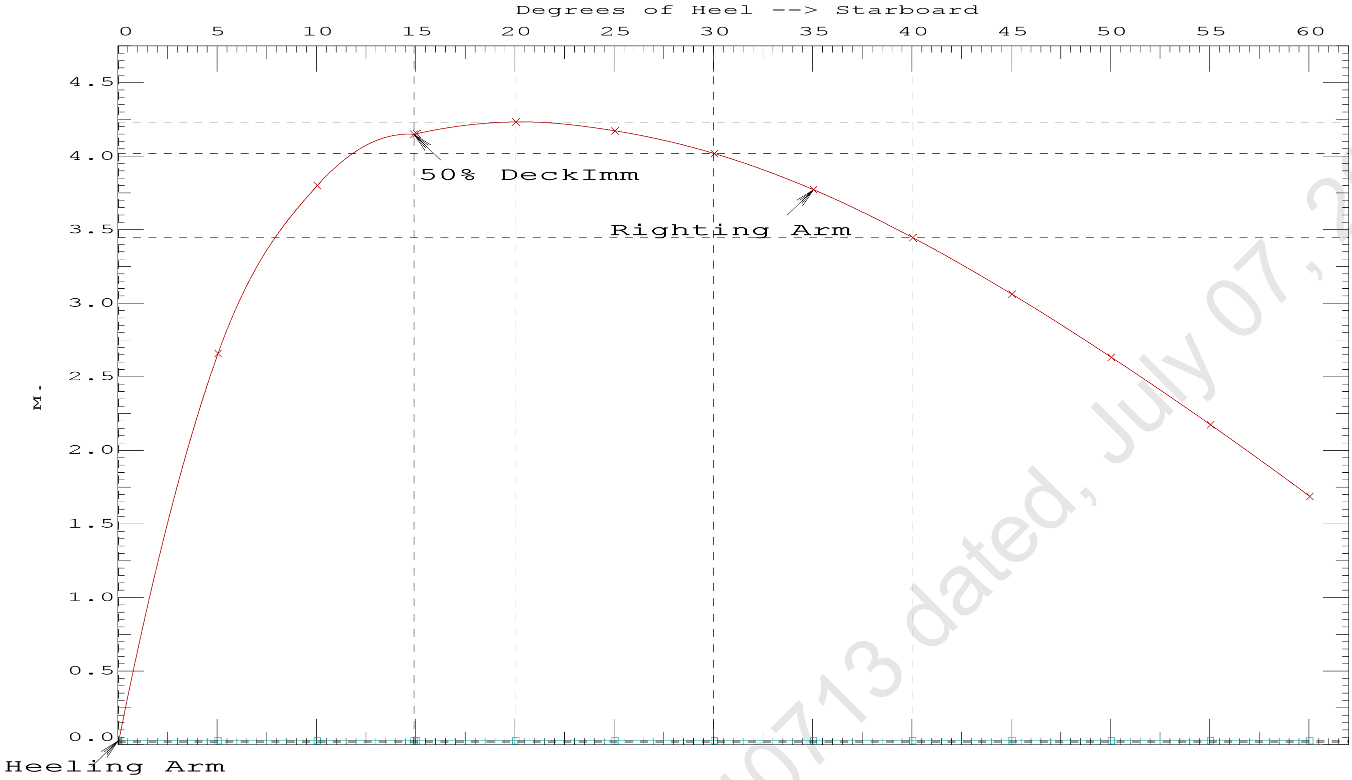
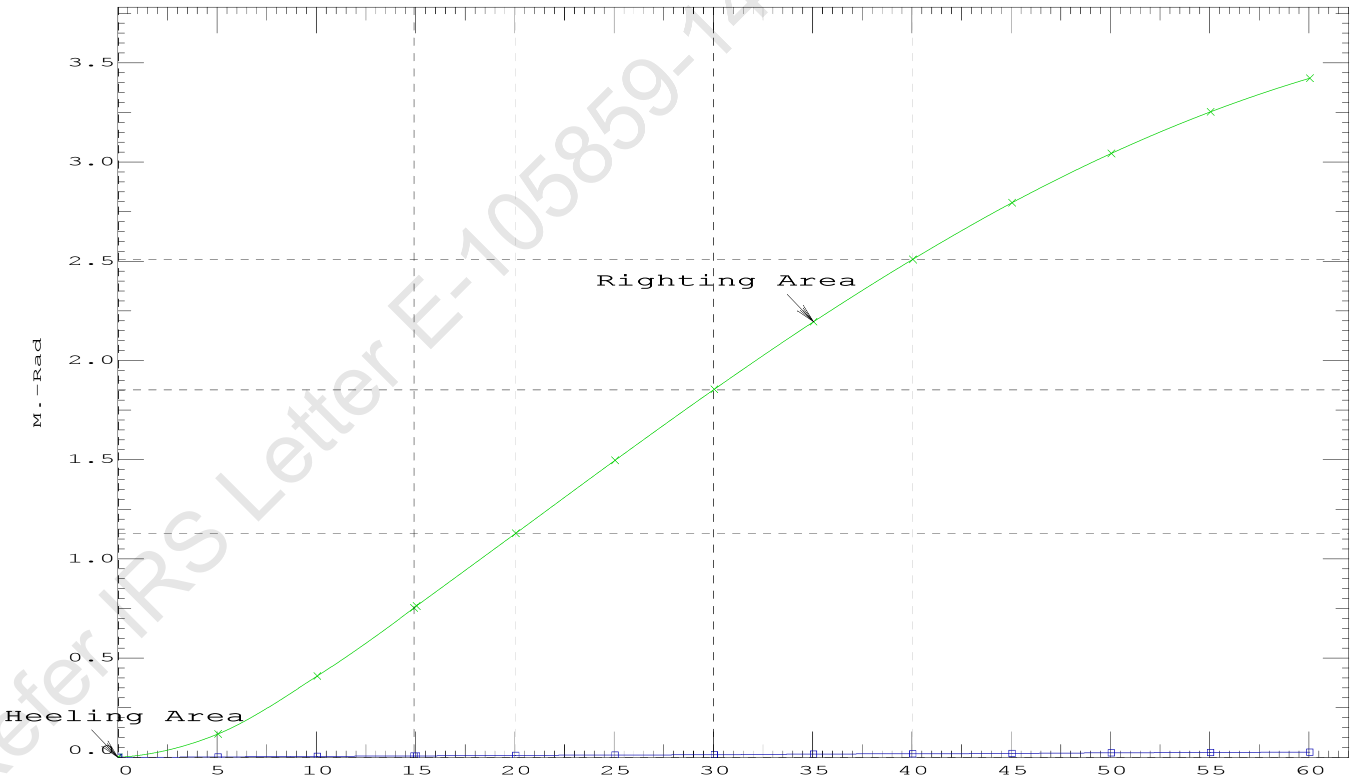
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 00 : Lightship

Lightship condition





Condition 1 : Lightship + Crane + Ballast

Condition Graphic - Draft: 0.889 @ 50.000f, 1.030 @ 0.000 Heel: zero

Tanks	14 WBTK6.P	15 WBTK6.C	16 WBTK6.S

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 1 : Lightship + Crane + Ballast

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 0.889 @ 50.00f, 0.959 @ 25.00f, 1.030 @ 0.00
Trim: Aft 0.141/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Total Fixed--------->			518.27	19.067f	0.000	3.903	
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTK6.S	1.000	1.025	39.99	47.660f	4.967s	1.915	41.84*
Total Tanks--------->			120.57	47.661f	0.000	1.912	125.54
Total Weight-------->			638.84	24.464f	0.000	3.527	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	638.82	24.455f	0.000	0.481	
Righting Arms:				0.000	0.000		

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	688.9	24.874f	0.000	193.82	20.526
		MT/cm		m.-MT/cm	GML	GMT
		7.06		24.37	190.78	17.480

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

117.6 Cu.M. (8%) SALT WATER

150.00 MT Crane

DRAFT AT FP (M) = 0.889
DRAFT AT MS (M) = 0.959
DRAFT AT AP (M) = 1.030
DRAFT AT FWD MARK (M) = 0.900
DRAFT AT AFT MARK (M) = 1.018

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 1 : Lightship + Crane + Ballast

HYDROSTATIC PROPERTIES
Trim: Aft 0.141/50.000, No Heel, VCG = 3.527

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
0.960	638.82	24.455f	0.481	7.06	24.874f	24.37	190.78	17.480

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 125.5 m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.464f TCG = 0.000 VCG = 3.527
Free Surface Adjustment: 0.197
Adjusted CG: LCG = 24.464f TCG = 0.000 VCG = 3.724

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.018	0.16a	0.00	638.85	0.000	-0.012	0.0000	8.626
1.018	0.16a	0.04s	638.84	0.000	0.000	-0.0000	8.588
1.007	0.16a	5.04s	638.84	0.000	1.521	0.0664	3.588
0.976	0.17a	8.63s	638.84	0.000	2.501	0.1928	0.000
0.948	0.18a	10.04s	638.84	0.000	2.769	0.2577	-1.412
0.761	0.21a	15.04s	638.84	0.000	3.283	0.5260	-6.412
0.493	0.25a	20.04s	638.84	0.000	3.444	0.8221	-11.412
0.475	0.25a	20.36s	638.84	0.000	3.445	0.8416	-11.736
0.218	0.31a	25.04s	638.84	0.000	3.349	1.1203	-16.412
-0.059	0.37a	30.04s	638.84	0.000	3.110	1.4032	-21.412
-0.334	0.43a	35.04s	638.84	0.000	2.790	1.6611	-26.412
-0.606	0.49a	40.04s	638.84	0.000	2.418	1.8887	-31.412
-0.874	0.55a	45.04s	638.84	0.000	2.009	2.0822	-36.412
-1.134	0.60a	50.04s	638.84	0.000	1.574	2.2388	-41.412
-1.385	0.65a	55.04s	638.84	0.000	1.120	2.3565	-46.412
-1.626	0.69a	60.04s	638.84	0.000	0.652	2.4339	-51.412

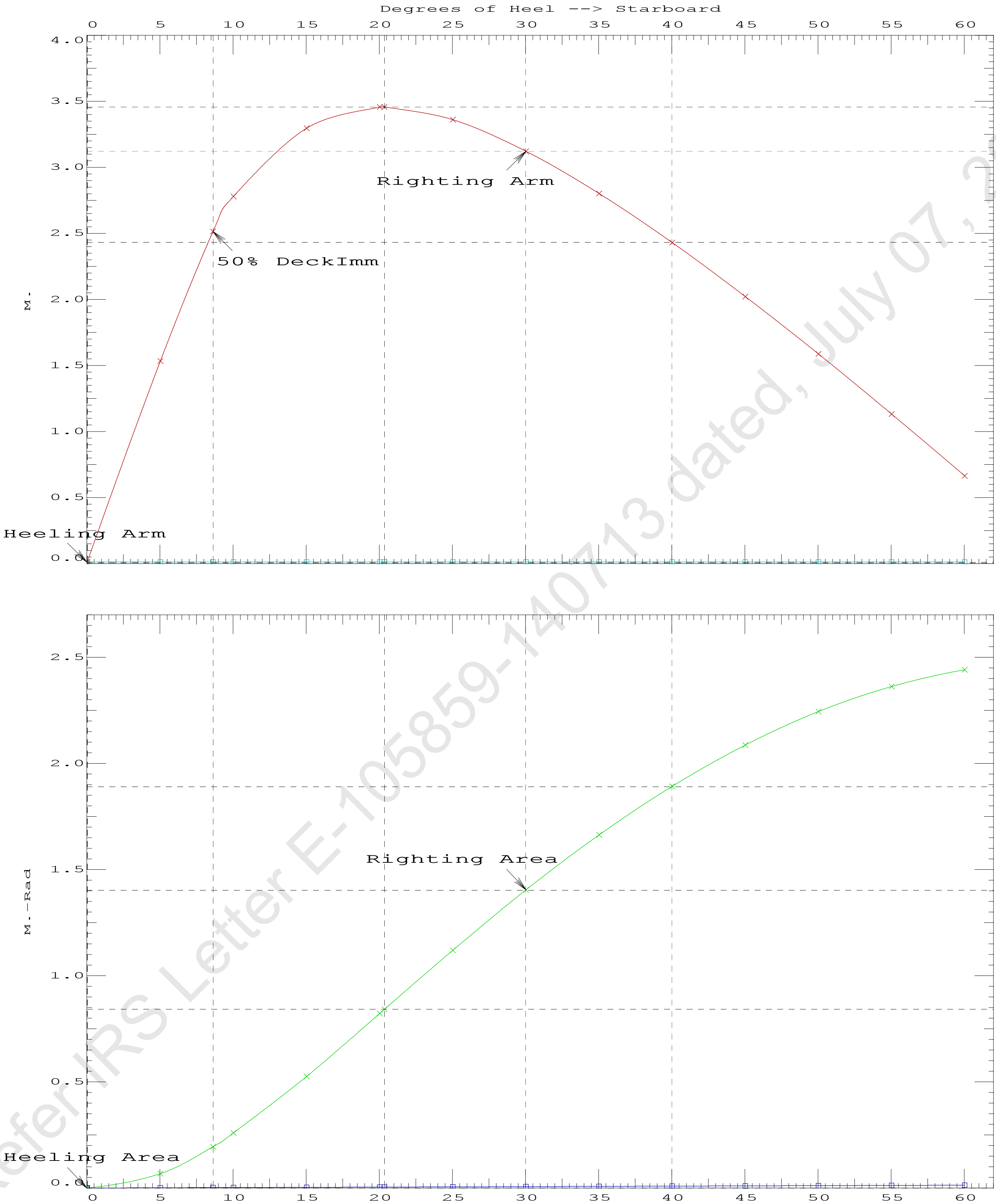
Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 125.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.89 (constant)

LIM	INTACT CRITERIA-IS CODE 2008PART B		Min/Max		Attained
(1)	Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.8416 P
(2)	Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3)	Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	8.59 P

PONTOON (YARD NO. CHISTI/009)
Condition 1 : Lightship + Crane + Ballast



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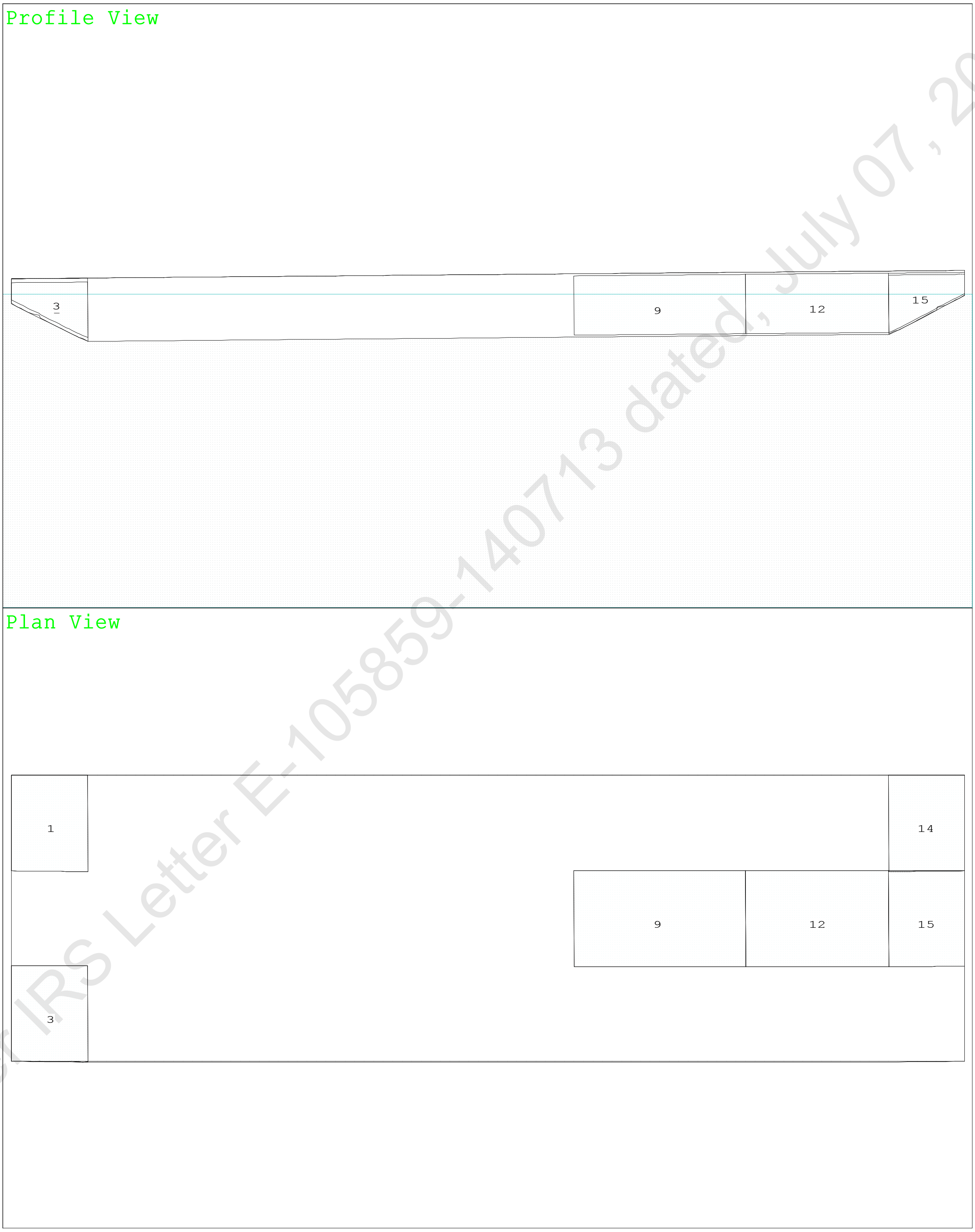
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Aft

CG - Draft: 1.940 @ 50.000f, 2.379 @ 0.000 Heel: port 1.14 deg.



Tanks 1 WBTK1.P 3 WBTK1.S 9 WBTK4.C 12 WBTK5.C 14 WBTK6.P 15 WBTK6.C

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.940 @ 50.00f, 2.160 @ 25.00f, 2.379 @ 0.00

Trim: Aft 0.438/50.000, Heel: Port 1.14 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	7.500a	0.000	33.320	
Deck Cargo			556.61	25.000f	0.000	3.500	
Total Fixed--------->			1,119.98	20.946f	0.000	4.887	
WBTK1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTK1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks--------->			409.19	32.782f	0.485p	1.662	340.04
Total Weight-------->			1,529.17	24.113f	0.130p	4.024	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,529.17	24.087f	0.188p	1.113

Righting Arms: 0.000 0.000p

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	748.0	24.899f	0.061p	103.58	9.190
		MT/cm		m.-MT/cm	GML	GMT
		7.67		30.79	100.67	6.278

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

399.2 Cu.M. (27%) SALT WATER

150.00 MT Crane 45.10 MT Crane Load

556.61 MT Deck Cargo

DRAFT AT FP (M) = 1.940
DRAFT AT MS (M) = 2.160
DRAFT AT AP (M) = 2.379
DRAFT AT FWD MARK (M) = 1.976
DRAFT AT AFT MARK (M) = 2.344

HYDROSTATIC PROPERTIES

Trim: Aft 0.438/50.000, Heel: Port 1.14 deg., VCG = 4.024

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.160	1,529.17	24.087f	1.113	7.67	24.899f	30.79	100.67	6.278

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 340.0 m.-MT

11-09-19 12:03:15 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.113f TCG = 0.130p VCG = 4.024
Free Surface Adjustment: 0.222
Adjusted CG: LCG = 24.115f TCG = 0.125p VCG = 4.247

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.366	0.50a	0.00	1,529.2	0.000	-0.136	0.0000	2.439
2.366	0.50a	1.23p	1,529.2	0.000	0.000	-0.0015	1.207
2.365	0.50a	2.44p	1,529.2	0.000	0.133	-0.0001	0.000
2.361	0.52a	6.23p	1,529.2	0.000	0.543	0.0224	-3.793
2.492	0.71a	11.23p	1,529.2	0.000	0.822	0.0837	-8.793
2.596	0.80a	13.68p	1,529.2	0.000	0.842	0.1194	-11.239
2.720	0.90a	16.23p	1,529.2	0.000	0.820	0.1567	-13.793
2.990	1.10a	21.23p	1,529.2	0.000	0.693	0.2241	-18.793
3.276	1.35a	26.23p	1,529.2	0.000	0.492	0.2763	-23.793
3.549	1.61a	31.23p	1,529.2	0.000	0.249	0.3089	-28.793
3.784	1.85a	35.93p	1,529.2	0.000	0.000	0.3192	-33.487
3.799	1.86a	36.23p	1,529.2	0.000	-0.017	0.3192	-33.793
4.021	2.10a	41.23p	1,529.2	0.000	-0.293	0.3058	-38.793
4.213	2.33a	46.23p	1,529.2	0.000	-0.573	0.2680	-43.793
4.372	2.54a	51.23p	1,529.2	0.000	-0.853	0.2058	-48.793
4.496	2.73a	56.23p	1,529.2	0.000	-1.130	0.1192	-53.793
4.582	2.89a	61.23p	1,529.2	0.000	-1.399	0.0088	-58.793

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

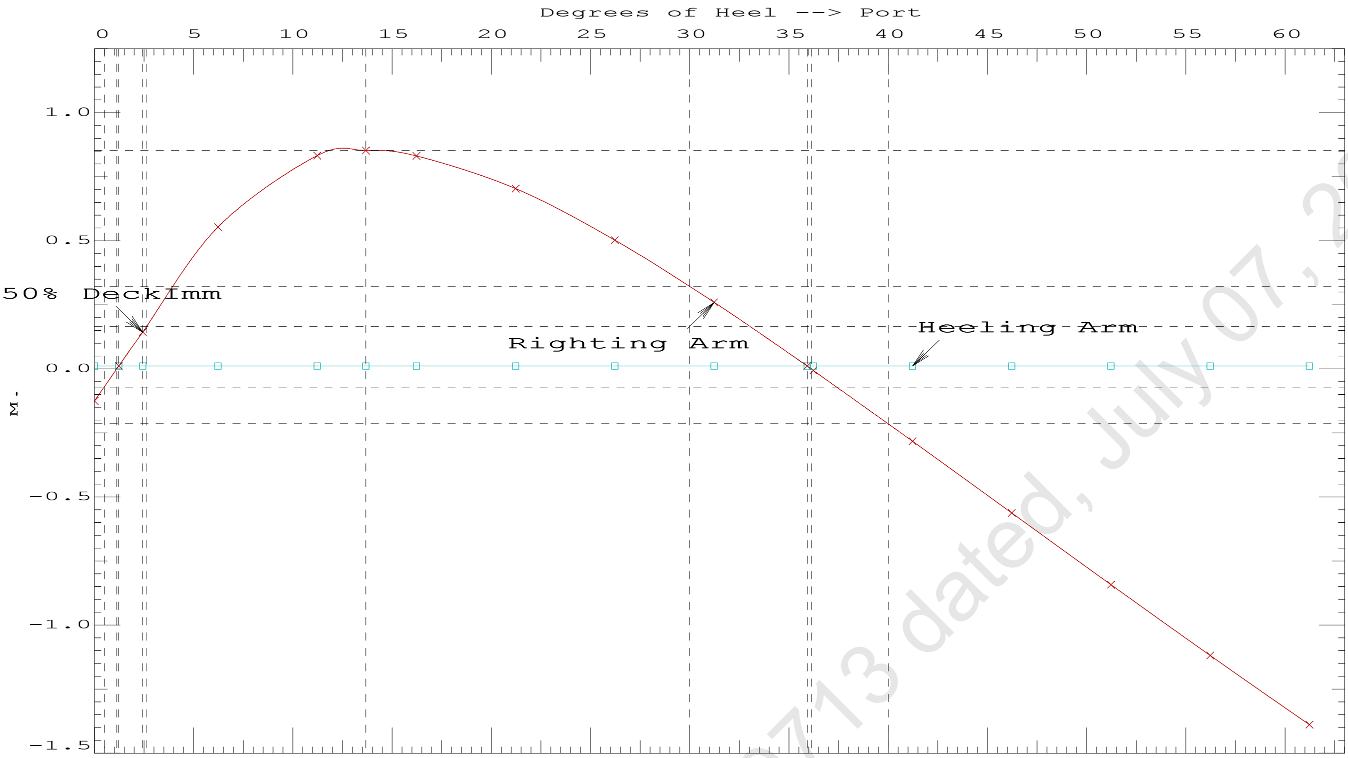
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 340.0 m.-MT was applied to artificially modify the CG.

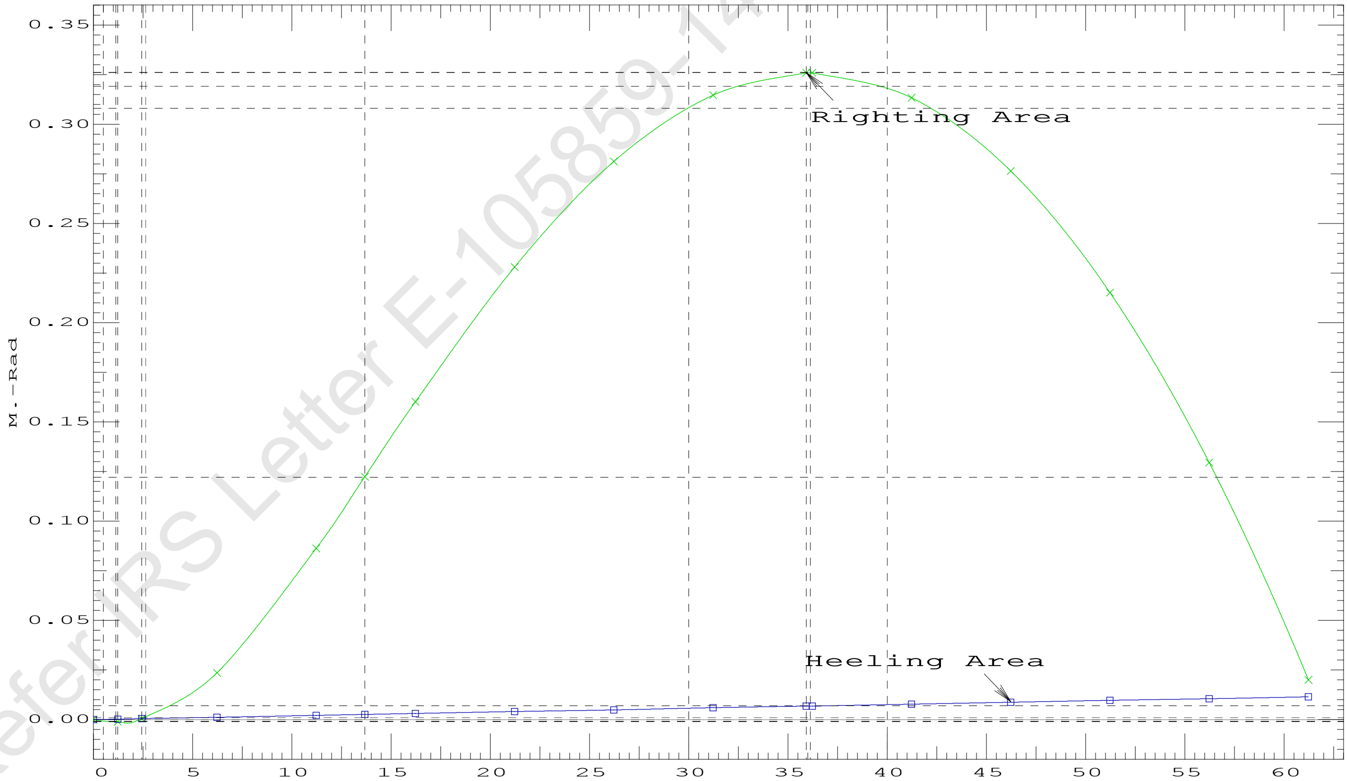
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Port heeling moment = 16.42 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1194 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	34.69 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	1.21 P

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Aft





11-09-19 12:03:15 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.113f TCG = 0.130p VCG = 4.024
Free Surface Adjustment: 0.222
Adjusted CG: LCG = 24.115f TCG = 0.125p VCG = 4.247

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.366	0.50a	1.14p	1,529.2	0.000	0.000	0.0000
2.360	0.52a	6.14p	1,529.2	0.000	0.544	0.0237
2.488	0.70a	11.14p	1,529.2	0.000	0.830	0.0869
2.596	0.80a	13.68p	1,529.2	0.000	0.852	0.1244
2.716	0.89a	16.14p	1,529.2	0.000	0.831	0.1606
2.985	1.10a	21.14p	1,529.2	0.000	0.706	0.2290
3.271	1.34a	26.14p	1,529.2	0.000	0.506	0.2824
3.545	1.60a	31.14p	1,529.2	0.000	0.263	0.3162
3.793	1.85a	36.11p	1,529.2	0.000	0.000	0.3278
3.795	1.86a	36.14p	1,529.2	0.000	-0.002	0.3278
4.017	2.10a	41.14p	1,529.2	0.000	-0.278	0.3156
4.210	2.33a	46.14p	1,529.2	0.000	-0.558	0.2792
4.369	2.54a	51.14p	1,529.2	0.000	-0.839	0.2182
4.494	2.73a	56.14p	1,529.2	0.000	-1.115	0.1329
4.581	2.88a	61.14p	1,529.2	0.000	-1.385	0.0238

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 340.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

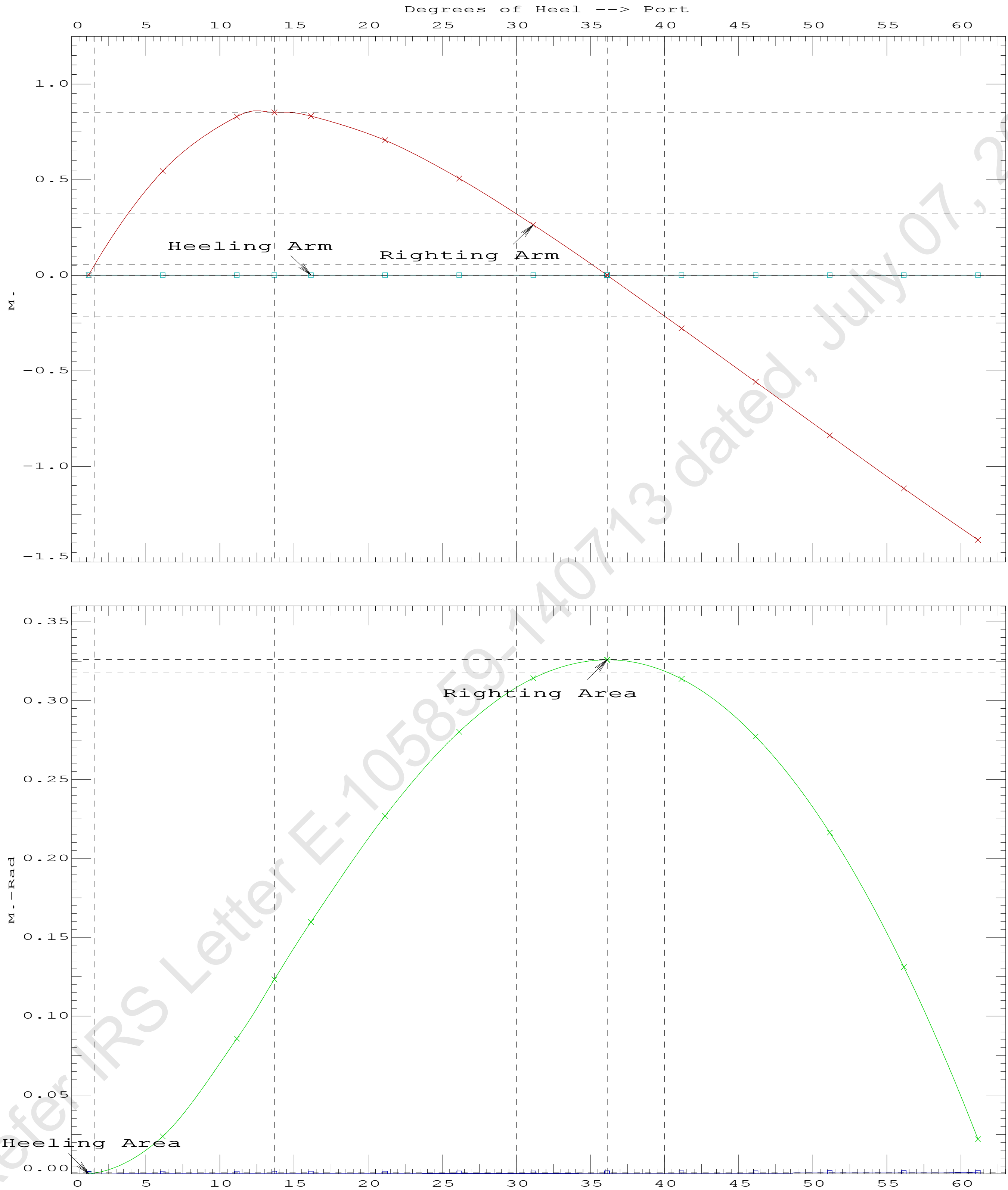
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	1.14 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	130280 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		822.349 P

11-09-19 12:03:15 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Aft



11-09-19 12:02:40 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 45.10T @ 12m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.941 @ 50.00f, 2.160 @ 25.00f, 2.379 @ 0.00
Trim: Aft 0.438/50.000, Heel: Port 1.14 deg.

Part			Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	7.500a	0.000	33.320	
Deck Cargo			556.61	25.000f	0.000	3.500	
Total Fixed--------->			1,119.98	20.946f	0.000	4.887	
	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
WBTK1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTK1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks--------->			409.19	32.782f	0.485p	1.662	340.04
Total Weight-------->			1,529.17	24.113f	0.130p	4.024	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,529.17	24.087f	0.189p	1.113	

Righting Arms:	0.000	0.001p
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000p

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	748.0	24.899f	0.061p	103.58	9.190
		MT/cm		m.-MT/cm	GML	GMT
		7.67		30.79	100.67	6.278

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = -1.14

```
                       After hook load drop
             WEIGHT and DISPLACEMENT and WATERPLANE STATUS
        USK draft: 2.119 @ 50.00f, 2.101 @ 25.00f, 2.084 @ 0.00
             Trim: Fwd 0.035/50.000,  Heel: Port 1.00 deg.
Part----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          368.27  25.000f  0.000   3.000
Crane                               150.00   4.500f  0.000   6.120
Deck Cargo                          556.61  25.000f  0.000   3.500
  Total Fixed--------->           1,074.88  22.139f  0.000   3.694
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
WBTK1.P        1.000    1.025        39.99   2.340f  4.967p  1.915     41.85*
WBTK1.S        1.000    1.025        39.99   2.340f  4.967s  1.915     41.85*
WBTK4.C        1.000    1.025       135.61  34.000f  0.000   1.500     94.17*
WBTK5.C        1.000    1.025       113.02  42.250f  0.000   1.500     78.48*
WBTK6.P        1.000    1.025        39.99  47.660f  4.967p  1.915     41.84*
WBTK6.C        1.000    1.025        40.60  47.664f  0.000   1.904     41.85*
  Total Tanks--------->             409.19  32.782f  0.485p  1.662    340.04
  Total Weight-------->           1,484.07  25.074f  0.134p  3.134
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025     1,484.06  25.074f  0.170p  1.078
------------------------------------------------------------------------------
               Righting Arms:                  0.001   0.000p
Part--------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025        749.8  25.018f  0.045p   107.53   9.483
                                     MT/cm--------m.-MT/cm-------GML-----GMT
                                      7.69             31.31   105.48   7.427
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.
```

```
limmarg: 2481257.50; t3: 30.0; t: 15.00
limmarg: 1018668.06; t3: 15.0; t: 7.50
limmarg: 323763.28; t3: 7.5; t: 3.75
limmarg: 101281.34; t3: 3.8; t: 1.88
limmarg: 38305.79; t3: 1.9; t: 0.94
limmarg: 17222.28; t3: 1.0; t: 0.47
limmarg: 10443.67; t3: 0.5; t: 0.23
limmarg: 7154.40; t3: 0.3; t: 0.12
limmarg: 6158.73; t3: 0.2; t: 0.06
limmarg: 5739.79; t3: 0.1; t: 0.03
limmarg: 5041.56; t3: 0.1; t: 0.01
```

Theta3 is 0.07 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.694

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.071	0.04f	1.14p	1,484.2	0.000	-0.020	0.0000
2.072	0.04f	1.00p	1,484.1	0.000	0.000	-0.0000
2.072	0.04f	0.07s	1,483.9	0.000	0.144	0.0013
2.065	0.04f	4.00s	1,484.1	0.000	0.669	0.0292
2.051	0.05f	9.00s	1,484.1	0.000	1.274	0.1145
2.139	0.06f	14.00s	1,484.1	0.000	1.501	0.2383
2.242	0.07f	17.59s	1,484.1	0.000	1.533	0.3338
2.287	0.08f	19.00s	1,484.1	0.000	1.528	0.3716
2.443	0.10f	24.00s	1,484.1	0.000	1.445	0.5023
2.580	0.12f	29.00s	1,484.1	0.000	1.290	0.6222

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Res. Area Ratio from abs -1.1 deg to abs 0.1	>	1.000	51.416 P

11-09-19 12:02:40 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.694

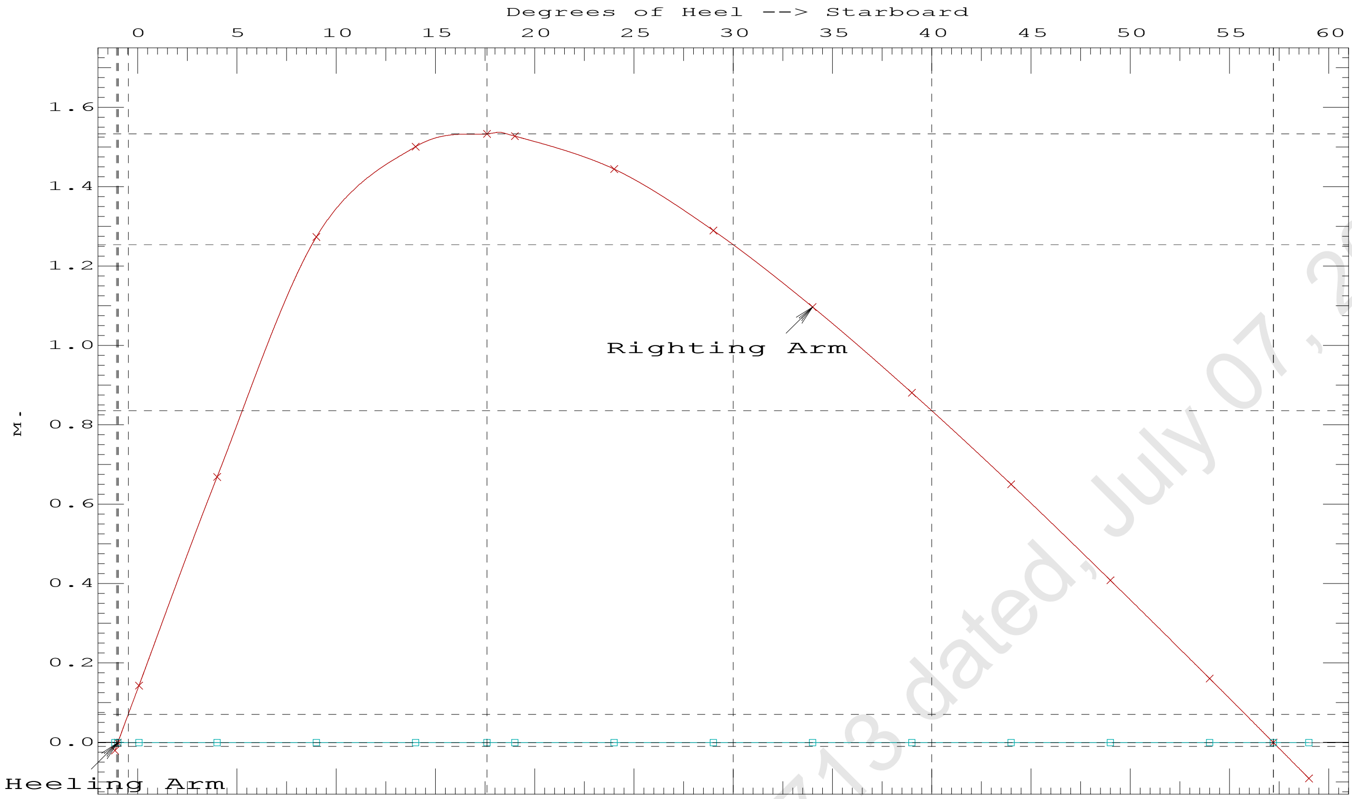
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.071	0.04f	1.14p	1,484.2	0.000	-0.020	0.0000
2.072	0.04f	1.00p	1,484.1	0.000	0.000	-0.0000
2.072	0.04f	0.07s	1,483.9	0.000	0.144	0.0013
2.065	0.04f	4.00s	1,484.1	0.000	0.669	0.0292
2.051	0.05f	9.00s	1,484.1	0.000	1.274	0.1145
2.139	0.06f	14.00s	1,484.1	0.000	1.501	0.2383
2.242	0.07f	17.59s	1,484.1	0.000	1.533	0.3338
2.287	0.08f	19.00s	1,484.1	0.000	1.528	0.3716
2.443	0.10f	24.00s	1,484.1	0.000	1.445	0.5023
2.580	0.12f	29.00s	1,484.1	0.000	1.290	0.6222
2.697	0.14f	34.00s	1,484.1	0.000	1.097	0.7266
2.794	0.16f	39.00s	1,484.1	0.000	0.882	0.8131
2.869	0.18f	44.00s	1,484.1	0.000	0.650	0.8801
2.922	0.19f	49.00s	1,484.1	0.000	0.409	0.9264
2.953	0.21f	54.00s	1,484.1	0.000	0.161	0.9513
2.961	0.22f	57.21s	1,484.1	0.000	0.000	0.9558
2.961	0.22f	59.00s	1,484.1	0.000	-0.090	0.9544

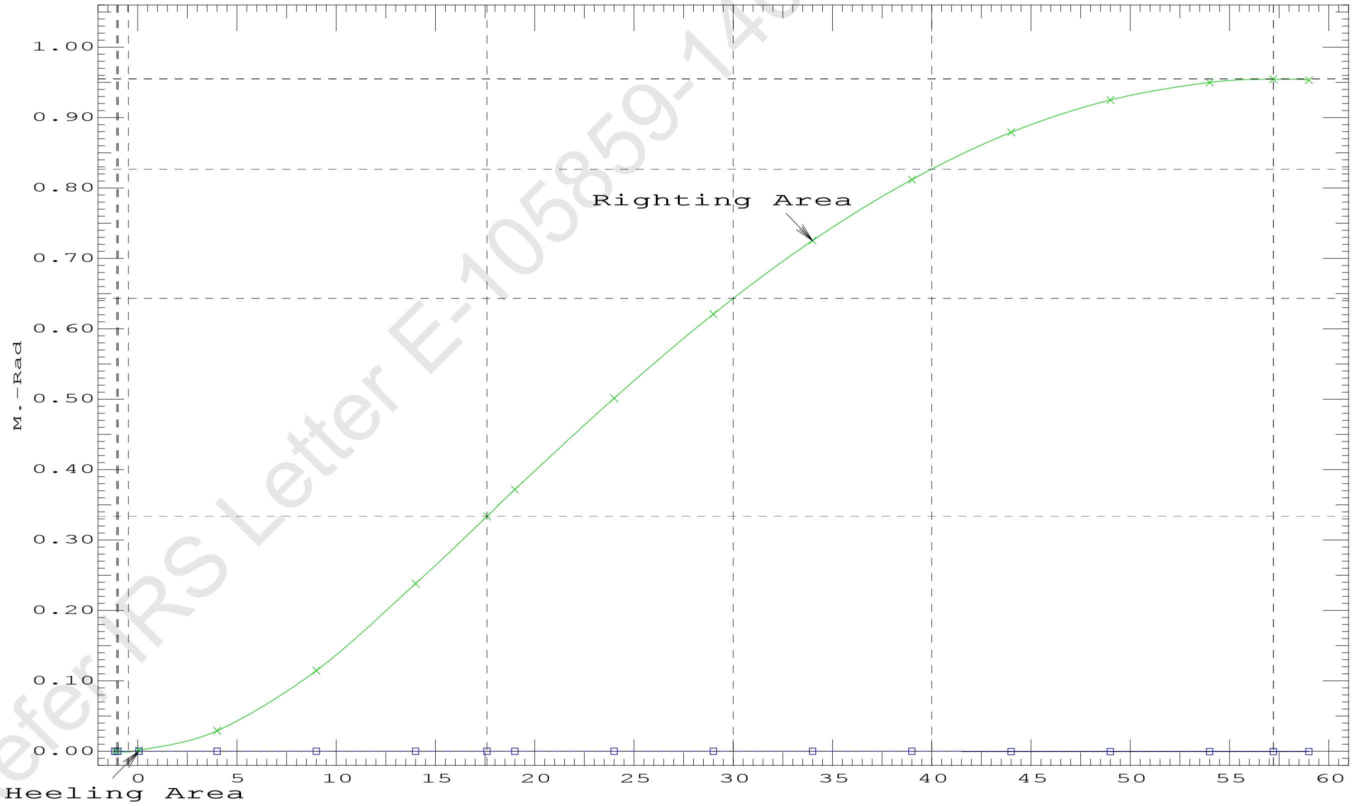
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

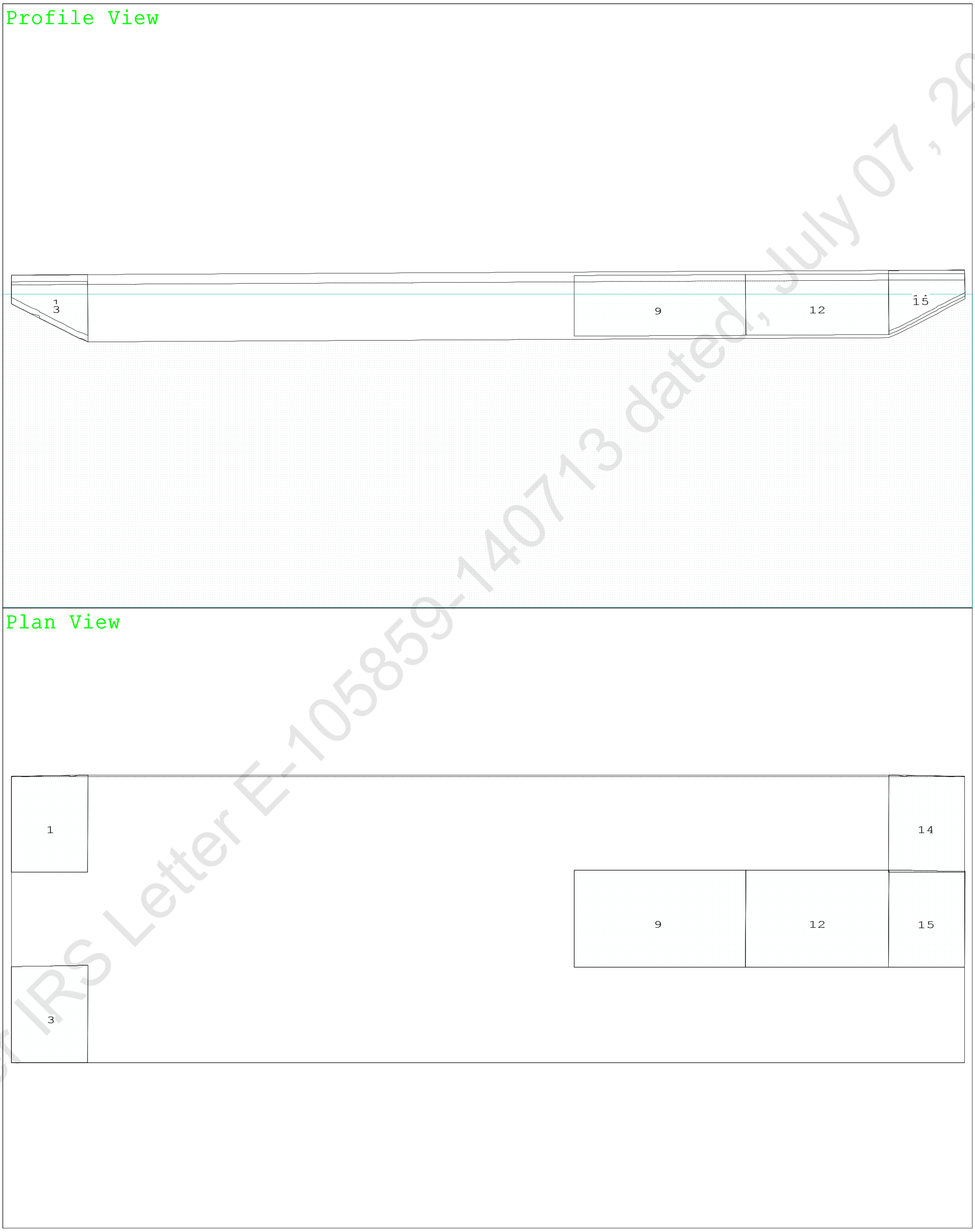
LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -1.1 deg to RAzero	>	1.000		36815.5 P
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	57.14 P


Degrees of Heel --> Starboard
0
5
10
15
20
25
30
35
40
45
50
55
60
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
M.
Righting Arm
Heeling Arm


1.00
0.90
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
M.-Rad
Righting Area
Heeling Area
0
5
10
15
20
25
30
35
40
45
50
55
60

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd

CG - Draft: 2.028 @ 50.000f, 2.291 @ 0.000 Heel: stbd 1.96 deg.



Tanks 1 WBTK1.P 3 WBTK1.S 9 WBTK4.C 12 WBTK5.C 14 WBTK6.P 15 WBTK6.C

11-09-19 12:04:22 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.028 @ 50.00f, 2.160 @ 25.00f, 2.291 @ 0.00
Trim: Aft 0.263/50.000, Heel: Stbd 1.96 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	4.500f	12.000s	33.320	
Deck Cargo			556.61	25.000f	0.000	3.500	
Total Fixed--------->			1,119.98	21.429f	0.483s	4.887	
WBTK1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTK1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks--------->			409.19	32.782f	0.485p	1.662	340.04
Total Weight-------->			1,529.17	24.467f	0.224s	4.024	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,529.17	24.451f	0.324s	1.114

Righting Arms: 0.000 0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	749.2	24.939f	0.091s	104.00	9.202
		MT/cm		m.-MT/cm	GML	GMT
		7.68		30.92	101.09	6.291

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

399.2 Cu.M. (27%) SALT WATER

150.00 MT Crane 45.10 MT Crane Load
556.61 MT Deck Cargo

DRAFT AT FP (M) = 2.028
DRAFT AT MS (M) = 2.160
DRAFT AT AP (M) = 2.291
DRAFT AT FWD MARK (M) = 2.049
DRAFT AT AFT MARK (M) = 2.270

HYDROSTATIC PROPERTIES
Trim: Aft 0.263/50.000, Heel: Stbd 1.96 deg., VCG = 4.024

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.160	1,529.17	24.451f	1.114	7.68	24.939f	30.92	101.09	6.291

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 340.0 m.-MT

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.467f TCG = 0.224s VCG = 4.024
Free Surface Adjustment: 0.222
Adjusted CG: LCG = 24.468f TCG = 0.216s VCG = 4.246

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.279	0.30a	0.00	1,529.2	0.000	-0.227	0.0000	2.775
2.278	0.30a	2.05s	1,529.2	0.000	0.000	-0.0041	0.720
2.277	0.30a	2.77s	1,529.2	0.000	0.079	-0.0036	-0.000
2.271	0.32a	7.05s	1,529.2	0.000	0.539	0.0196	-4.280
2.392	0.44a	12.05s	1,529.2	0.000	0.756	0.0789	-9.280
2.453	0.48a	13.67s	1,529.2	0.000	0.765	0.1003	-10.894
2.597	0.56a	17.05s	1,529.2	0.000	0.730	0.1450	-14.280
2.836	0.68a	22.05s	1,529.2	0.000	0.593	0.2038	-19.280
3.081	0.83a	27.05s	1,529.2	0.000	0.388	0.2472	-24.280
3.305	0.99a	32.05s	1,529.2	0.000	0.144	0.2706	-29.280
3.419	1.07a	34.81s	1,529.2	0.000	0.000	0.2741	-32.035
3.507	1.14a	37.05s	1,529.2	0.000	-0.120	0.2718	-34.280
3.682	1.29a	42.05s	1,529.2	0.000	-0.393	0.2495	-39.280
3.829	1.42a	47.05s	1,529.2	0.000	-0.669	0.2032	-44.280
3.947	1.54a	52.05s	1,529.2	0.000	-0.944	0.1328	-49.280
4.033	1.65a	57.05s	1,529.2	0.000	-1.214	0.0386	-54.280
4.087	1.74a	62.05s	1,529.2	0.000	-1.478	-0.0790	-59.280

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 340.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 16.50 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1003	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	32.76	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	0.72	P

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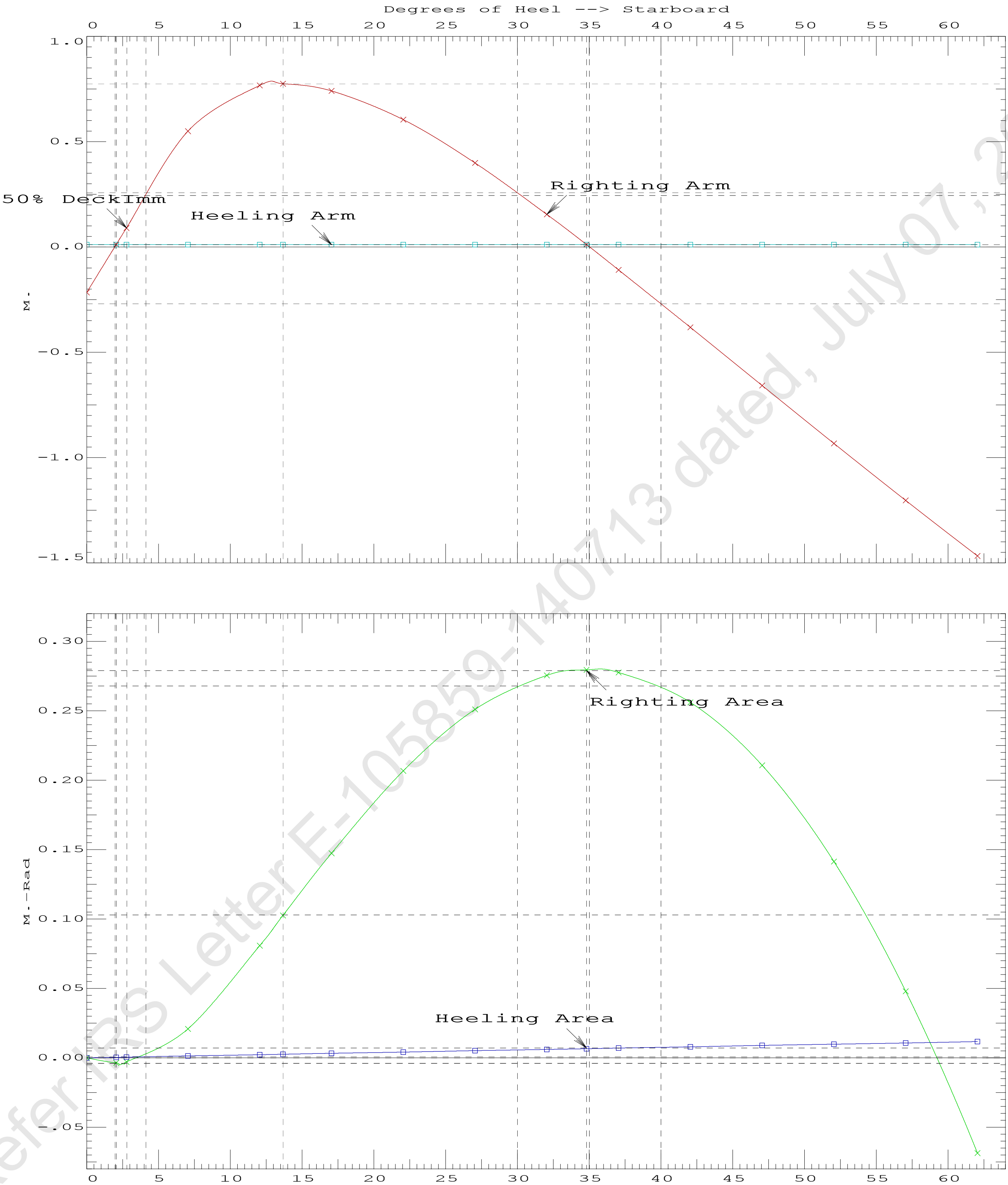
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Stbd



11-09-19 12:04:22 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.467f TCG = 0.224s VCG = 4.024
Free Surface Adjustment: 0.222
Adjusted CG: LCG = 24.468f TCG = 0.216s VCG = 4.246

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.278	0.30a	1.96s	1,529.2	0.000	0.000	0.0000
2.270	0.32a	6.96s	1,529.2	0.000	0.540	0.0236
2.389	0.44a	11.96s	1,529.2	0.000	0.765	0.0834
2.453	0.48a	13.67s	1,529.2	0.000	0.775	0.1065
2.593	0.55a	16.96s	1,529.2	0.000	0.742	0.1505
2.831	0.68a	21.96s	1,529.2	0.000	0.607	0.2105
3.076	0.83a	26.96s	1,529.2	0.000	0.403	0.2551
3.301	0.99a	31.96s	1,529.2	0.000	0.159	0.2799
3.427	1.08a	35.01s	1,529.2	0.000	0.000	0.2841
3.503	1.14a	36.96s	1,529.2	0.000	-0.104	0.2823
3.679	1.28a	41.96s	1,529.2	0.000	-0.377	0.2614
3.827	1.42a	46.96s	1,529.2	0.000	-0.653	0.2165
3.945	1.54a	51.96s	1,529.2	0.000	-0.928	0.1474
4.032	1.65a	56.96s	1,529.2	0.000	-1.199	0.0546
4.086	1.74a	61.96s	1,529.2	0.000	-1.462	-0.0616

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

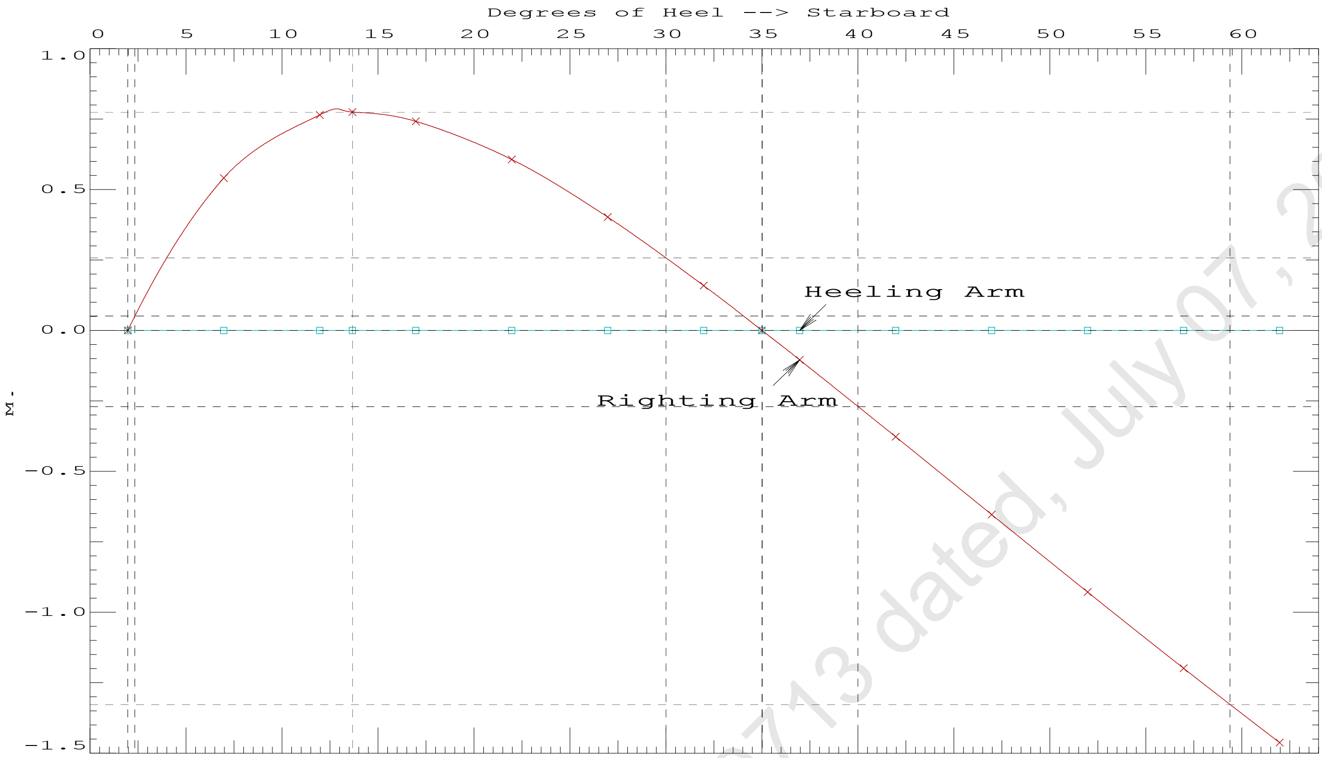
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 340.0 m.-MT was applied to artificially modify the CG.

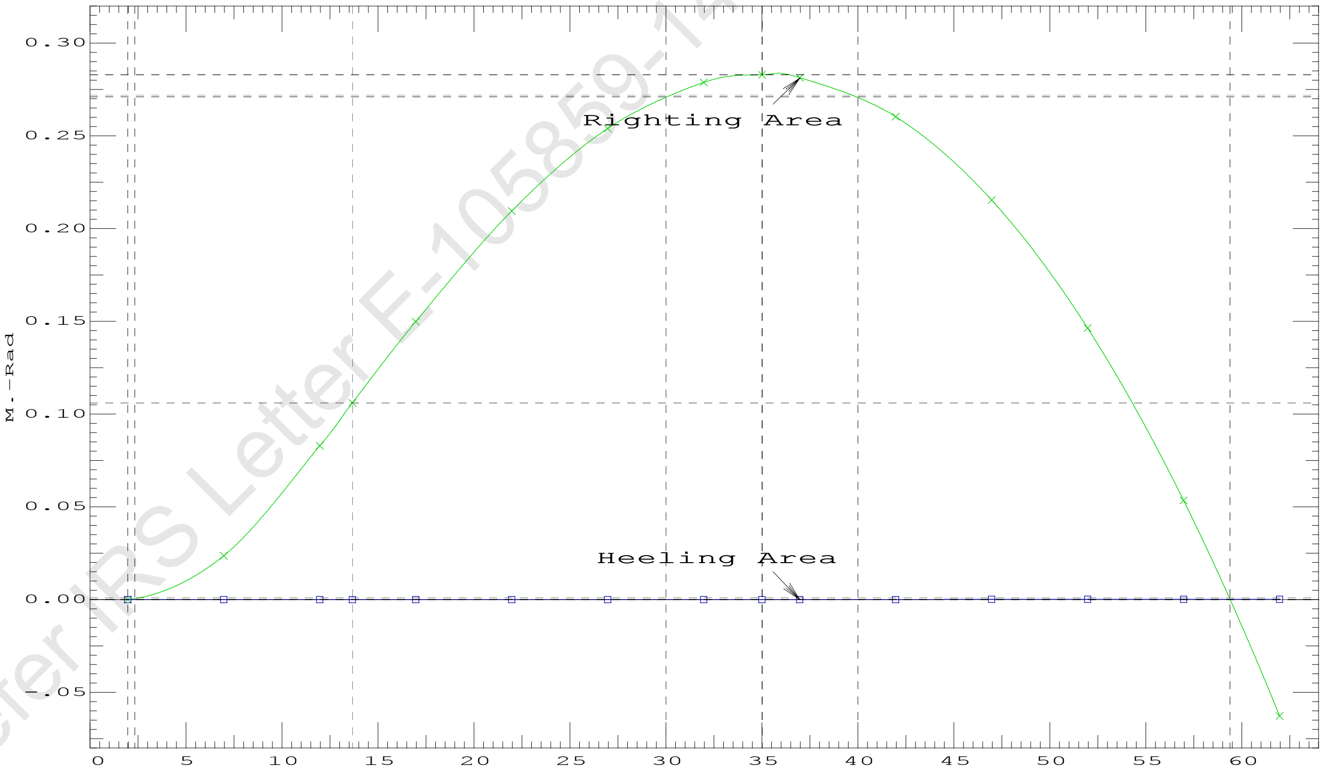
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.10

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	1.96 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	1184971 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		7533.71 P

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd





11-09-19 12:04:04 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.10T @ 12m Radius Port
Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.028 @ 50.00f, 2.160 @ 25.00f, 2.291 @ 0.00
Trim: Aft 0.263/50.000, Heel: Stbd 1.96 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	4.500f	12.000s	33.320	
Deck Cargo			556.61	25.000f	0.000	3.500	
Total Fixed--------->			1,119.98	21.429f	0.483s	4.887	
WBTK1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTK1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks--------->			409.19	32.782f	0.485p	1.662	340.04
Total Weight-------->			1,529.17	24.467f	0.224s	4.024	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,529.17	24.451f	0.324s	1.114	

Righting Arms:		0.000	0.000s			
Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	749.2	24.939f	0.091s	104.00	9.202
		MT/cm		m.-MT/cm	GML	GMT
		7.68		30.92	101.09	6.291

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 1.96

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11-09-19 12:04:04    Cybermarine Technologies PTE. Ltd.
GHS 16.68C              PONTOON (YARD NO. CHISTI/009)

                              After hook load drop
                   WEIGHT and DISPLACEMENT and WATERPLANE STATUS
               USK draft: 2.119 @ 50.00f, 2.101 @ 25.00f, 2.084 @ 0.00
                   Trim: Fwd 0.035/50.000,  Heel: Port 1.00 deg.
Part----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          368.27  25.000f  0.000   3.000
Crane                               150.00   4.500f  0.000   6.120
Deck Cargo                          556.61  25.000f  0.000   3.500
  Total Fixed--------->           1,074.88  22.139f  0.000   3.694
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
WBTK1.P        1.000    1.025        39.99   2.340f  4.967p  1.915     41.85*
WBTK1.S        1.000    1.025        39.99   2.340f  4.967s  1.915     41.85*
WBTK4.C        1.000    1.025       135.61  34.000f  0.000   1.500     94.17*
WBTK5.C        1.000    1.025       113.02  42.250f  0.000   1.500     78.48*
WBTK6.P        1.000    1.025        39.99  47.660f  4.967p  1.915     41.84*
WBTK6.C        1.000    1.025        40.60  47.664f  0.000   1.904     41.85*
  Total Tanks--------->             409.19  32.782f  0.485p  1.662    340.04
  Total Weight-------->           1,484.07  25.074f  0.134p  3.134
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                        1.025 1,484.11  25.074f  0.170p  1.078
------------------------------------------------------------------------------
               Righting Arms:                 0.001   0.000p
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          749.8  25.018f  0.045p  107.53   9.483
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        7.69             31.31  105.47   7.427
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.
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limmarg: 4862.08; t3: 30.00; t: 15.00
limmarg: 1791.32; t3: 15.00; t: 7.50
limmarg: 379.23; t3: 7.50; t: 3.75
limmarg: -13.76; t3: 3.75; t: 1.88
limmarg: 149.11; t3: 5.68; t: 0.94
limmarg: 59.24; t3: 4.74; t: 0.47
limmarg: 21.81; t3: 4.27; t: 0.23
limmarg: 5.32; t3: 4.04; t: 0.12
limmarg: -2.80; t3: 3.92; t: 0.06
limmarg: -0.13; t3: 3.96; t: 0.03
limmarg: 4.63; t3: 4.03; t: 0.01
```

Theta3 is 4.03 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.694

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.070	0.04f	1.96s	1,484.1	0.000	-0.398	0.0000
2.072	0.04f	1.00p	1,484.1	0.000	0.000	-0.0103
2.065	0.04f	4.03p	1,484.1	0.000	0.405	0.0005
2.057	0.04f	6.00p	1,484.1	0.000	0.666	0.0189
2.075	0.05f	11.00p	1,484.1	0.000	1.136	0.0986
2.194	0.07f	16.00p	1,484.0	0.000	1.270	0.2060
2.249	0.07f	17.83p	1,484.0	0.000	1.277	0.2468
2.352	0.08f	21.00p	1,484.1	0.000	1.256	0.3171
2.500	0.10f	26.00p	1,484.1	0.000	1.148	0.4228
2.630	0.13f	31.00p	1,484.1	0.000	0.987	0.5163

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs -2 deg to abs 4	>	1.000	1.046 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.694

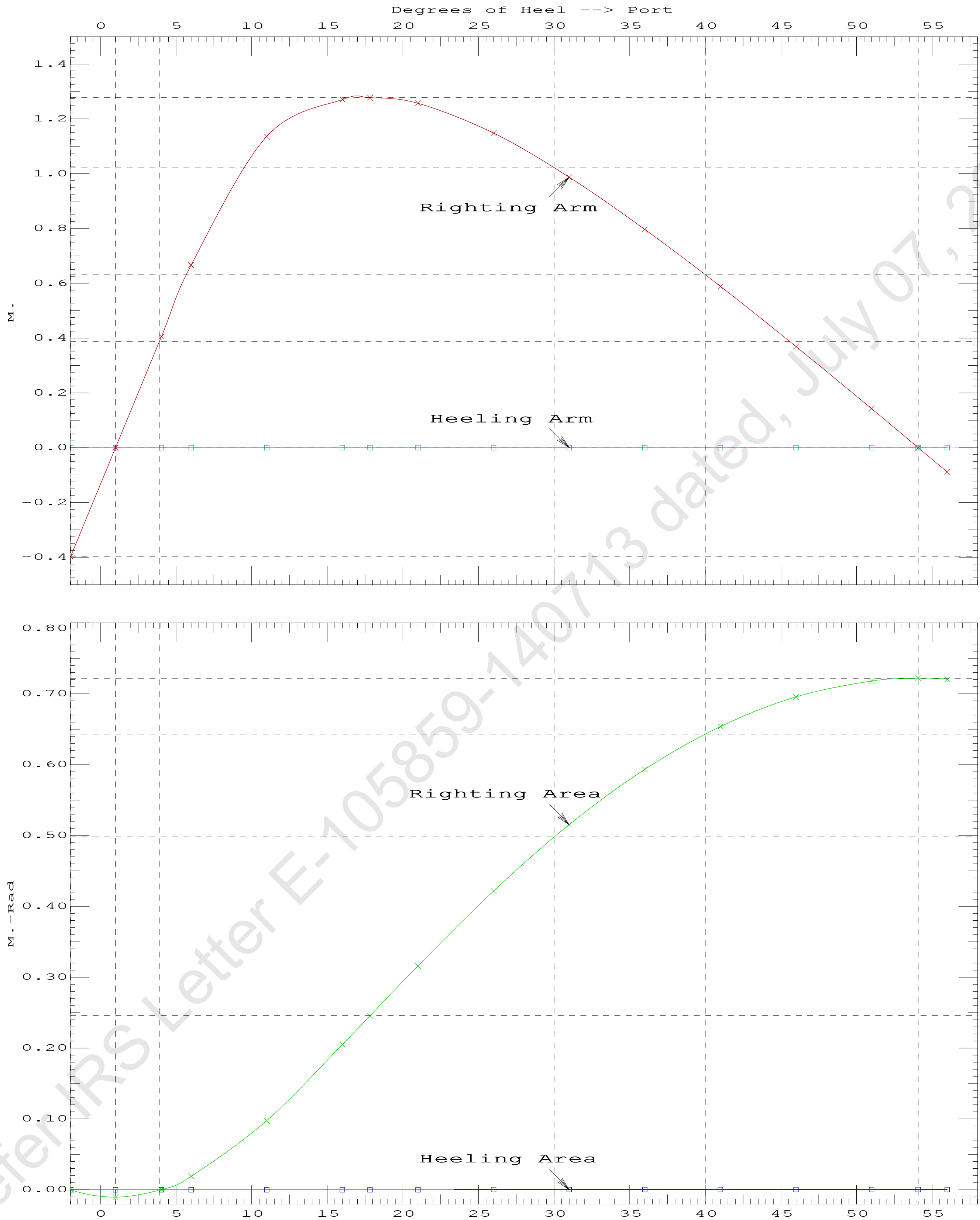
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.070	0.04f	1.96s	1,484.1	0.000	-0.398	0.0000
2.072	0.04f	1.00p	1,484.1	0.000	0.000	-0.0103
2.065	0.04f	4.03p	1,484.1	0.000	0.405	0.0005
2.057	0.04f	6.00p	1,484.1	0.000	0.666	0.0189
2.075	0.05f	11.00p	1,484.1	0.000	1.136	0.0986
2.194	0.07f	16.00p	1,484.0	0.000	1.270	0.2060
2.249	0.07f	17.83p	1,484.0	0.000	1.277	0.2468
2.352	0.08f	21.00p	1,484.1	0.000	1.256	0.3171
2.500	0.10f	26.00p	1,484.1	0.000	1.148	0.4228
2.630	0.13f	31.00p	1,484.1	0.000	0.987	0.5163
2.739	0.14f	36.00p	1,484.1	0.000	0.796	0.5944
2.827	0.16f	41.00p	1,484.1	0.000	0.588	0.6549
2.893	0.18f	46.00p	1,484.1	0.000	0.369	0.6968
2.938	0.20f	51.00p	1,484.1	0.000	0.142	0.7191
2.954	0.21f	54.07p	1,484.1	0.000	0.000	0.7229
2.960	0.21f	56.00p	1,484.1	0.000	-0.089	0.7214

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

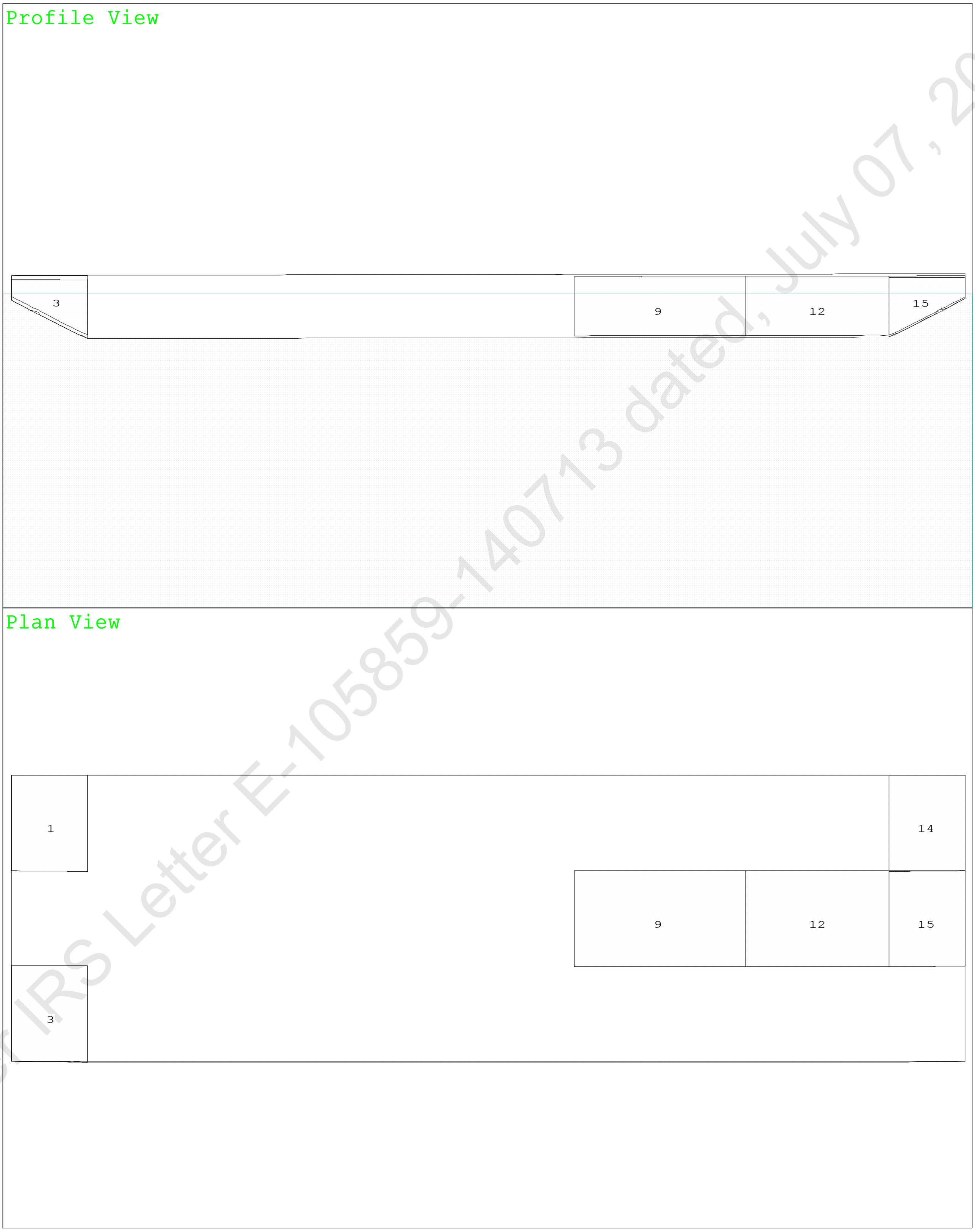
LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs -2 deg to RAzero	>	1.000		71.413 P
(2) Angle from abs 4 deg to RAzero	>	20.00	deg	50.04 P


Degrees of Heel --> Port
Righting Arm
Heeling Arm
M.
Righting Area
Heeling Area
M.-Rad

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Fwd

CG - Draft: 2.116 @ 50.000f, 2.204 @ 0.000 Heel: port 1.13 deg.



Tanks						
	1 WBTK1.P	3 WBTK1.S	9 WBTK4.C	12 WBTK5.C	14 WBTK6.P	15 WBTK6.C

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.116 @ 50.00f, 2.160 @ 25.00f, 2.204 @ 0.00
Trim: Aft 0.088/50.000, Heel: Port 1.13 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	16.500f	0.000	33.320	
Deck Cargo			556.61	25.000f	0.000	3.500	
Total Fixed--------->			1,119.98	21.912f	0.000	4.887	
WBTK1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTK1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks--------->			409.19	32.782f	0.485p	1.662	340.04
Total Weight-------->			1,529.17	24.821f	0.130p	4.024	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,529.17	24.816f	0.188p	1.109	

Righting Arms:				0.000	0.000p		
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane---->	1.025	750.3	24.988f	0.049p	104.54	9.217	
		MT/cm		m.-MT/cm	GML	GMT	
		7.69		31.08	101.62	6.301	

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

399.2 Cu.M. (27%) SALT WATER

150.00 MT Crane 45.10 MT Crane Load
556.61 MT Deck Cargo

DRAFT AT FP (M) = 2.116
DRAFT AT MS (M) = 2.160
DRAFT AT AP (M) = 2.204
DRAFT AT FWD MARK (M) = 2.123
DRAFT AT AFT MARK (M) = 2.197

HYDROSTATIC PROPERTIES
Trim: Aft 0.088/50.000, Heel: Port 1.13 deg., VCG = 4.024

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.160	1,529.17	24.816f	1.109	7.69	24.988f	31.08	101.62	6.301

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 340.0 m.-MT

11-09-19 12:05:24 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.821f TCG = 0.130p VCG = 4.024
Free Surface Adjustment: 0.222
Adjusted CG: LCG = 24.821f TCG = 0.125p VCG = 4.247

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
2.192	0.10a	0.00	1,529.2	0.000	-0.136	0.0000	3.113
2.192	0.10a	1.23p	1,529.2	0.000	0.000	-0.0015	1.886
2.189	0.10a	3.11p	1,529.2	0.000	0.208	0.0020	0.000
2.178	0.10a	6.23p	1,529.2	0.000	0.549	0.0225	-3.114
2.241	0.14a	11.23p	1,529.2	0.000	0.838	0.0847	-8.114
2.313	0.16a	13.68p	1,529.2	0.000	0.858	0.1214	-10.572
2.403	0.18a	16.23p	1,529.2	0.000	0.836	0.1592	-13.114
2.603	0.22a	21.23p	1,529.2	0.000	0.713	0.2281	-18.114
2.807	0.27a	26.23p	1,529.2	0.000	0.514	0.2822	-23.114
2.991	0.32a	31.23p	1,529.2	0.000	0.271	0.3168	-28.114
3.152	0.38a	36.23p	1,529.2	0.000	0.005	0.3289	-33.114
3.155	0.38a	36.31p	1,529.2	0.000	0.000	0.3289	-33.201
3.290	0.43a	41.23p	1,529.2	0.000	-0.272	0.3173	-38.114
3.403	0.47a	46.23p	1,529.2	0.000	-0.554	0.2813	-43.114
3.490	0.51a	51.23p	1,529.2	0.000	-0.836	0.2206	-48.114
3.550	0.55a	56.23p	1,529.2	0.000	-1.115	0.1354	-53.114
3.582	0.58a	61.23p	1,529.2	0.000	-1.387	0.0262	-58.114

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 340.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Port heeling moment = 16.60 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1214 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	35.09 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	1.89 P

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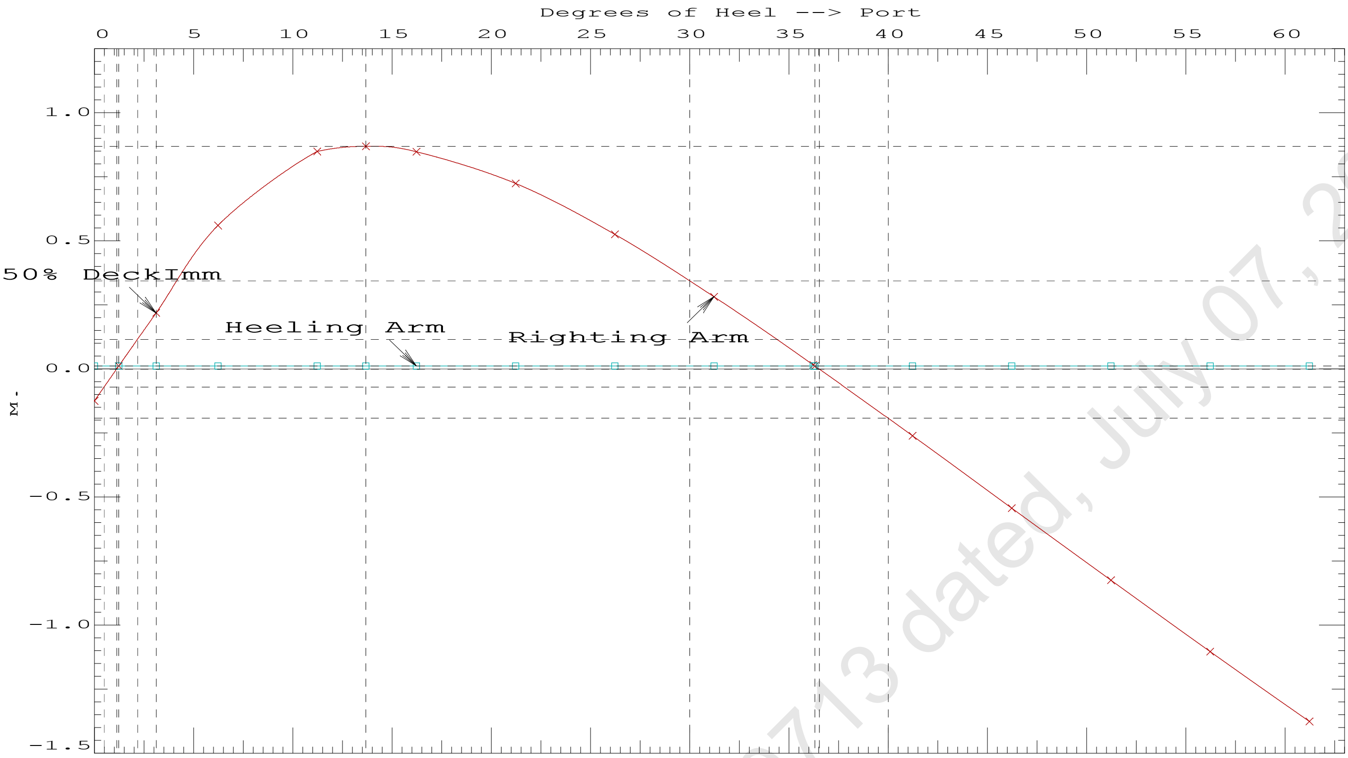
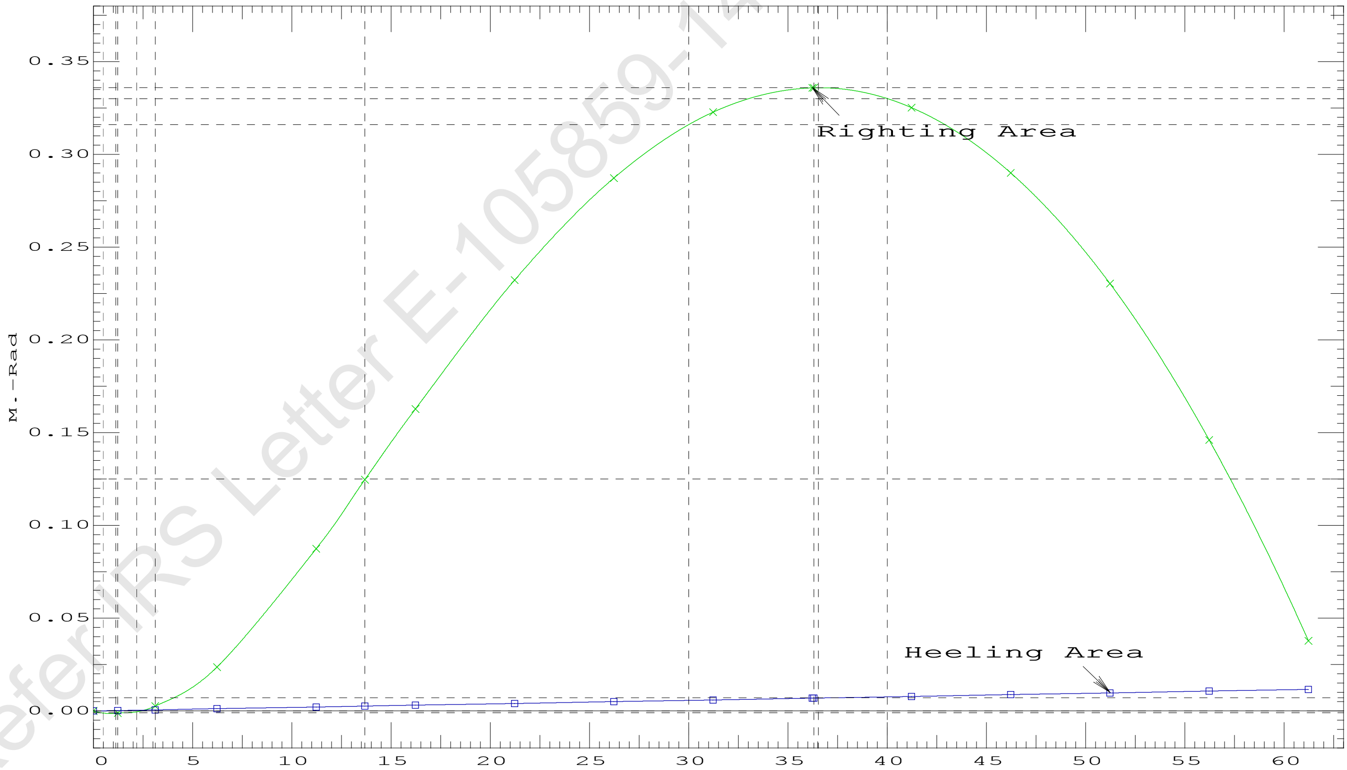
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Fwd





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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.821f TCG = 0.130p VCG = 4.024
Free Surface Adjustment: 0.222
Adjusted CG: LCG = 24.821f TCG = 0.125p VCG = 4.247

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.192	0.10a	1.14p	1,529.2	0.000	0.000	0.0000
2.178	0.10a	6.14p	1,529.2	0.000	0.549	0.0240
2.238	0.14a	11.14p	1,529.2	0.000	0.846	0.0866
2.313	0.16a	13.69p	1,529.2	0.000	0.868	0.1251
2.400	0.18a	16.14p	1,529.2	0.000	0.848	0.1618
2.599	0.22a	21.14p	1,529.2	0.000	0.726	0.2317
2.804	0.27a	26.14p	1,529.2	0.000	0.528	0.2870
2.988	0.32a	31.14p	1,529.2	0.000	0.285	0.3228
3.150	0.38a	36.14p	1,529.2	0.000	0.020	0.3363
3.160	0.38a	36.50p	1,529.2	0.000	0.000	0.3363
3.288	0.42a	41.14p	1,529.2	0.000	-0.258	0.3260
3.401	0.47a	46.14p	1,529.2	0.000	-0.539	0.2912
3.489	0.51a	51.14p	1,529.2	0.000	-0.821	0.2318
3.549	0.55a	56.14p	1,529.2	0.000	-1.100	0.1480
3.582	0.58a	61.14p	1,529.2	0.000	-1.372	0.0401

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 340.0 m.-MT was applied to artificially modify the CG.

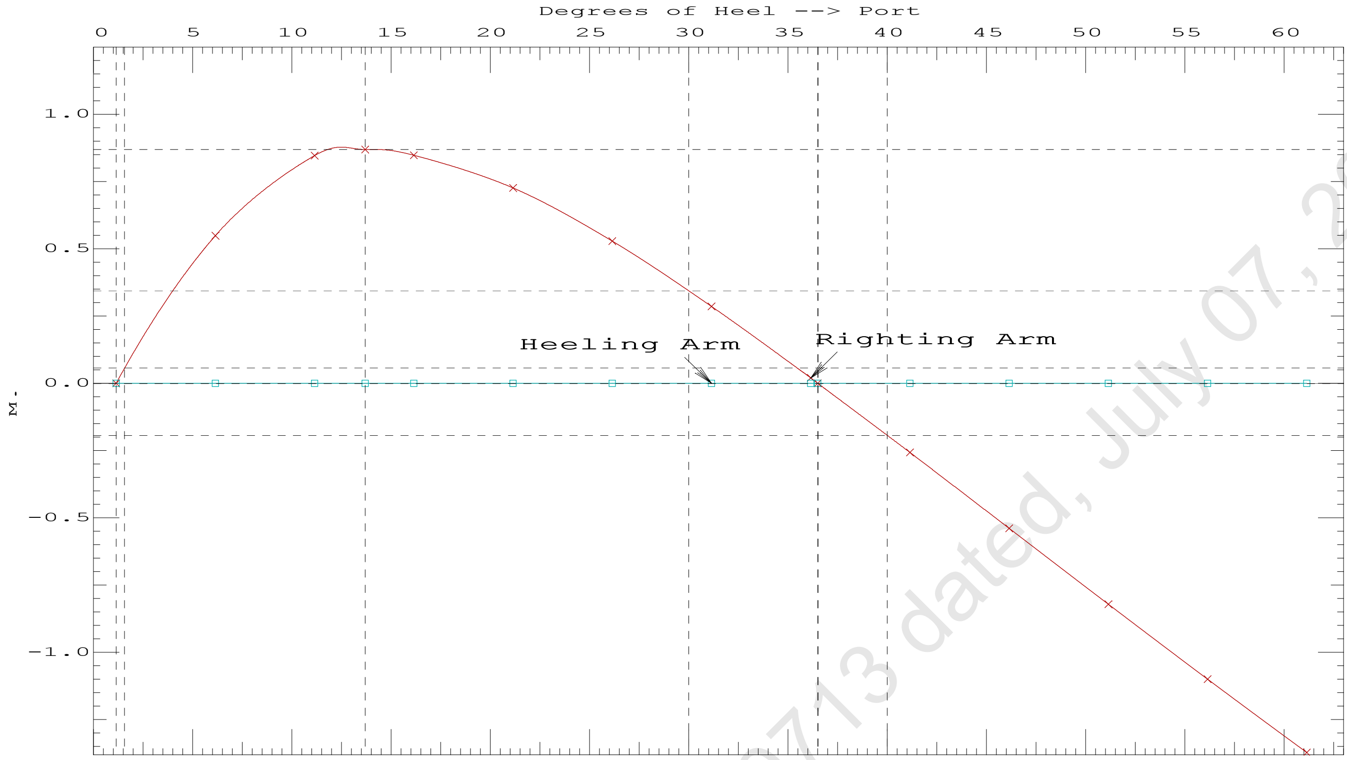
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

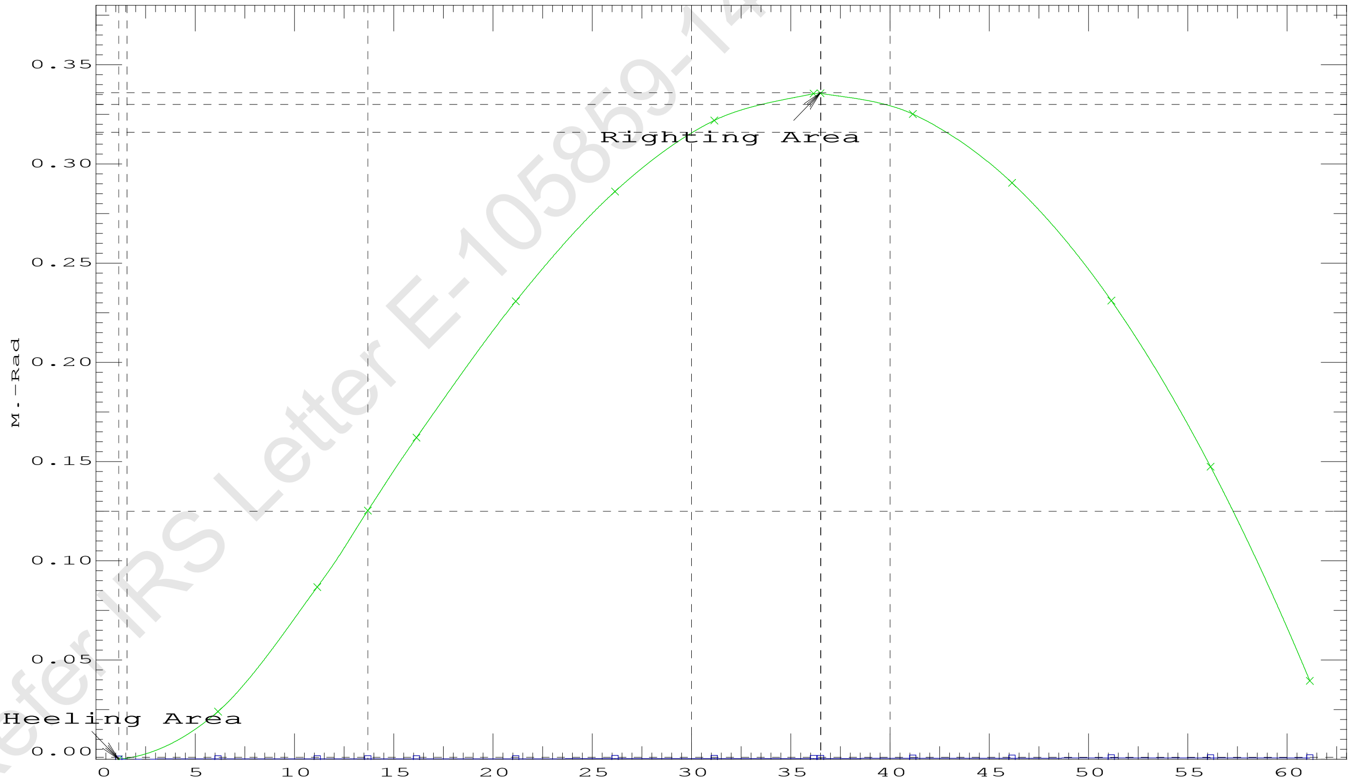
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	1.14 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	132736 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		834.409 P

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo

Crane Lifting 45.1T @ 12m Radius Fwd





Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 45.10T @ 12m Radius Fwd
Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.116 @ 50.00f, 2.160 @ 25.00f, 2.204 @ 0.00
Trim: Aft 0.088/50.000, Heel: Port 1.14 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	16.500f	0.000	33.320	
Deck Cargo			556.61	25.000f	0.000	3.500	
Total Fixed--------->			1,119.98	21.912f	0.000	4.887	
WBTK1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTK1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks--------->			409.19	32.782f	0.485p	1.662	340.04
Total Weight-------->			1,529.17	24.821f	0.130p	4.024	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,529.17	24.816f	0.188p	1.109

Righting Arms:	0.000	0.001p
External Arms:	0.000	0.001
Residual Righting Arms:	0.000	0.000p

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	750.3	24.987f	0.049p	104.54	9.217
		MT/cm		m.-MT/cm	GML	GMT
		7.69		31.08	101.62	6.301

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = -1.14

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GHS 16.68C                PONTOON (YARD NO. CHISTI/009)

                                After hook load drop
                      WEIGHT and DISPLACEMENT and WATERPLANE STATUS
                  USK draft: 2.119 @ 50.00f, 2.101 @ 25.00f, 2.084 @ 0.00
                      Trim: Fwd 0.035/50.000,  Heel: Port 1.00 deg.
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                               368.27  25.000f  0.000   3.000
Crane                                    150.00   4.500f  0.000   6.120
Deck Cargo                               556.61  25.000f  0.000   3.500
  Total Fixed--------->                1,074.88  22.139f  0.000   3.694
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
WBTK1.P        1.000    1.025             39.99   2.340f  4.967p  1.915     41.85*
WBTK1.S        1.000    1.025             39.99   2.340f  4.967s  1.915     41.85*
WBTK4.C        1.000    1.025            135.61  34.000f  0.000   1.500     94.17*
WBTK5.C        1.000    1.025            113.02  42.250f  0.000   1.500     78.48*
WBTK6.P        1.000    1.025             39.99  47.660f  4.967p  1.915     41.84*
WBTK6.C        1.000    1.025             40.60  47.664f  0.000   1.904     41.85*
  Total Tanks--------->                  409.19  32.782f  0.485p  1.662    340.04
  Total Weight-------->                1,484.07  25.074f  0.134p  3.134
                                       Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025          1,484.07  25.075f  0.170p  1.078
------------------------------------------------------------------------------
               Righting Arms:                     0.000   0.000p
Part--------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025            749.8  25.019f  0.045p   107.53   9.483
                                        MT/cm--------m.-MT/cm-------GML-----GMT
                                          7.69             31.31   105.48   7.427
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 2694107.50; t3: 30.0; t: 15.00
limmarg: 1106057.38; t3: 15.0; t: 7.50
limmarg: 351544.22; t3: 7.5; t: 3.75
limmarg: 109977.95; t3: 3.8; t: 1.88
limmarg: 41600.31; t3: 1.9; t: 0.94
limmarg: 18708.19; t3: 1.0; t: 0.47
limmarg: 11348.13; t3: 0.5; t: 0.23
limmarg: 7776.71; t3: 0.3; t: 0.12
limmarg: 6695.63; t3: 0.2; t: 0.06
limmarg: 6240.76; t3: 0.1; t: 0.03
limmarg: 5482.63; t3: 0.1; t: 0.01

               Theta3 is 0.07 degrees, as demonstrated below.
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RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.694

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.071	0.04f	1.14p	1,484.2	0.000	-0.019	0.0000
2.072	0.04f	1.00p	1,484.1	0.000	0.000	-0.0000
2.072	0.04f	0.07s	1,483.9	0.000	0.144	0.0013
2.065	0.04f	4.00s	1,484.1	0.000	0.669	0.0292
2.051	0.05f	9.00s	1,484.1	0.000	1.274	0.1145
2.139	0.06f	14.00s	1,484.1	0.000	1.501	0.2383
2.242	0.07f	17.59s	1,484.1	0.000	1.533	0.3338
2.287	0.08f	19.00s	1,484.1	0.000	1.528	0.3716
2.443	0.10f	24.00s	1,484.1	0.000	1.445	0.5023
2.580	0.12f	29.00s	1,484.1	0.000	1.290	0.6222

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Res. Area Ratio from abs -1.1 deg to abs 0.1	>	1.000	55.826 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.694

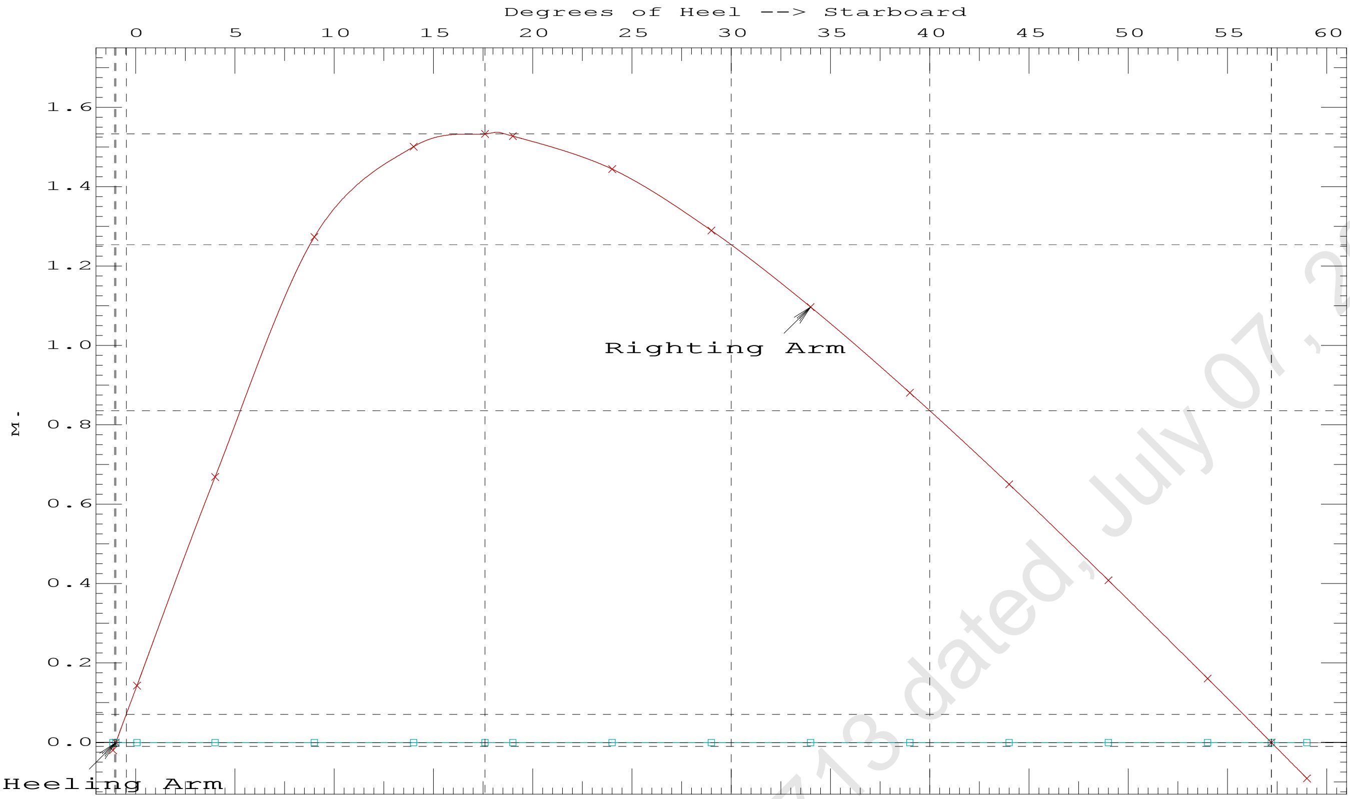
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.071	0.04f	1.14p	1,484.2	0.000	-0.019	0.0000
2.072	0.04f	1.00p	1,484.1	0.000	0.000	-0.0000
2.072	0.04f	0.07s	1,483.9	0.000	0.144	0.0013
2.065	0.04f	4.00s	1,484.1	0.000	0.669	0.0292
2.051	0.05f	9.00s	1,484.1	0.000	1.274	0.1145
2.139	0.06f	14.00s	1,484.1	0.000	1.501	0.2383
2.242	0.07f	17.59s	1,484.1	0.000	1.533	0.3338
2.287	0.08f	19.00s	1,484.1	0.000	1.528	0.3716
2.443	0.10f	24.00s	1,484.1	0.000	1.445	0.5023
2.580	0.12f	29.00s	1,484.1	0.000	1.290	0.6222
2.697	0.14f	34.00s	1,484.1	0.000	1.097	0.7266
2.794	0.16f	39.00s	1,484.1	0.000	0.882	0.8131
2.869	0.18f	44.00s	1,484.1	0.000	0.650	0.8801
2.922	0.19f	49.00s	1,484.1	0.000	0.409	0.9264
2.953	0.21f	54.00s	1,484.1	0.000	0.161	0.9513
2.961	0.22f	57.21s	1,484.1	0.000	0.000	0.9558
2.961	0.22f	59.00s	1,484.1	0.000	-0.090	0.9544

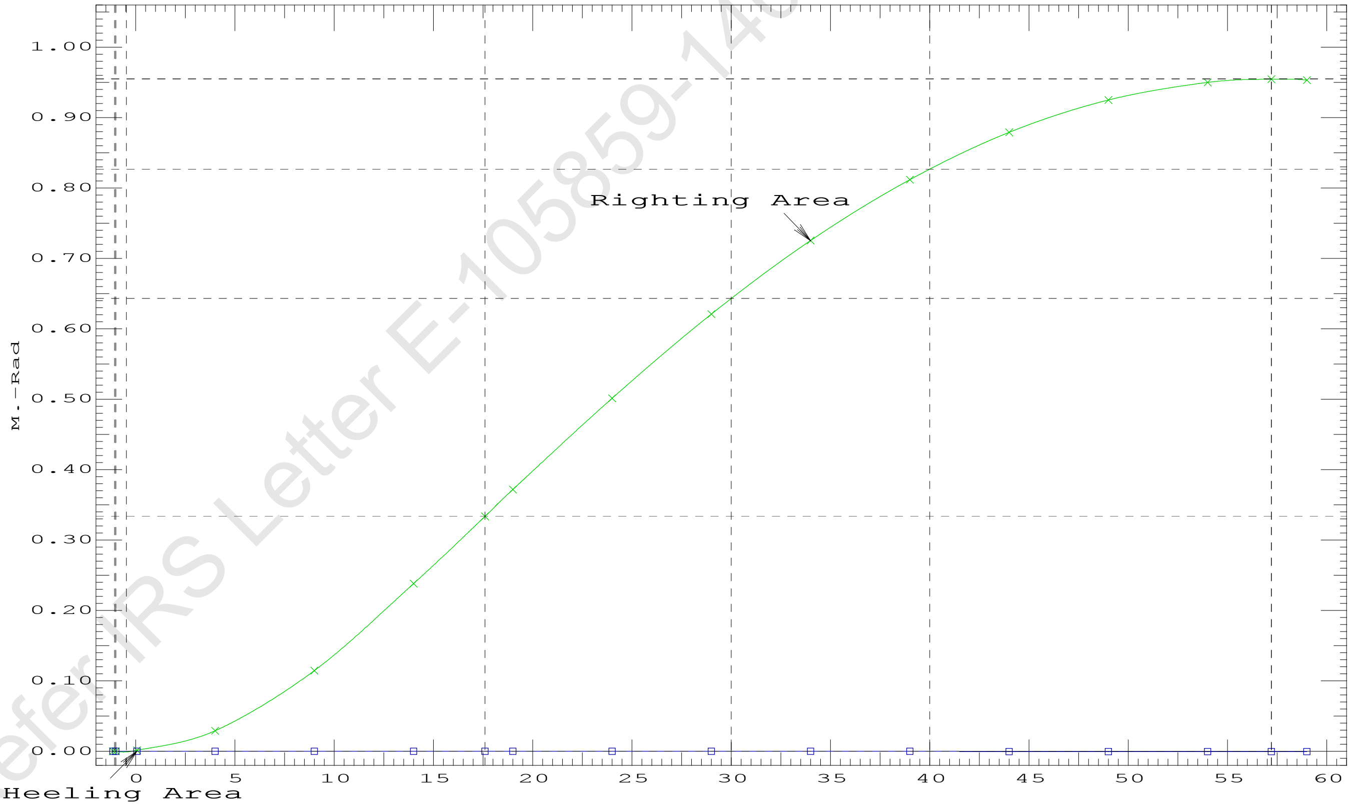
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -1.1 deg to RAzero	>	1.000		39973.5 P
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	57.14 P


Degrees of Heel --> Starboard
0
5
10
15
20
25
30
35
40
45
50
55
60
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
M.
Righting Arm
Heeling Arm

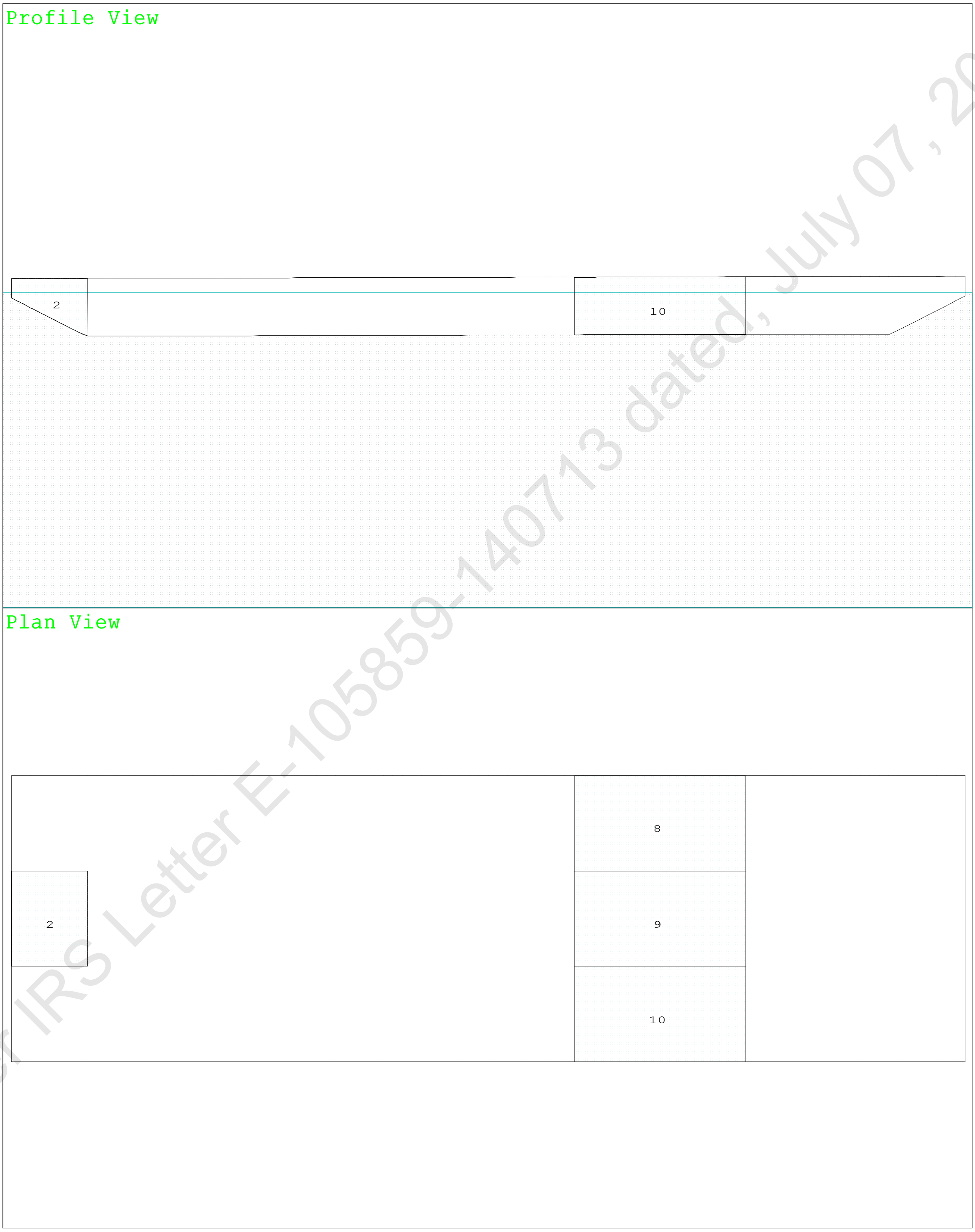

1.00
0.90
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
M.-Rad
Righting Area
Heeling Area
0
5
10
15
20
25
30
35
40
45
50
55
60

PONTOON (YARD NO. CHISTI/009)

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition with Max. draft 2.212m

Condition Graphic - Draft: 2.156 @ 50.000f, 2.268 @ 0.000 Heel: zero



Tanks 2 WBTK1.C 8 WBTK4.P 9 WBTK4.C 10 WBTK4.S

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition with Max. draft 2.212m

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.156 @ 50.00f, 2.212 @ 25.00f, 2.268 @ 0.00
Trim: Aft 0.113/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Deck Cargo			605.72	25.000f	0.000	3.500	
Total Fixed--------->			1,123.99	22.264f	0.000	3.686	
WBTK1.C	1.000	1.025	40.60	2.336f	0.000	1.904	41.85*
WBTK4.P	1.000	1.025	134.48	34.000f	4.980p	1.511	94.17*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK4.S	1.000	1.025	134.48	34.000f	4.980s	1.511	94.17*
Total Tanks--------->			445.16	31.112f	0.000	1.544	324.37
Total Weight-------->			1,569.15	24.774f	0.000	3.078	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,569.15	24.770f	0.000	1.135	

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	750.6	24.995f	0.000	102.05	9.002
		MT/cm		m.-MT/cm	GML	GMT
		7.69		31.42	100.10	7.059

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

434.3 Cu.M. (30%) SALT WATER

150.00 MT Crane 605.72 MT Deck Cargo

DRAFT AT FP (M) = 2.156
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.268
DRAFT AT FWD MARK (M) = 2.165
DRAFT AT AFT MARK (M) = 2.259

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition with Max. draft 2.212m

HYDROSTATIC PROPERTIES
Trim: Aft 0.113/50.000, No Heel, VCG = 3.078

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,569.15	24.770f	1.135	7.69	24.995f	31.42	100.10	7.059

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 324.4 m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.774f TCG = 0.000 VCG = 3.078
Free Surface Adjustment: 0.207
Adjusted CG: LCG = 24.775f TCG = 0.000 VCG = 3.285

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.256	0.13a	0.00	1,569.2	0.000	-0.010	0.0000	2.867
2.256	0.13a	0.08s	1,569.2	0.000	0.000	-0.0000	2.786
2.254	0.13a	2.87s	1,569.1	0.000	0.344	0.0084	0.000
2.247	0.13a	5.08s	1,569.2	0.000	0.616	0.0269	-2.214
2.298	0.17a	10.08s	1,569.2	0.000	1.031	0.1002	-7.214
2.465	0.22a	15.08s	1,569.2	0.000	1.119	0.1964	-12.214
2.485	0.23a	15.59s	1,569.1	0.000	1.120	0.2063	-12.720
2.679	0.27a	20.08s	1,569.2	0.000	1.079	0.2932	-17.214
2.908	0.33a	25.08s	1,569.2	0.000	0.968	0.3832	-22.214
3.123	0.40a	30.08s	1,569.2	0.000	0.806	0.4610	-27.214
3.315	0.46a	35.08s	1,569.2	0.000	0.614	0.5231	-32.214
3.483	0.52a	40.08s	1,569.2	0.000	0.405	0.5677	-37.214
3.624	0.58a	45.08s	1,569.2	0.000	0.186	0.5935	-42.214
3.720	0.63a	49.21s	1,569.2	0.000	0.000	0.6003	-46.340
3.738	0.64a	50.08s	1,569.2	0.000	-0.040	0.6000	-47.214
3.823	0.69a	55.08s	1,569.2	0.000	-0.268	0.5866	-52.214
3.879	0.73a	60.08s	1,569.2	0.000	-0.496	0.5532	-57.214

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 324.4 m.-MT was applied to artificially modify the CG.

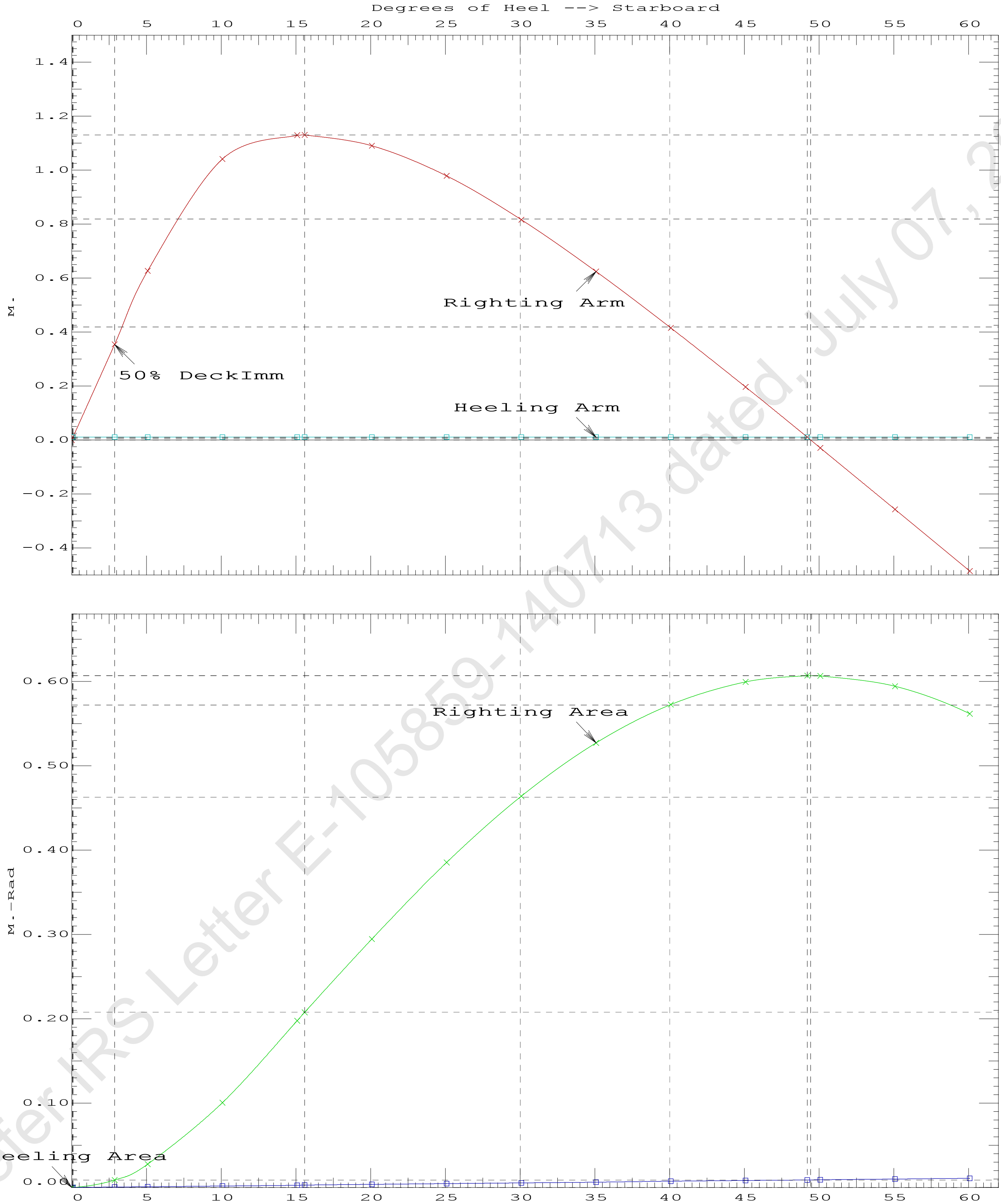
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 16.28 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.2063	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	49.13	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	2.79	P

PONTOON (YARD NO. CHISTI/009)

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition with Max. draft 2.212m



PONTOON (YARD NO. CHISTI/009)

Condition 6 : Lightship + Max.Deck Cargo

Max. deck cargo loading condition

Condition Graphic - Draft: 2.212 Trim: zero Heel: zero

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11-09-19 12:06:08    Cybermarine Technologies PTE. Ltd.
GHS 16.68C                PONTOON (YARD NO. CHISTI/009)
```

Condition 6 : Lightship + Max.Deck Cargo

Max. deck cargo loading condition

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
                           Trim: zero,  Heel: zero
Part------------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
LIGHT SHIP                           368.27  25.000f  0.000   3.000
Deck Cargo                         1,200.88  25.000f  0.000   3.500
  Total Weight-------->            1,569.15  25.000f  0.000   3.383
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->                0.00                                0.00
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      1,569.15  25.000f  0.000   1.134
------------------------------------------------------------------------------
               Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         750.6  25.000f  0.000    102.05   9.209
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       7.69             31.32     99.81   6.961
Distances in METERS.-----------------------------------------Moments in m.-MT.

                              SUMMARY OF LOADING
         0.0 Cu.M.   (0%) SALT WATER

    1,200.88 MT  Deck Cargo

DRAFT AT FP (M) =    2.212
DRAFT AT MS (M) =    2.212
DRAFT AT AP (M) =    2.212
DRAFT AT FWD MARK (M) =    2.212
DRAFT AT AFT MARK (M) =    2.212
```

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 6 : Lightship + Max.Deck Cargo

Max. deck cargo loading condition

HYDROSTATIC PROPERTIES
No Trim, No Heel, VCG = 3.383

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,569.15	25.000f	1.134	7.69	25.000f	31.32	99.81	6.961

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.000f TCG = 0.000 VCG = 3.383

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
2.200	0.00	0.00	1,569.2	0.000	-0.005	0.0000	3.085
2.200	0.00	0.04s	1,569.1	0.000	0.000	-0.0000	3.042
2.196	0.00	3.09s	1,569.2	0.000	0.371	0.0098	-0.000
2.190	0.00	5.04s	1,569.1	0.000	0.608	0.0265	-1.958
2.221	0.00	10.04s	1,569.1	0.000	1.018	0.0989	-6.958
2.366	0.00	15.04s	1,569.2	0.000	1.100	0.1937	-11.958
2.369	0.00	15.11s	1,569.2	0.000	1.100	0.1951	-12.029
2.558	0.00	20.04s	1,569.1	0.000	1.053	0.2886	-16.958
2.763	0.00	25.04s	1,569.2	0.000	0.934	0.3758	-21.958
2.949	0.00	30.04s	1,569.2	0.000	0.764	0.4503	-26.958
3.114	0.00	35.04s	1,569.2	0.000	0.566	0.5085	-31.958
3.254	0.00	40.04s	1,569.1	0.000	0.350	0.5486	-36.958
3.370	0.00	45.04s	1,569.1	0.000	0.124	0.5693	-41.958
3.422	0.00	47.74s	1,569.2	0.000	0.000	0.5723	-44.658
3.460	0.00	50.04s	1,569.1	0.000	-0.107	0.5701	-46.958
3.524	0.00	55.04s	1,569.1	0.000	-0.340	0.5507	-51.958
3.561	0.00	60.04s	1,569.2	0.000	-0.573	0.5108	-56.958

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 8.20 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1951	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	47.70	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	3.04	P

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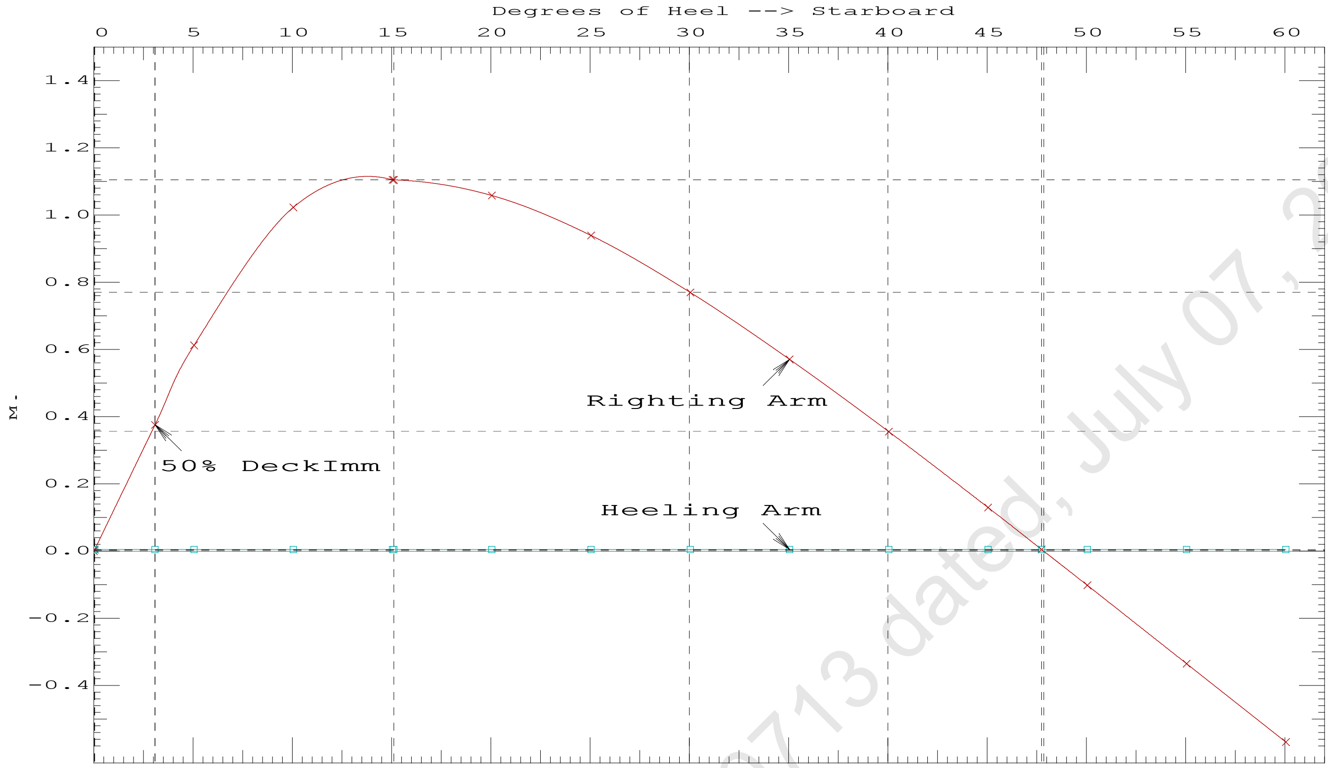
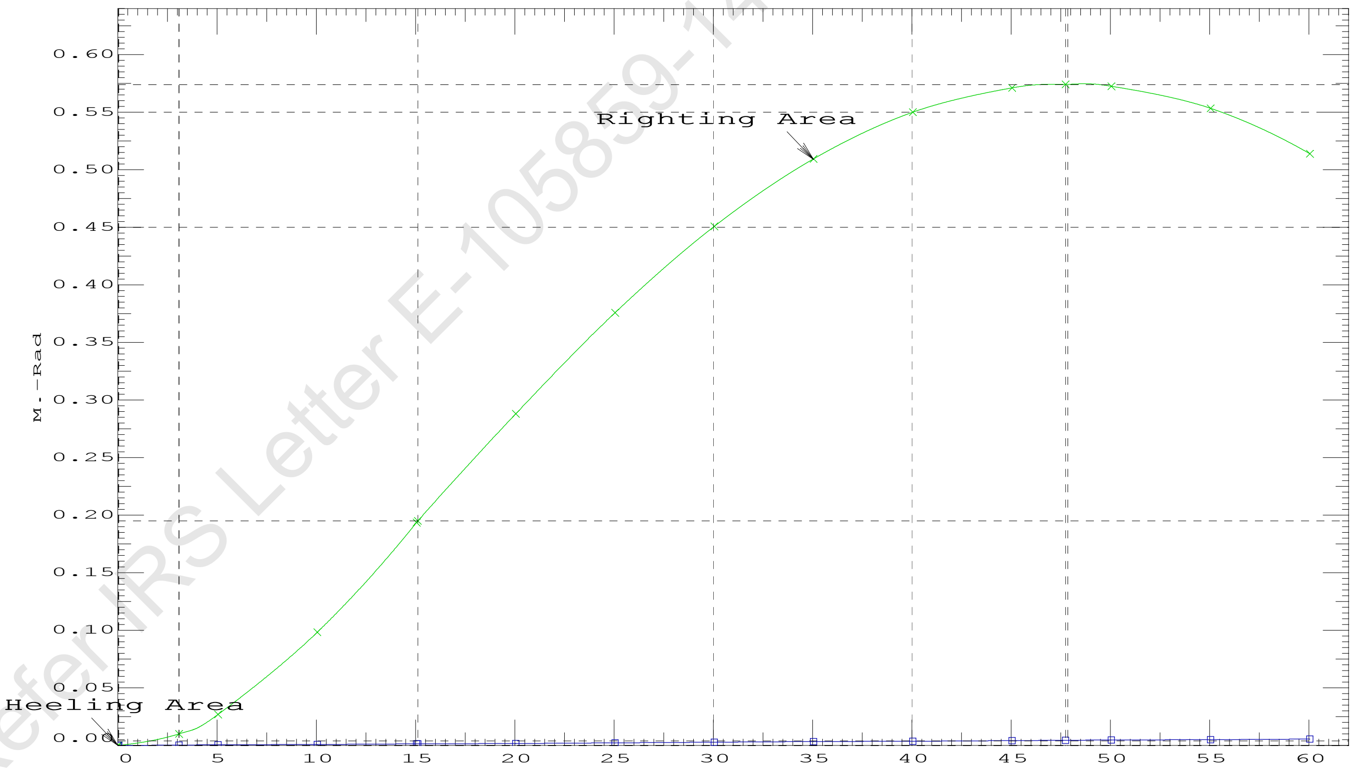
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 6 : Lightship + Max.Deck Cargo

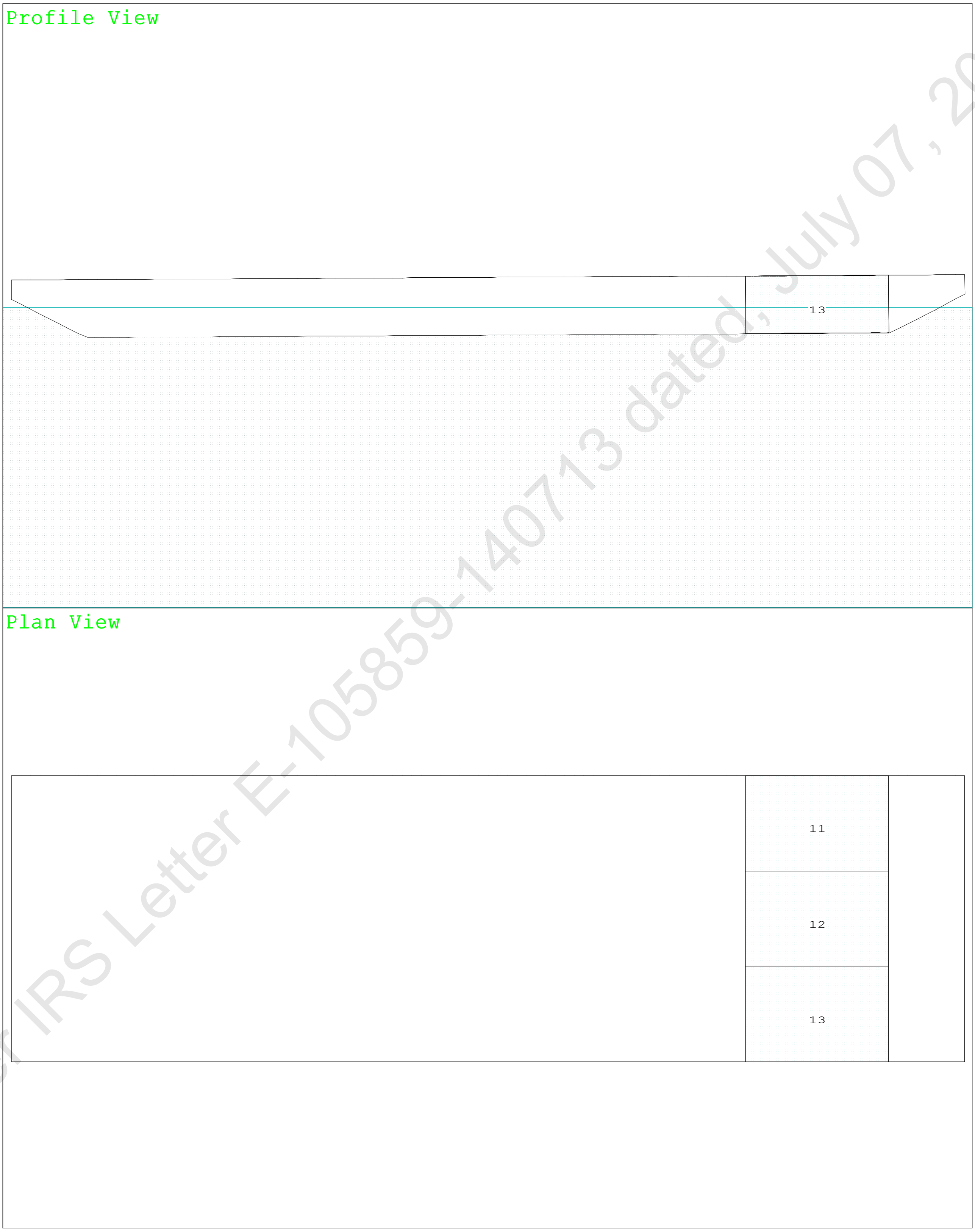
Max. deck cargo loading condition





Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.308 @ 50.000f, 1.589 @ 0.000 Heel: zero



Tanks	11 WBTK5.P	12 WBTK5.C	13 WBTK5.S

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 7 : Lightship + Load at Aft + Ballast

Crane Lifting 36.6T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.308 @ 50.00f, 1.449 @ 25.00f, 1.589 @ 0.00

Trim: Aft 0.281/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	14.580a	0.000	76.930	
Total Fixed--------->			654.87	14.962f	0.000	8.323	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks--------->			337.18	42.250f	0.000	1.507	235.45
Total Weight-------->			992.05	24.237f	0.000	6.006	
			Displ(MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	992.05	24.207f	0.000	0.737	

Part	SpGr	WPA	LCF	TCF	BML	BMT
Righting Arms:			0.000	0.000		
Total Waterplane---->	1.025	717.7	24.738f	0.000	141.11	13.665
		MT/cm	**m.-MT/cm**		**GML**	**GMT**
		7.36	26.95		135.84	8.395

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

329.0 Cu.M. (22%) SALT WATER

250.00 MT Crane 36.60 MT Crane Load

DRAFT AT FP (M) = 1.308
DRAFT AT MS (M) = 1.449
DRAFT AT AP (M) = 1.589
DRAFT AT FWD MARK (M) = 1.331
DRAFT AT AFT MARK (M) = 1.567

HYDROSTATIC PROPERTIES

Trim: Aft 0.281/50.000, No Heel, VCG = 6.006

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.450	992.05	24.207f	0.737	7.36	24.738f	26.95	135.84	8.395

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 235.4 m.-MT

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.237f TCG = 0.000 VCG = 6.006
Free Surface Adjustment: 0.237
Adjusted CG: LCG = 24.238f TCG = 0.000 VCG = 6.244

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.577	0.32a	0.00	992.05	0.000	-0.014	0.0000	5.497
1.577	0.32a	0.09s	992.05	0.000	0.000	-0.0000	5.405
1.565	0.32a	5.09s	992.05	0.000	0.738	0.0322	0.405
1.563	0.32a	5.50s	992.05	0.000	0.798	0.0376	0.000
1.532	0.33a	10.09s	992.05	0.000	1.458	0.1282	-4.595
1.494	0.43a	15.09s	992.05	0.000	1.859	0.2753	-9.595
1.492	0.44a	15.47s	992.05	0.000	1.860	0.2876	-9.973
1.459	0.58a	20.09s	992.05	0.000	1.710	0.4347	-14.595
1.416	0.73a	25.09s	992.05	0.000	1.359	0.5700	-19.595
1.365	0.88a	30.09s	992.05	0.000	0.922	0.6702	-24.595
1.305	1.03a	35.09s	992.05	0.000	0.442	0.7300	-29.595
1.246	1.16a	39.48s	992.05	0.000	0.000	0.7471	-33.987
1.237	1.17a	40.09s	992.05	0.000	-0.062	0.7467	-34.595
1.160	1.31a	45.09s	992.05	0.000	-0.577	0.7190	-39.595
1.075	1.44a	50.09s	992.05	0.000	-1.094	0.6461	-44.595
0.980	1.56a	55.09s	992.05	0.000	-1.608	0.5282	-49.595
0.876	1.65a	60.09s	992.05	0.000	-2.112	0.3658	-54.595

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 13.95 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.2876	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	39.39	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	5.41	P

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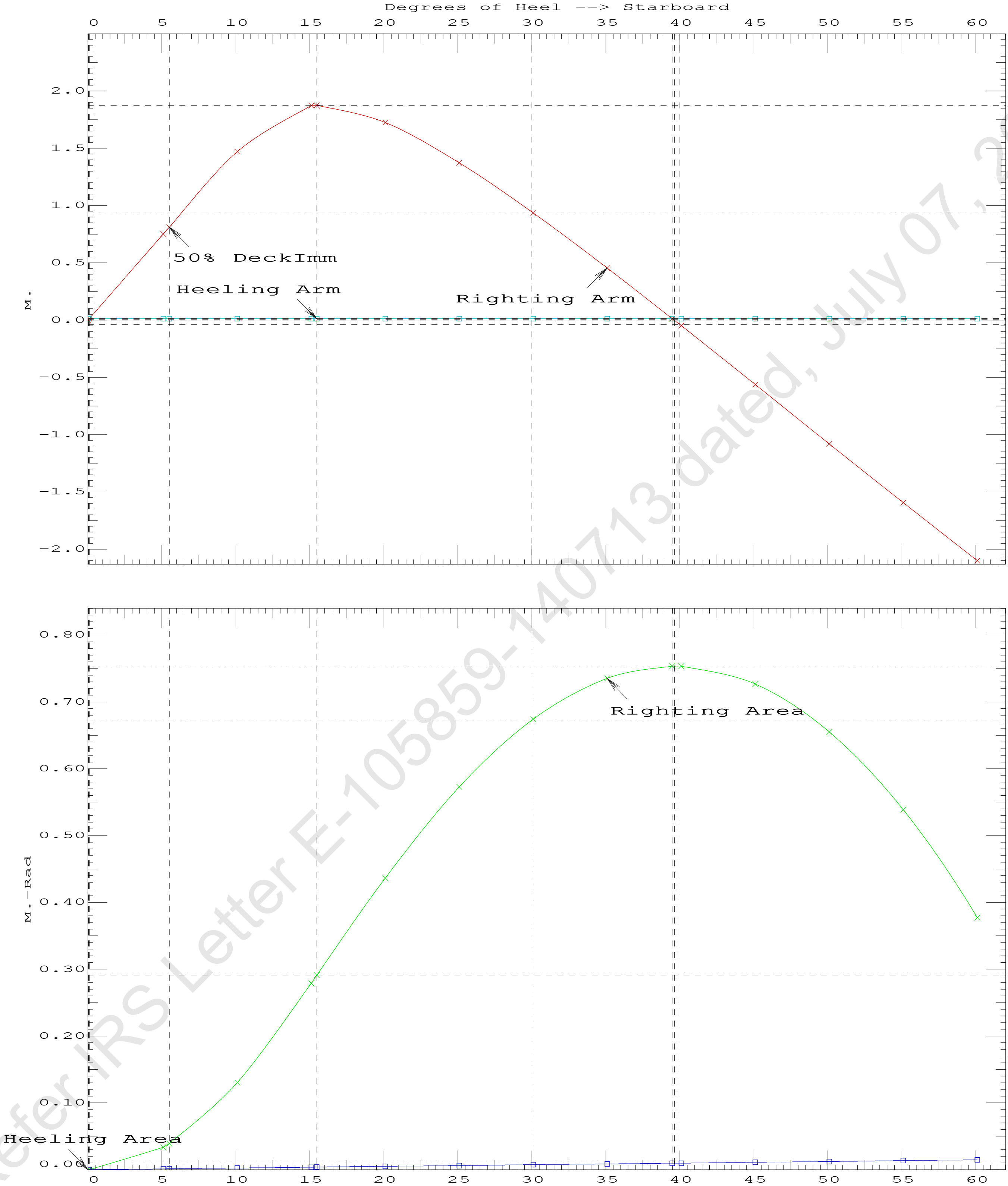
GHS 16.68C PONTOON (YARD NO. CHISTI/009)

Condition 7 : Lightship + Load at Aft + Ballast

Crane Lifting 36.6T @ 18m Radius Aft



11-09-19 12:06:40 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.237f TCG = 0.000 VCG = 6.006
Free Surface Adjustment: 0.237
Adjusted CG: LCG = 24.238f TCG = 0.000 VCG = 6.244

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.577	0.32a	0.01p	992.05	0.000	0.000	0.0000
1.565	0.32a	5.01p	992.05	0.000	0.738	0.0322
1.533	0.33a	10.01p	992.05	0.000	1.459	0.1282
1.495	0.43a	15.01p	992.05	0.000	1.871	0.2757
1.492	0.44a	15.49p	992.05	0.000	1.873	0.2915
1.460	0.58a	20.01p	992.05	0.000	1.728	0.4363
1.417	0.73a	25.01p	992.05	0.000	1.379	0.5742
1.365	0.88a	30.01p	992.05	0.000	0.943	0.6762
1.306	1.03a	35.01p	992.05	0.000	0.463	0.7378
1.244	1.16a	39.61p	992.05	0.000	0.000	0.7565
1.238	1.17a	40.01p	992.05	0.000	-0.040	0.7564
1.162	1.31a	45.01p	992.05	0.000	-0.555	0.7305
1.076	1.44a	50.01p	992.05	0.000	-1.072	0.6595
0.982	1.55a	55.01p	992.05	0.000	-1.586	0.5435
0.878	1.65a	60.01p	992.05	0.000	-2.091	0.3830

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

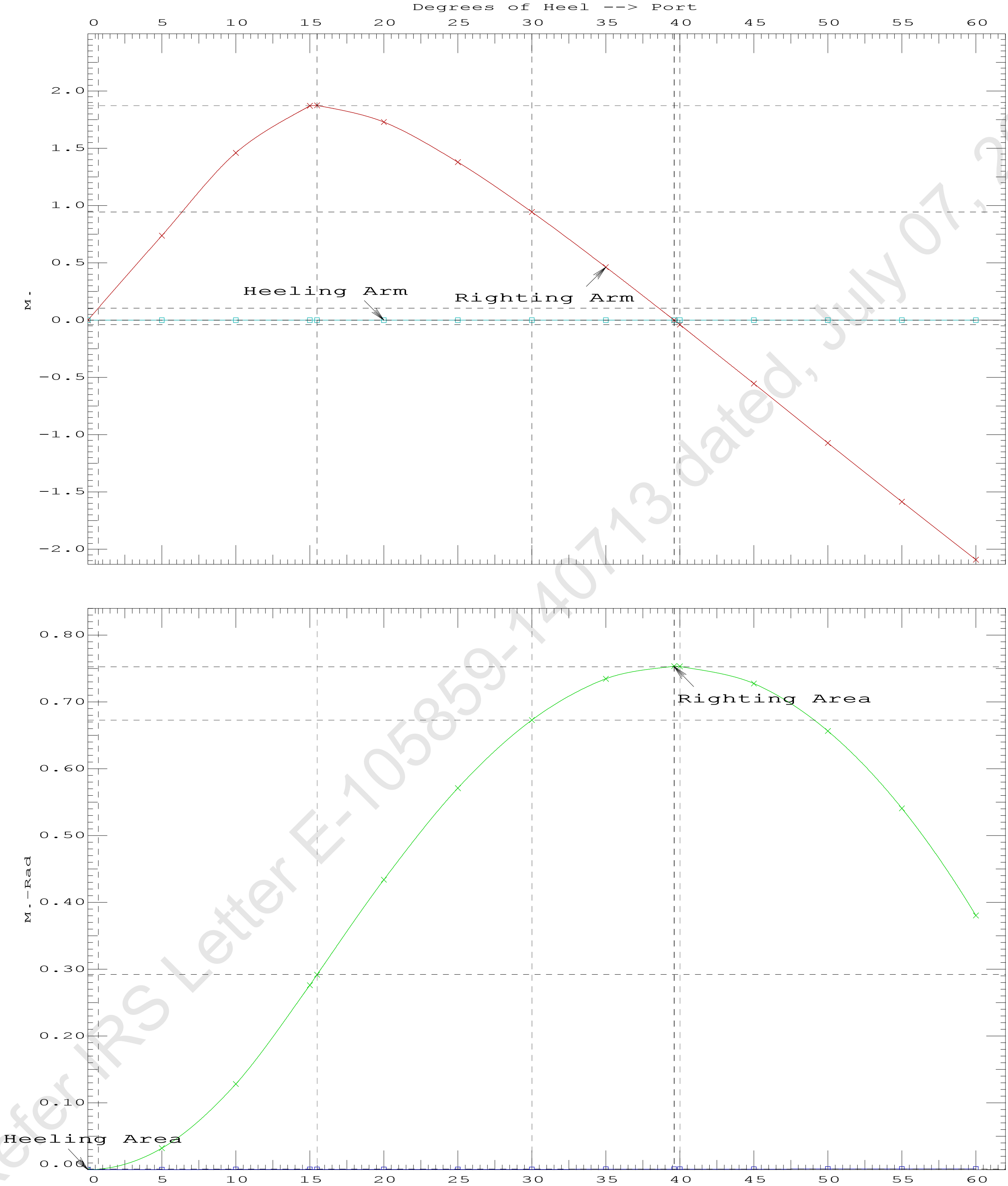
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	185827 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1086.80 P

Condition 7 : Lightship + Load at Aft + Ballast

Crane Lifting 36.6T @ 18m Radius Aft



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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.60T @ 18m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.308 @ 50.00f, 1.449 @ 25.00f, 1.589 @ 0.00
Trim: Aft 0.281/50.000, Heel: Port 0.01 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	14.580a	0.000	76.930	
Total Fixed--------->			654.87	14.962f	0.000	8.323	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks--------->			337.18	42.250f	0.000	1.507	235.45
Total Weight-------->			992.05	24.237f	0.000	6.006	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	991.93	24.207f	0.002p	0.737	

Righting Arms:	0.000	-0.001s
External Arms:	0.000	-0.001p
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	716.8	24.742f	0.007p	140.62	13.638
		MT/cm		m.-MT/cm	GML	GMT
		7.35		26.85	135.35	8.369

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = -0.01

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GHS 16.68C               PONTOON (YARD NO. CHISTI/009)

                      After hook load drop
           WEIGHT and DISPLACEMENT and WATERPLANE STATUS
      USK draft: 1.526 @ 50.00f, 1.399 @ 25.00f, 1.272 @ 0.00
             Trim: Fwd 0.255/50.000,  Heel: 0.00 deg.
Part----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                         368.27  25.000f  0.000   3.000
Crane                              250.00   4.500f  0.000   6.120
  Total Fixed--------->            618.27  16.711f  0.000   4.262
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
WBTK5.P        1.000    1.025      112.08  42.251f  4.980p  1.511     78.49*
WBTK5.C        1.000    1.025      113.02  42.250f  0.000   1.500     78.48*
WBTK5.S        1.000    1.025      112.08  42.251f  4.980s  1.511     78.49*
  Total Tanks--------->            337.18  42.250f  0.000   1.507   235.45
  Total Weight-------->            955.45  25.724f  0.000   3.290
                                 Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025      955.39  25.738f  0.001   0.710
------------------------------------------------------------------------------
               Righting Arms:                0.001f  0.001s
               External Arms:                0.000   0.001
      Residual Righting Arms:                0.001f  0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         714.6  25.239f  0.002s   144.65  14.124
                                     MT/cm--------m.-MT/cm-------GML-----GMT
                                      7.32            27.15    142.07  11.545
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: 843451584.00; t3: 30.0; t: 15.00
limmarg: 295994528.00; t3: 15.0; t: 7.50
limmarg: 76989072.00; t3: 7.5; t: 3.75
limmarg: 19230878.00; t3: 3.8; t: 1.88
limmarg: 5040704.00; t3: 1.9; t: 0.94
limmarg: 1259850.38; t3: 1.0; t: 0.47
limmarg: 384005.44; t3: 0.5; t: 0.23
limmarg: 98928.77; t3: 0.3; t: 0.12
limmarg: 43488.90; t3: 0.2; t: 0.06
limmarg: 26054.27; t3: 0.1; t: 0.03
limmarg: 6203.59; t3: 0.1; t: 0.01

                 Theta3 is 0.07 degrees, as demonstrated below.
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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 16.711f TCG = 0.000 VCG = 4.262

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.260	0.29f	0.01p	955.45	0.000	-0.002	0.0000
1.260	0.29f	0.00	955.45	0.000	0.000	-0.0000
1.260	0.29f	0.07s	955.46	0.000	0.014	0.0000
1.249	0.29f	5.00s	955.43	0.000	1.035	0.0452
1.208	0.30f	10.00s	955.45	0.000	2.045	0.1797
1.078	0.38f	15.00s	955.45	0.000	2.726	0.3903
0.926	0.49f	19.33s	955.45	0.000	2.842	0.6034
0.902	0.51f	20.00s	955.45	0.000	2.839	0.6366
0.717	0.64f	25.00s	955.45	0.000	2.731	0.8821
0.526	0.76f	30.00s	955.45	0.000	2.522	1.1121

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 618.27 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	63.036 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 16.711f TCG = 0.000 VCG = 4.262

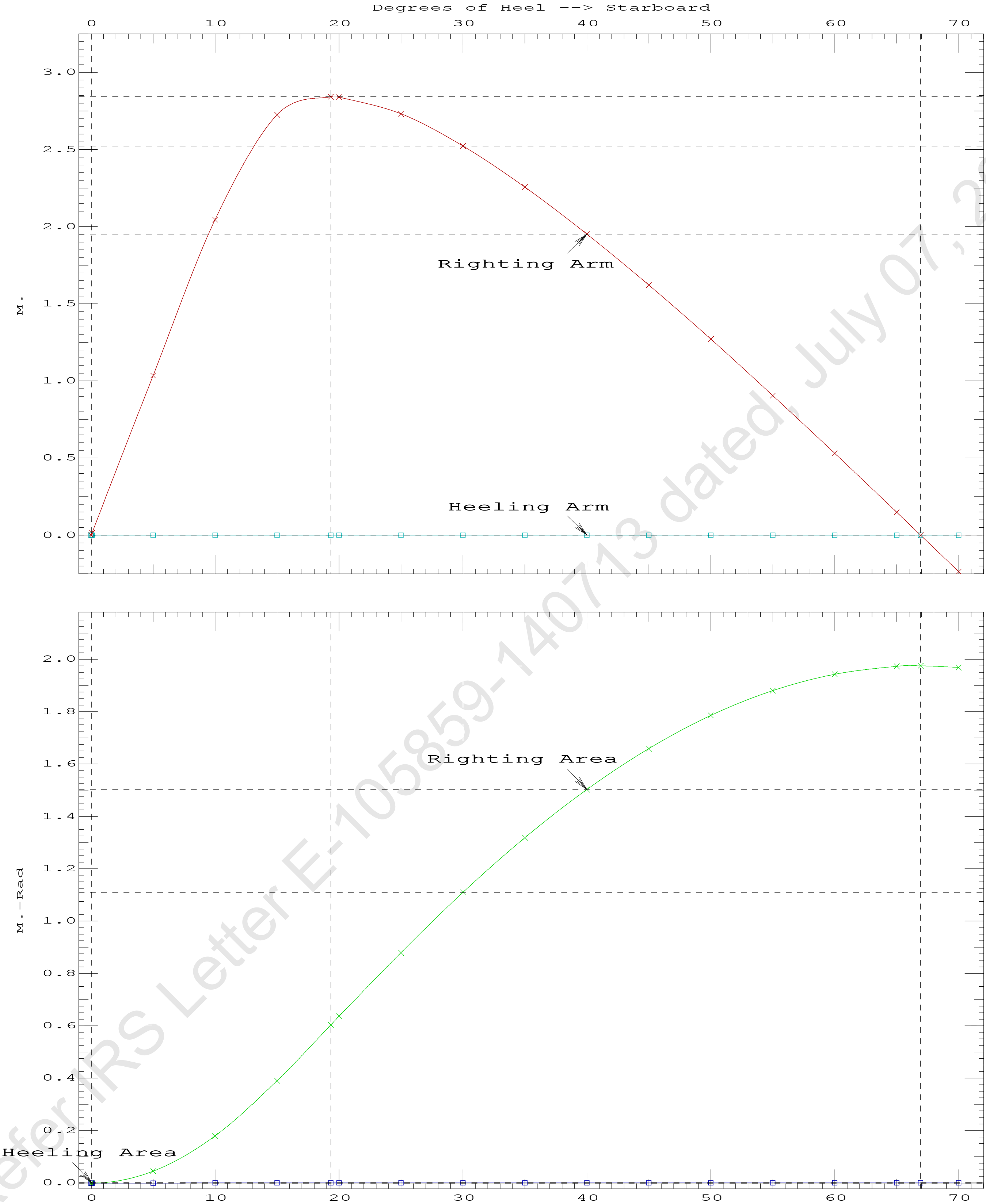
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.260	0.29f	0.01p	955.45	0.000	-0.002	0.0000
1.260	0.29f	0.00	955.45	0.000	0.000	-0.0000
1.260	0.29f	0.07s	955.46	0.000	0.014	0.0000
1.249	0.29f	5.00s	955.43	0.000	1.035	0.0452
1.208	0.30f	10.00s	955.45	0.000	2.045	0.1797
1.078	0.38f	15.00s	955.45	0.000	2.726	0.3903
0.926	0.49f	19.33s	955.45	0.000	2.842	0.6034
0.902	0.51f	20.00s	955.45	0.000	2.839	0.6366
0.717	0.64f	25.00s	955.45	0.000	2.731	0.8821
0.526	0.76f	30.00s	955.45	0.000	2.522	1.1121
0.328	0.89f	35.00s	955.45	0.000	2.256	1.3210
0.127	1.01f	40.00s	955.45	0.000	1.952	1.5048
-0.077	1.13f	45.00s	955.45	0.000	1.621	1.6609
-0.281	1.24f	50.00s	955.45	0.000	1.270	1.7872
-0.483	1.34f	55.00s	955.45	0.000	0.905	1.8822
-0.682	1.43f	60.00s	955.45	0.000	0.529	1.9448
-0.874	1.51f	65.00s	955.45	0.000	0.147	1.9744
-0.946	1.53f	66.92s	955.45	0.000	0.000	1.9769
-1.059	1.57f	70.00s	955.45	0.000	-0.237	1.9705

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 618.27 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	66.85 P


Degrees of Heel --> Starboard
0
10
20
30
40
50
60
70
M.
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Righting Arm
Heeling Arm
M.-Rad
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Righting Area
Heeling Area

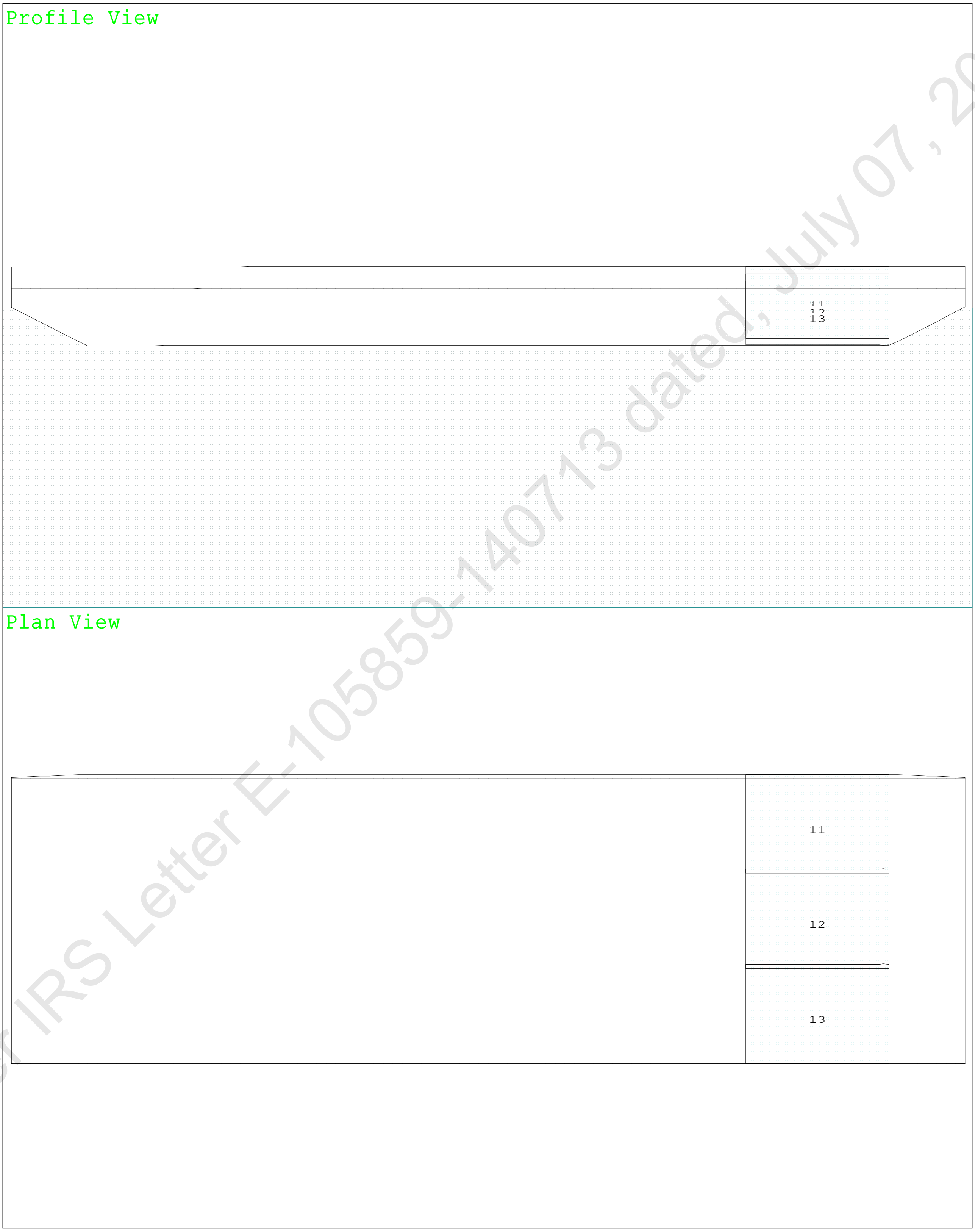
11-09-19 12:07:15 Cybermarine Technologies PTE. Ltd.

GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 8 : Lightship + Load at Stbd + Ballast

Crane Lifting 36.6T @ 18m Radius Stbd

CG - Draft: 1.434 @ 50.000f, 1.456 @ 0.000 Heel: stbd 4.37 deg.



Tanks	
	11 WBTK5.P 12 WBTK5.C 13 WBTK5.S

11-09-19 12:07:15 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.434 @ 50.00f, 1.445 @ 25.00f, 1.456 @ 0.00
Trim: Aft 0.022/50.000, Heel: Stbd 4.37 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	4.500f	18.000s	76.930	
Total Fixed--------->			654.87	16.028f	1.006s	8.323	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks--------->			337.18	42.250f	0.000	1.507	235.45
Total Weight-------->			992.05	24.941f	0.664s	6.006	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	992.06	24.938f	1.063s	0.775	

Righting Arms: 0.000 0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	719.2	24.979f	0.232s	141.48	13.762
		MT/cm	m.-MT/cm		GML	GMT
		7.37		27.03	136.23	8.516

Distances in METERS.---Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

329.0 Cu.M. (22%) SALT WATER

250.00 MT Crane 36.60 MT Crane Load

DRAFT AT FP (M) = 1.434
DRAFT AT MS (M) = 1.445
DRAFT AT AP (M) = 1.456
DRAFT AT FWD MARK (M) = 1.436
DRAFT AT AFT MARK (M) = 1.454

HYDROSTATIC PROPERTIES

Trim: Aft 0.022/50.000, Heel: Stbd 4.37 deg., VCG = 6.006

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.445	992.06	24.938f	0.775	7.37	24.979f	27.03	136.23	8.516

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 235.4 m.-MT

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.941f TCG = 0.664s VCG = 6.006
Free Surface Adjustment: 0.237
Adjusted CG: LCG = 24.941f TCG = 0.646s VCG = 6.243

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.448	0.02a	0.00	992.05	0.000	-0.660	0.0000	6.032
1.439	0.02a	4.46s	992.05	0.000	0.000	-0.0257	1.573
1.432	0.02a	6.03s	992.05	0.000	0.235	-0.0225	0.000
1.406	0.03a	9.46s	992.05	0.000	0.736	0.0066	-3.427
1.333	0.03a	14.46s	992.05	0.000	1.234	0.0938	-8.427
1.309	0.03a	15.76s	992.05	0.000	1.249	0.1220	-9.732
1.239	0.04a	19.46s	992.05	0.000	1.147	0.2009	-13.427
1.136	0.05a	24.46s	992.05	0.000	0.834	0.2883	-18.427
1.025	0.07a	29.46s	992.05	0.000	0.430	0.3440	-23.427
0.911	0.08a	34.27s	992.05	0.000	0.000	0.3624	-28.240
0.906	0.08a	34.46s	992.05	0.000	-0.017	0.3623	-28.427
0.780	0.09a	39.46s	992.05	0.000	-0.485	0.3406	-33.427
0.649	0.10a	44.46s	992.05	0.000	-0.962	0.2775	-38.427
0.512	0.11a	49.46s	992.05	0.000	-1.438	0.1728	-43.427
0.372	0.12a	54.46s	992.05	0.000	-1.910	0.0267	-48.427
0.228	0.12a	59.46s	992.05	0.000	-2.370	-0.1601	-53.427
0.083	0.13a	64.46s	992.05	0.000	-2.814	-0.3864	-58.427

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

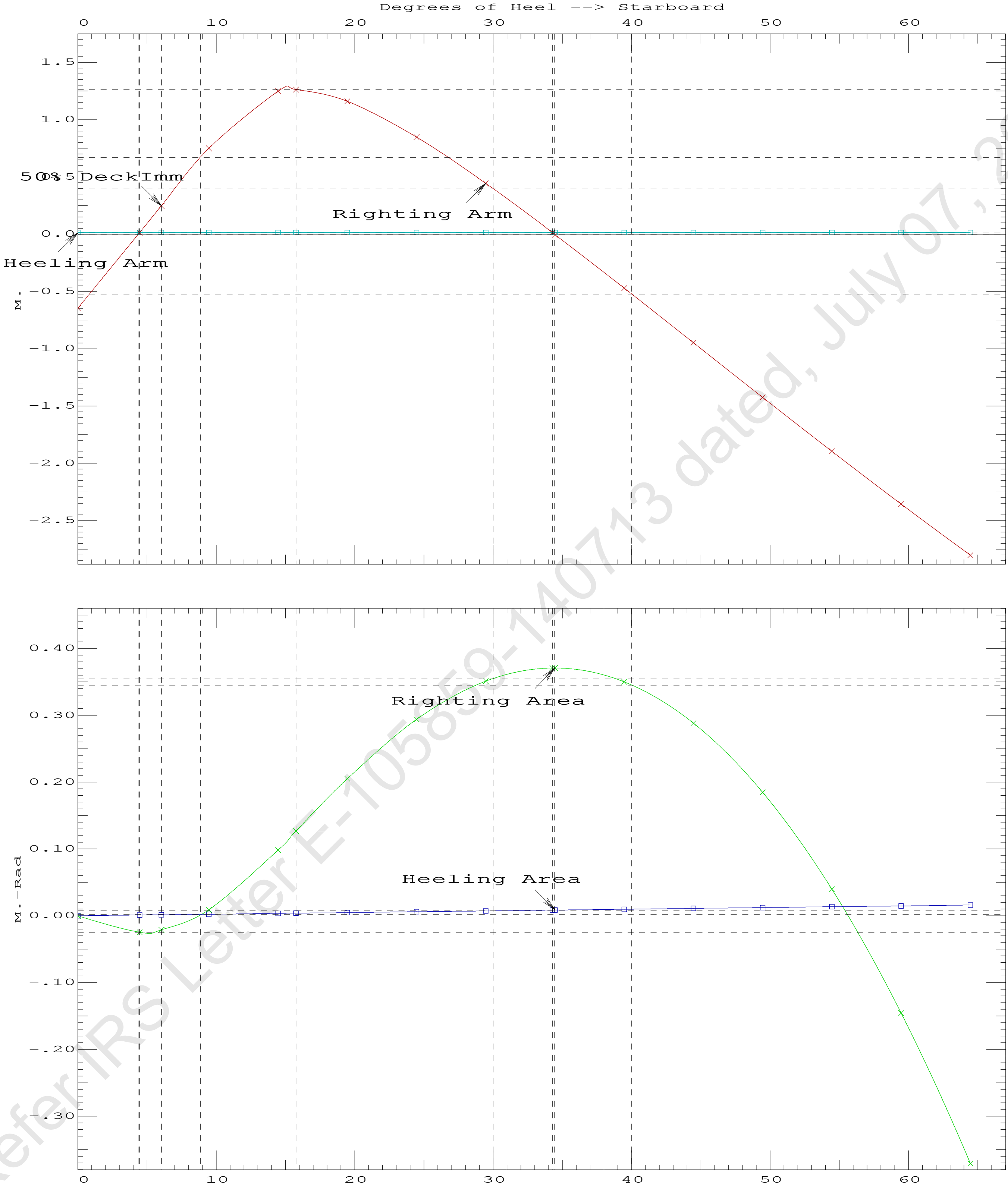
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 14.09 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1220	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	29.81	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	1.57	P

PONTOON (YARD NO. CHISTI/009)

Condition 8 : Lightship + Load at Stbd + Ballast

Crane Lifting 36.6T @ 18m Radius Stbd



11-09-19 12:07:15 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.941f TCG = 0.664s VCG = 6.006
Free Surface Adjustment: 0.237
Adjusted CG: LCG = 24.941f TCG = 0.646s VCG = 6.243

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.439	0.02a	4.37s	992.05	0.000	0.000	0.0000
1.448	0.03a	0.63p	992.05	0.000	0.740	0.0323
1.434	0.02a	5.63p	992.05	0.000	1.475	0.1289
1.394	0.03a	10.63p	992.05	0.000	2.180	0.2886
1.316	0.03a	15.38p	992.05	0.000	2.508	0.4852
1.312	0.03a	15.63p	992.05	0.000	2.507	0.4962
1.216	0.05a	20.63p	992.05	0.000	2.309	0.7102
1.111	0.06a	25.63p	992.05	0.000	1.923	0.8961
0.998	0.07a	30.63p	992.05	0.000	1.453	1.0440
0.878	0.08a	35.63p	992.05	0.000	0.938	1.1487
0.751	0.09a	40.63p	992.05	0.000	0.398	1.2071
0.656	0.10a	44.23p	992.05	0.000	0.000	1.2196
0.618	0.10a	45.63p	992.05	0.000	-0.157	1.2177
0.481	0.11a	50.63p	992.05	0.000	-0.717	1.1797
0.340	0.12a	55.63p	992.05	0.000	-1.276	1.0927

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.10

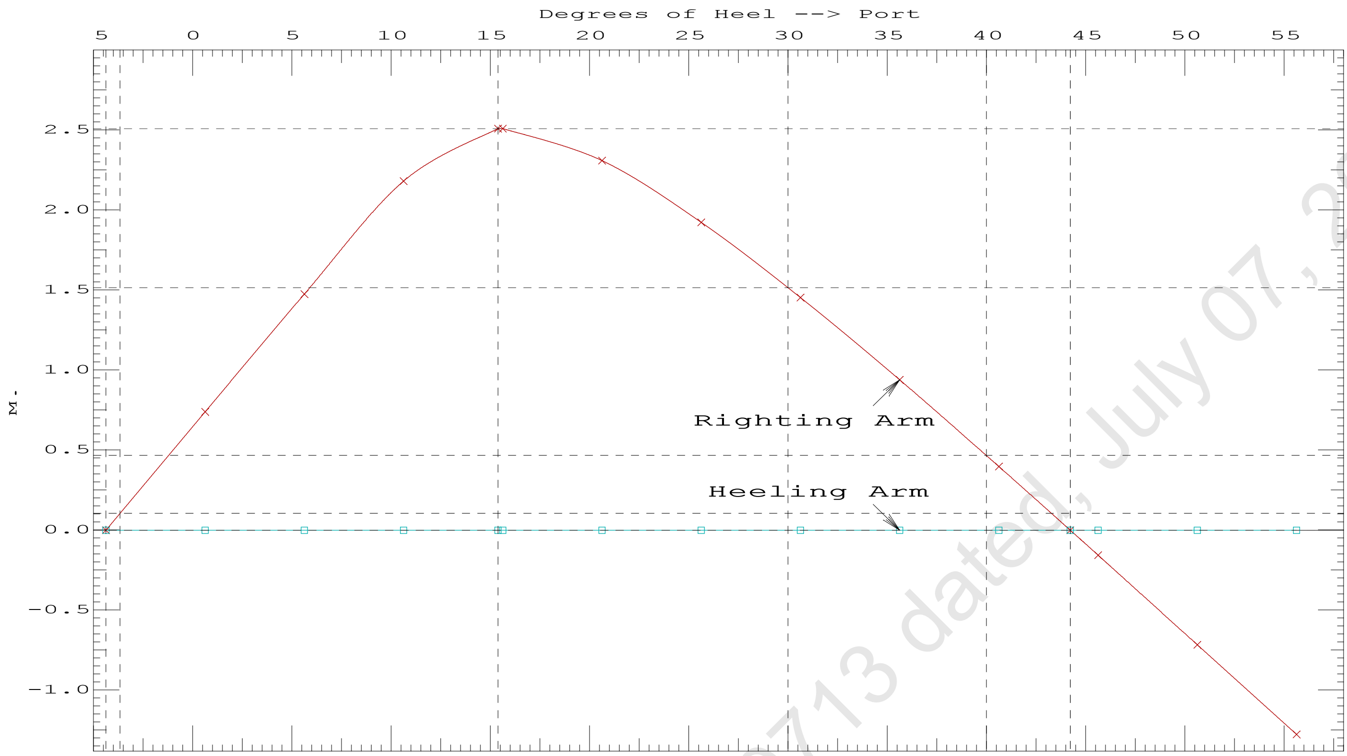
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

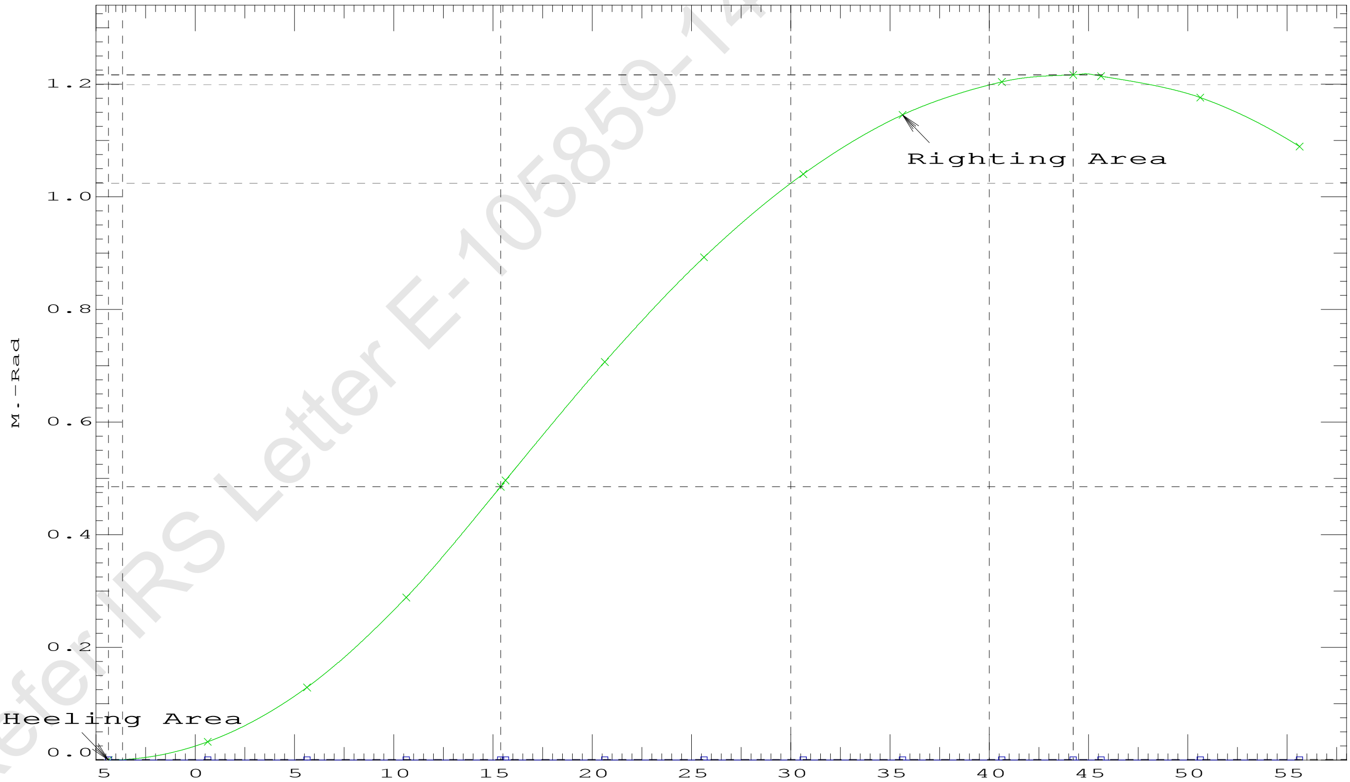
LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	4.37 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	2488214 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		14267.2 P

PONTOON (YARD NO. CHISTI/009)

Condition 8 : Lightship + Load at Stbd + Ballast

Crane Lifting 36.6T @ 18m Radius Stbd





Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.434 @ 50.00f, 1.445 @ 25.00f, 1.456 @ 0.00
Trim: Aft 0.022/50.000, Heel: Stbd 4.36 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	4.500f	18.000s	76.630	
Total Fixed--------->			654.87	16.028f	1.006s	8.306	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks--------->			337.18	42.250f	0.000	1.507	235.45
Total Weight-------->			992.05	24.941f	0.664s	5.995	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	992.05	24.939f	1.062s	0.775

Righting Arms: 0.000 0.000s

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	719.2	24.979f	0.232s	141.48	13.762

MT/cm	m.-MT/cm	GML	GMT
7.37	27.03	136.25	8.527

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 4.36

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.526 @ 50.00f, 1.399 @ 25.00f, 1.272 @ 0.00
Trim: Fwd 0.255/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Total Fixed--------->			618.27	16.711f	0.000	4.262	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks--------->			337.18	42.250f	0.000	1.507	235.45
Total Weight-------->			955.45	25.724f	0.000	3.290	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	955.39	25.737f	0.001	0.710	

Righting Arms:	0.000	-0.001s
External Arms:	0.000	-0.001
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	714.6	25.239f	0.002p	144.65	14.124
		MT/cm	m.-MT/cm		GML	GMT
		7.32		27.15	142.07	11.545

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 3136.34; t3: 30.00; t: 15.00
limmarg: 1035.71; t3: 15.00; t: 7.50
limmarg: 195.40; t3: 7.50; t: 3.75
limmarg: -26.21; t3: 3.75; t: 1.88
limmarg: 69.40; t3: 5.68; t: 0.94
limmarg: 17.94; t3: 4.74; t: 0.47
limmarg: -4.31; t3: 4.27; t: 0.23
limmarg: 7.71; t3: 4.53; t: 0.12
limmarg: 2.07; t3: 4.41; t: 0.06
limmarg: -0.69; t3: 4.35; t: 0.03
limmarg: 3.00; t3: 4.43; t: 0.01

Theta3 is 4.43 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 16.711f TCG = 0.000 VCG = 4.262

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.252	0.29f	4.36s	955.43	0.000	-0.903	0.0000
1.260	0.29f	0.00	955.43	0.000	0.000	-0.0344
1.251	0.29f	4.43p	955.45	0.000	0.916	0.0010
1.249	0.29f	5.00p	955.45	0.000	1.035	0.0108
1.208	0.30f	10.00p	955.45	0.000	2.045	0.1453
1.078	0.38f	15.00p	955.45	0.000	2.726	0.3558
0.926	0.49f	19.33p	955.46	0.000	2.842	0.5689
0.902	0.51f	20.00p	955.45	0.000	2.839	0.6022
0.717	0.64f	25.00p	955.45	0.000	2.731	0.8477
0.526	0.76f	30.00p	955.45	0.000	2.522	1.0776

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 618.27 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM	STABILITY CRITERION		Min/Max	Attained
	(1) Res. Area Ratio from abs -4.4 deg to abs 4.4	>	1.000	1.030 P

11-09-19 12:06:57 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 16.711f TCG = 0.000 VCG = 4.262

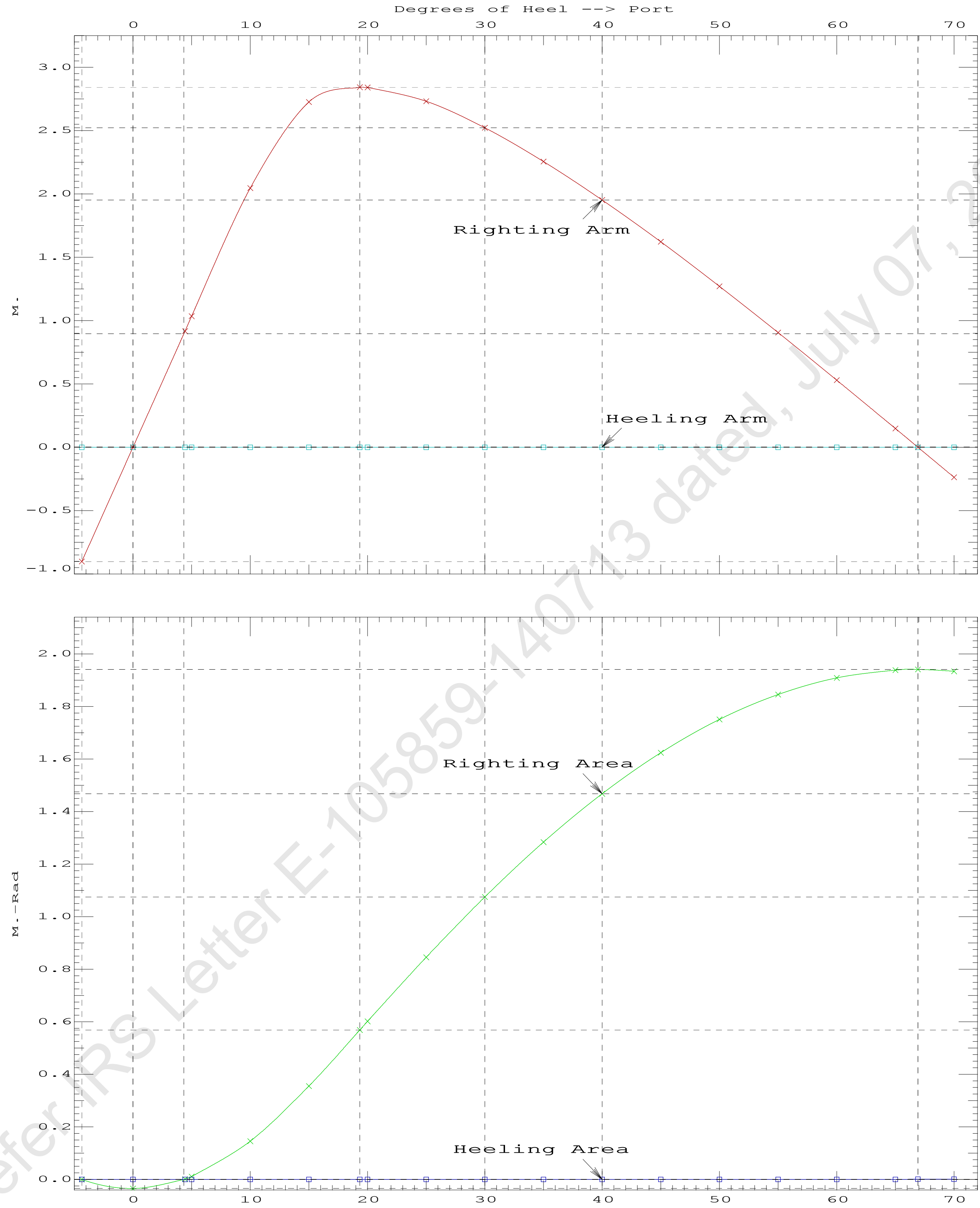
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.252	0.29f	4.36s	955.43	0.000	-0.903	0.0000
1.260	0.29f	0.00	955.43	0.000	0.000	-0.0344
1.251	0.29f	4.43p	955.45	0.000	0.916	0.0010
1.249	0.29f	5.00p	955.45	0.000	1.035	0.0108
1.208	0.30f	10.00p	955.45	0.000	2.045	0.1453
1.078	0.38f	15.00p	955.45	0.000	2.726	0.3558
0.926	0.49f	19.33p	955.46	0.000	2.842	0.5689
0.902	0.51f	20.00p	955.45	0.000	2.839	0.6022
0.717	0.64f	25.00p	955.45	0.000	2.731	0.8477
0.526	0.76f	30.00p	955.45	0.000	2.522	1.0776
0.328	0.89f	35.00p	955.45	0.000	2.256	1.2865
0.127	1.01f	40.00p	955.45	0.000	1.952	1.4704
-0.077	1.13f	45.00p	955.45	0.000	1.621	1.6265
-0.281	1.24f	50.00p	955.45	0.000	1.270	1.7527
-0.483	1.34f	55.00p	955.45	0.000	0.905	1.8477
-0.682	1.43f	60.00p	955.45	0.000	0.529	1.9104
-0.874	1.51f	65.00p	955.45	0.000	0.147	1.9400
-0.946	1.53f	66.92p	955.45	0.000	0.000	1.9424
-1.059	1.57f	70.00p	955.45	0.000	-0.237	1.9361

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 618.27 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

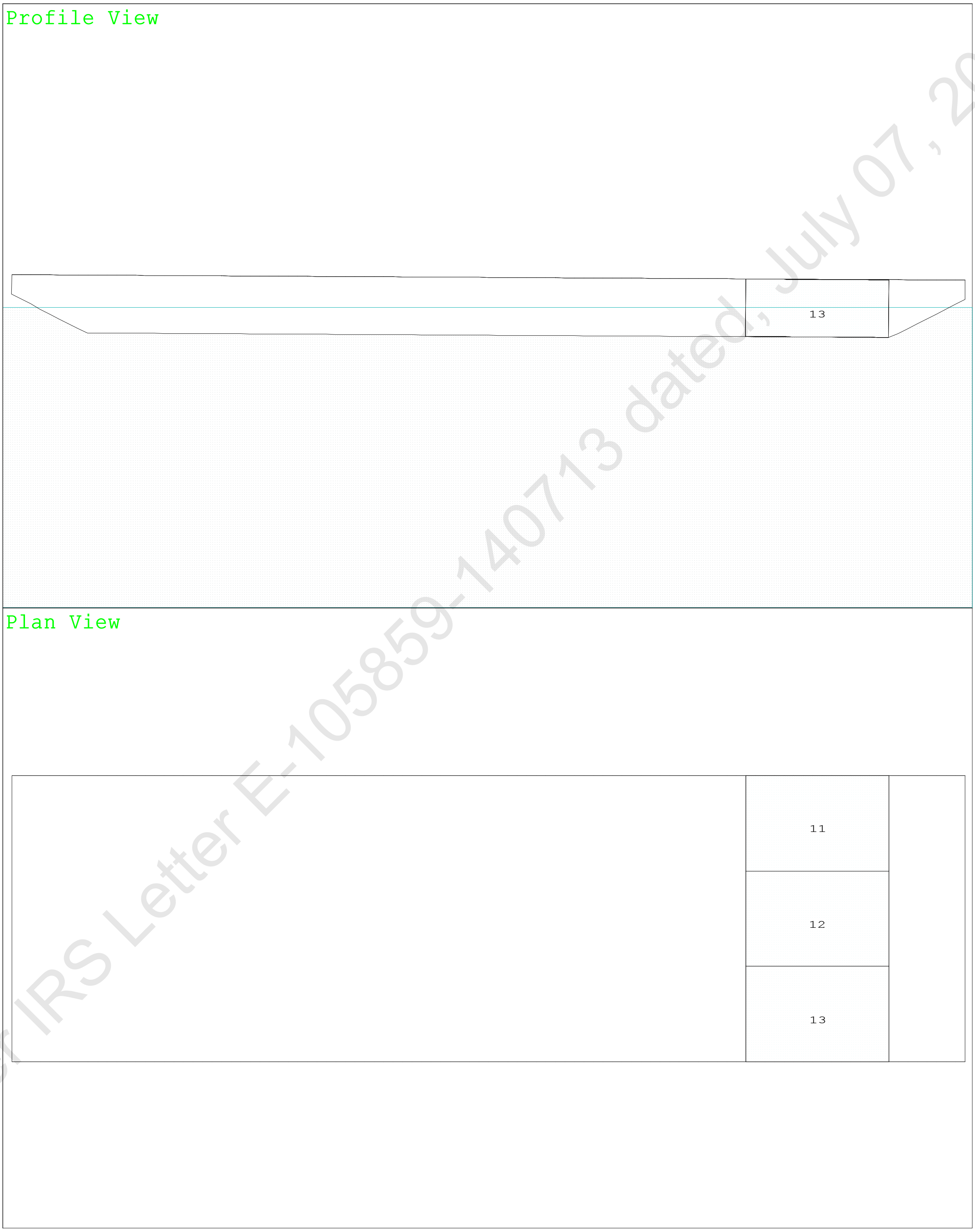
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -4.4 deg to RAzero	>	1.000		57.528 P
(2) Angle from abs 4.4 deg to RAzero	>	20.00	deg	62.49 P


Degrees of Heel --> Port
Righting Arm
Heeling Arm
M.
Righting Area
Heeling Area
M.-Rad

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

Condition Graphic - Draft: 1.592 @ 50.000f, 1.305 @ 0.000 Heel: zero



Tanks 11 WBTK5.P 12 WBTK5.C 13 WBTK5.S

11-09-19 12:07:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.592 @ 50.00f, 1.449 @ 25.00f, 1.305 @ 0.00
Trim: Fwd 0.287/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	27.250f	0.000	76.930	
Total Fixed--------->			654.87	17.300f	0.000	8.323	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks--------->			337.18	42.250f	0.000	1.507	235.45
Total Weight-------->			992.05	25.780f	0.000	6.006	
			Displ(MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	992.05	25.810f	0.000	0.737	

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	717.7	25.268f	0.000	141.11	13.665
		MT/cm		**m.-MT/cm**	**GML**	**GMT**
		7.36		26.95	135.84	8.395

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

329.0 Cu.M. (22%) SALT WATER

250.00 MT Crane 36.60 MT Crane Load

DRAFT AT FP (M) = 1.592
DRAFT AT MS (M) = 1.449
DRAFT AT AP (M) = 1.305
DRAFT AT FWD MARK (M) = 1.569
DRAFT AT AFT MARK (M) = 1.328

HYDROSTATIC PROPERTIES
Trim: Fwd 0.287/50.000, No Heel, VCG = 6.006

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.450	992.05	25.810f	0.737	7.36	25.268f	26.95	135.84	8.395

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 235.4 m.-MT

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.780f TCG = 0.000 VCG = 6.006
Free Surface Adjustment: 0.237
Adjusted CG: LCG = 25.779f TCG = 0.000 VCG = 6.244

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.293	0.33f	0.00	992.02	0.000	-0.014	0.0000	5.484
1.293	0.33f	0.09s	992.05	0.000	0.000	-0.0000	5.390
1.282	0.33f	5.09s	992.05	0.000	0.738	0.0322	0.390
1.280	0.33f	5.48s	992.05	0.000	0.796	0.0374	0.000
1.241	0.34f	10.09s	992.05	0.000	1.458	0.1282	-4.610
1.115	0.44f	15.09s	992.05	0.000	1.858	0.2752	-9.610
1.103	0.45f	15.47s	992.05	0.000	1.860	0.2874	-9.986
0.949	0.59f	20.09s	992.05	0.000	1.709	0.4345	-14.610
0.773	0.75f	25.09s	992.05	0.000	1.358	0.5698	-19.610
0.588	0.90f	30.09s	992.05	0.000	0.921	0.6699	-24.610
0.397	1.05f	35.09s	992.05	0.000	0.440	0.7296	-29.610
0.225	1.18f	39.48s	992.05	0.000	0.000	0.7466	-33.992
0.201	1.20f	40.09s	992.05	0.000	-0.063	0.7462	-34.610
0.002	1.34f	45.09s	992.05	0.000	-0.578	0.7184	-39.610
-0.196	1.47f	50.09s	992.05	0.000	-1.095	0.6454	-44.610
-0.392	1.59f	55.09s	992.05	0.000	-1.609	0.5274	-49.610
-0.583	1.69f	60.09s	992.05	0.000	-2.113	0.3650	-54.610

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 14.31 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.2874	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	39.38	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	5.39	P

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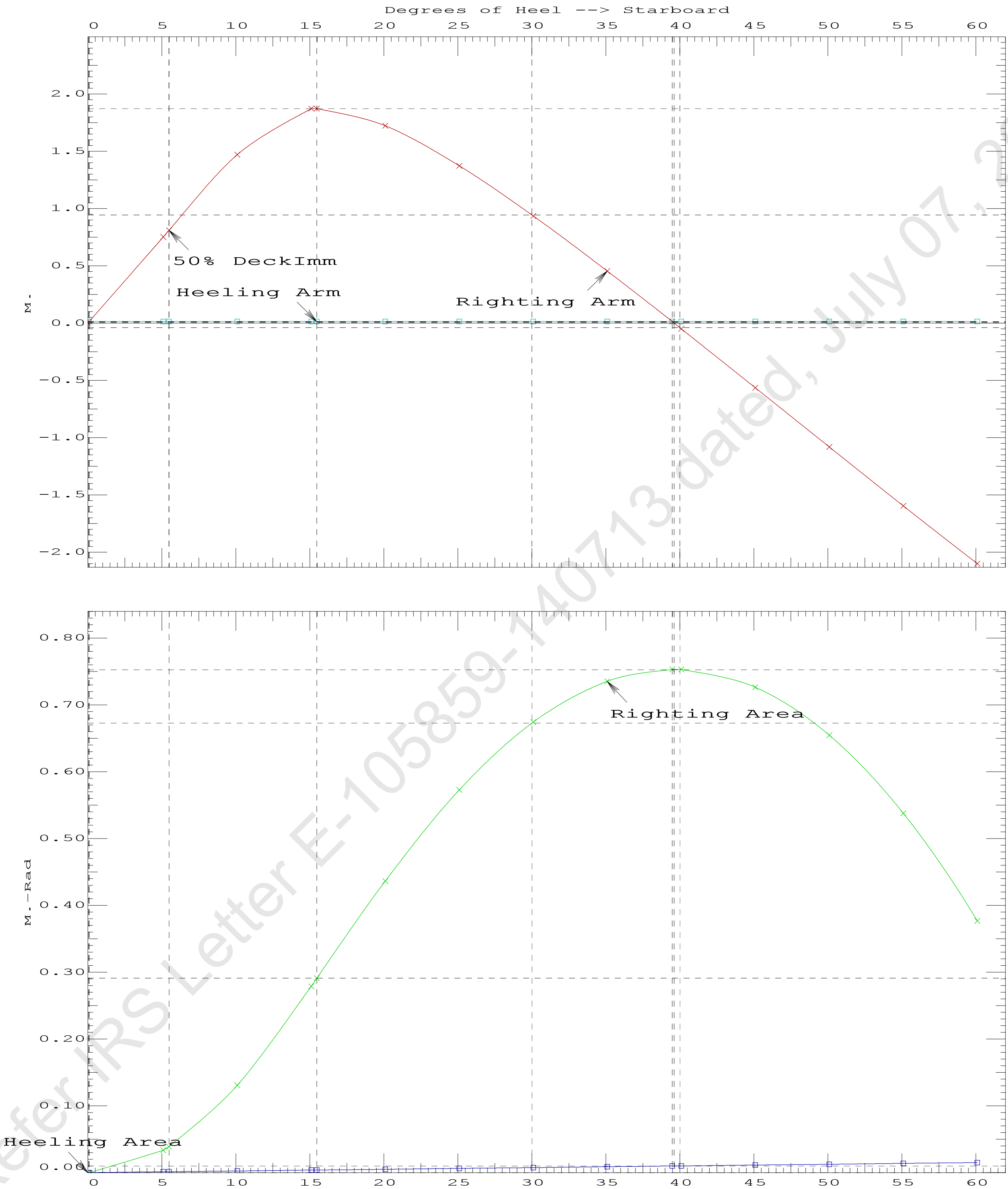
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 9 : Lightship + Load at Fwd + Ballast

Crane Lifting 36.60T @ 18m Radius Fwd



11-09-19 12:07:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.780f TCG = 0.000 VCG = 6.006
Free Surface Adjustment: 0.237
Adjusted CG: LCG = 25.779f TCG = 0.000 VCG = 6.244

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.293	0.33f	0.01p	991.93	0.000	0.000	0.0000
1.282	0.33f	5.01p	992.01	0.000	0.738	0.0322
1.242	0.34f	10.01p	992.05	0.000	1.459	0.1282
1.118	0.44f	15.01p	992.05	0.000	1.870	0.2757
1.102	0.45f	15.49p	992.05	0.000	1.873	0.2914
0.952	0.59f	20.01p	992.05	0.000	1.727	0.4362
0.776	0.74f	25.01p	992.05	0.000	1.378	0.5741
0.591	0.90f	30.01p	992.05	0.000	0.942	0.6760
0.400	1.05f	35.01p	992.05	0.000	0.462	0.7376
0.220	1.19f	39.60p	992.05	0.000	0.000	0.7563
0.204	1.20f	40.01p	992.05	0.000	-0.041	0.7561
0.006	1.34f	45.01p	992.05	0.000	-0.555	0.7302
-0.192	1.47f	50.01p	992.05	0.000	-1.073	0.6592
-0.388	1.59f	55.01p	992.05	0.000	-1.586	0.5432
-0.579	1.69f	60.01p	992.05	0.000	-2.091	0.3826

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

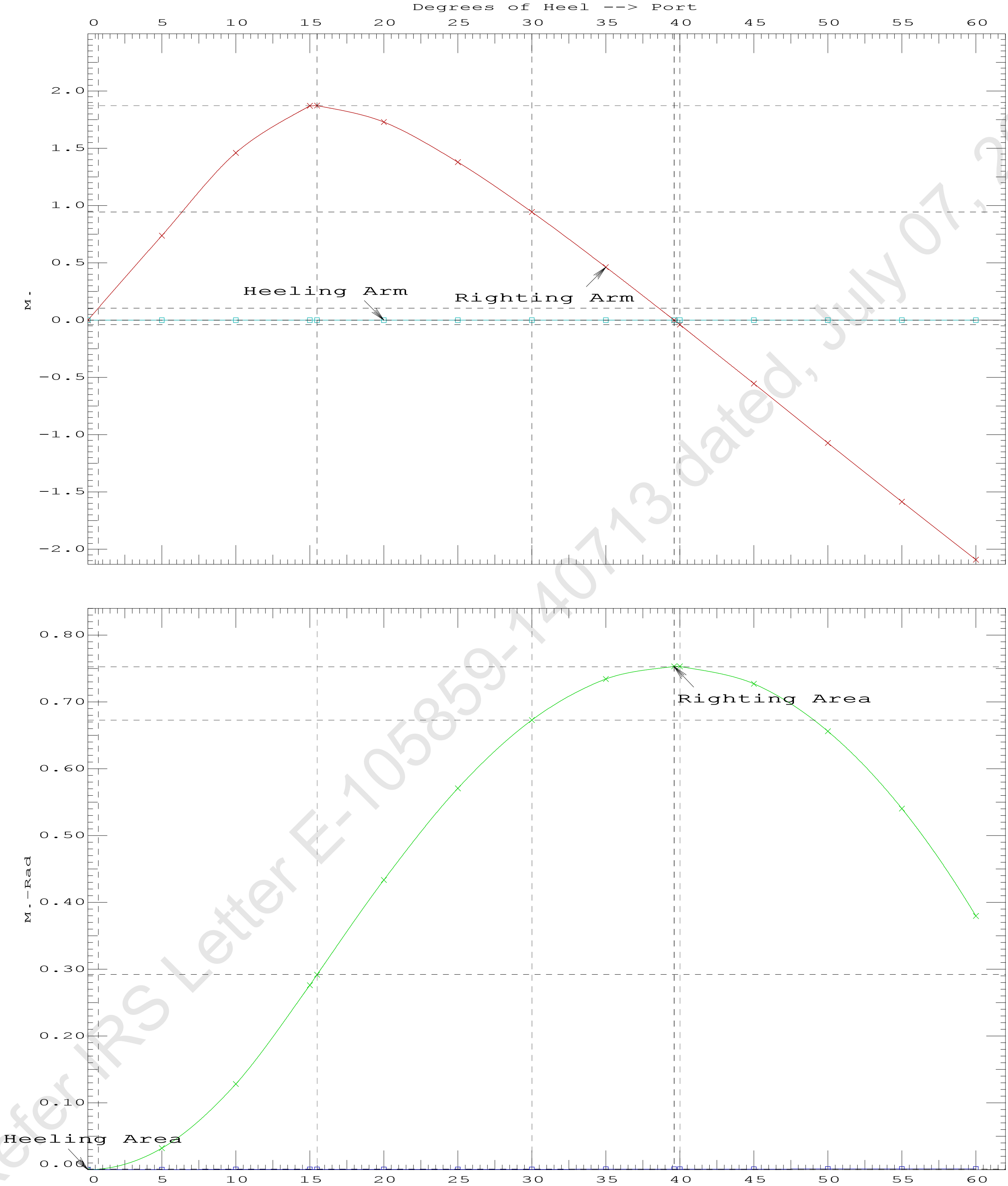
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	185754 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1086.57 P

PONTOON (YARD NO. CHISTI/009)

Condition 9 : Lightship + Load at Fwd + Ballast

Crane Lifting 36.60T @ 18m Radius Fwd



11-09-19 12:07:28 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

```
                  Crane in operation with load in place
               WEIGHT and DISPLACEMENT and WATERPLANE STATUS
          USK draft: 2.311 @ 50.00f, 2.185 @ 25.00f, 2.060 @ 0.00
               Trim: Fwd 0.251/50.000,  Heel: Port 0.01 deg.
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                               368.27  25.000f  0.000   3.000
Crane                                    250.00   4.500f  0.000   6.120
Crane Load                                36.60  27.250f  0.000  76.930
Deck Cargo                               556.61  25.000f  0.000   3.500
  Total Fixed--------->                1,211.48  20.838f  0.000   6.107
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
WBTK5.P        1.000    1.025            112.08  42.251f  4.980p  1.511     78.49*
WBTK5.C        1.000    1.025            113.02  42.250f  0.000   1.500     78.48*
WBTK5.S        1.000    1.025            112.08  42.251f  4.980s  1.511     78.49*
  Total Tanks--------->                  337.18  42.250f  0.000   1.507    235.45
  Total Weight-------->                1,548.66  25.500f  0.000   5.106
                                       Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025          1,548.66  25.520f  0.001p  1.122
------------------------------------------------------------------------------
               Righting Arms:                      0.000  -0.001s
               External Arms:                      0.000  -0.001
      Residual Righting Arms:                      0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          750.5  25.010f  0.000    103.35   9.175
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                          7.69            30.78     99.37   5.191
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                           Angle Theta1 = -0.01
```

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.248 @ 50.00f, 2.138 @ 25.00f, 2.027 @ 0.00
Trim: Fwd 0.221/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			368.27	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Deck Cargo			556.61	25.000f	0.000	3.500	
Total Fixed--------->			1,174.88	20.638f	0.000	3.901	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks--------->			337.18	42.250f	0.000	1.507	235.45
Total Weight-------->			1,512.06	25.457f	0.000	3.367	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	1,512.06	25.468f	0.001	1.097	

Righting Arms:		0.000	0.000s		
Part---SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane----> 1.025	750.4	25.011f	0.000	105.82	9.394
	MT/cm		m.-MT/cm	GML	GMT
	7.69		31.31	103.55	7.123

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 499596896.00; t3: 30.0; t: 15.00
limmarg: 205885616.00; t3: 15.0; t: 7.50
limmarg: 60331776.00; t3: 7.5; t: 3.75
limmarg: 15171372.00; t3: 3.8; t: 1.88
limmarg: 3983760.50; t3: 1.9; t: 0.94
limmarg: 995345.69; t3: 1.0; t: 0.47
limmarg: 302904.19; t3: 0.5; t: 0.23
limmarg: 78089.45; t3: 0.3; t: 0.12
limmarg: 34330.20; t3: 0.2; t: 0.06
limmarg: 20556.21; t3: 0.1; t: 0.03
limmarg: 4872.61; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.638f TCG = 0.000 VCG = 3.901

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.016	0.25f	0.01p	1,512.1	0.000	-0.001	0.0000
2.015	0.25f	0.00s	1,512.1	0.000	0.000	-0.0000
2.016	0.25f	0.07s	1,512.1	0.000	0.009	0.0000
2.003	0.26f	5.00s	1,512.1	0.000	0.634	0.0277
1.985	0.33f	10.00s	1,512.2	0.000	1.119	0.1053
2.066	0.42f	15.00s	1,512.1	0.000	1.251	0.2112
2.098	0.45f	16.34s	1,512.1	0.000	1.255	0.2406

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GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

2.195	0.52f	20.00s	1,512.1	0.000	1.226	0.3202
2.316	0.64f	25.00s	1,512.1	0.000	1.109	0.4229
2.416	0.77f	30.00s	1,512.1	0.000	0.935	0.5125

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1174.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

```
LIM--------------------STABILITY CRITERION------------Min/Max--------Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1  >    1.000        49.726 P
-----------------------------------------------------------------------------
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11-09-19 12:07:28 Cybermarine Technologies PTE. Ltd.
GHS 16.68C **PONTOON (YARD NO. CHISTI/009)**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.638f TCG = 0.000 VCG = 3.901

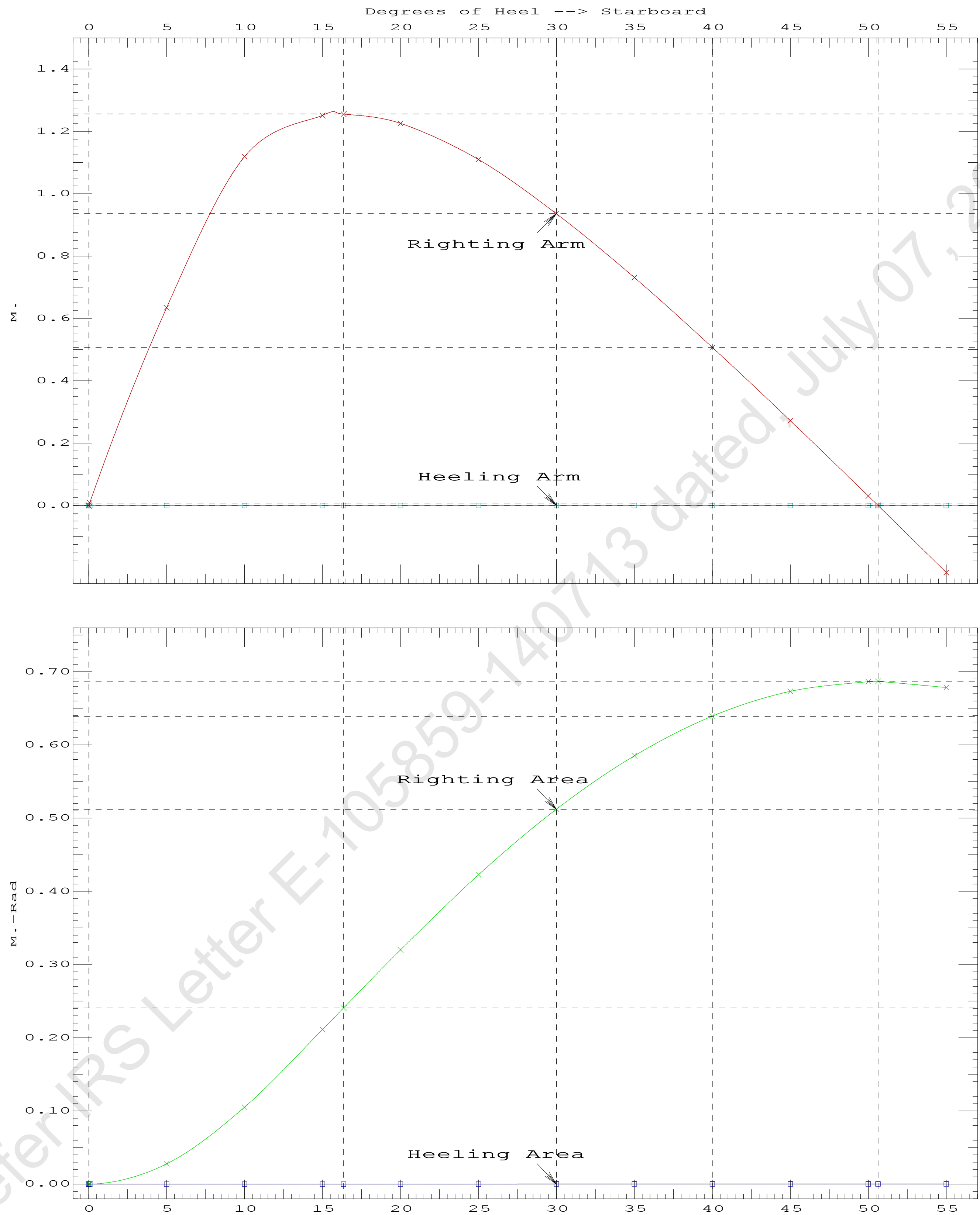
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
2.016	0.25f	0.01p	1,512.1	0.000	-0.001	0.0000
2.015	0.25f	0.00s	1,512.1	0.000	0.000	-0.0000
2.016	0.25f	0.07s	1,512.1	0.000	0.009	0.0000
2.003	0.26f	5.00s	1,512.1	0.000	0.634	0.0277
1.985	0.33f	10.00s	1,512.2	0.000	1.119	0.1053
2.066	0.42f	15.00s	1,512.1	0.000	1.251	0.2112
2.098	0.45f	16.34s	1,512.1	0.000	1.255	0.2406
2.195	0.52f	20.00s	1,512.1	0.000	1.226	0.3202
2.316	0.64f	25.00s	1,512.1	0.000	1.109	0.4229
2.416	0.77f	30.00s	1,512.1	0.000	0.935	0.5125
2.497	0.90f	35.00s	1,512.1	0.000	0.730	0.5854
2.557	1.02f	40.00s	1,512.1	0.000	0.507	0.6395
2.597	1.14f	45.00s	1,512.1	0.000	0.272	0.6736
2.617	1.25f	50.00s	1,512.1	0.000	0.030	0.6868
2.618	1.26f	50.62s	1,512.1	0.000	0.000	0.6870
2.617	1.35f	55.00s	1,512.1	0.000	-0.215	0.6787

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1174.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		6696577 P
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	50.55 P


Degrees of Heel --> Starboard
0
5
10
15
20
25
30
35
40
45
50
55
M.
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Righting Arm
Heeling Arm
M.-Rad
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
Righting Area
Heeling Area

LIGHTSHIP PARTICULARS

Lightship Weight	=	368.27	T
VCG	=	3.000	M above Baseline
LCG	=	25.000	M fwd of AP
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved Draft Survey Report of CHISTI-Y/012.)

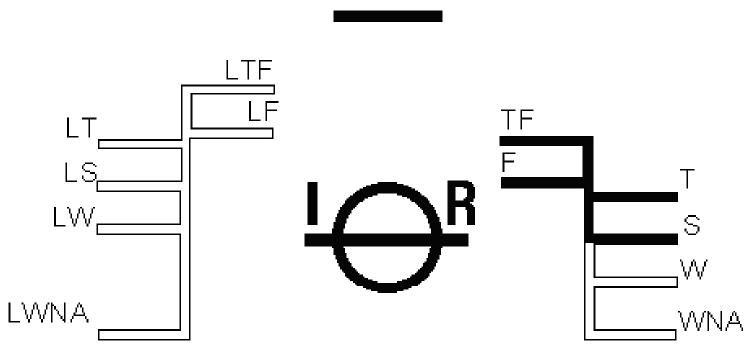
MEMORANDUM OF FREEBOARDS
17 MAR 2020

	Type B freeboard		
Chisti Marine Engg.Works	Freeboard Length	:	48.00 m
Yard No. CHISTI/012	Summer WL Extreme Draught - load line amidships	:	2.212 m
K.B.25			

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of * ~~Wood~~ / *steel Deck *at side / *~~continued to side~~. (*Delete words not applicable).

Freeboards from Deck Line			**Seasonal, Fresh Water, or Timber allowances**	
Tropical Fresh Water	713	**mm.**	97	**mm. above summer**
Fresh Water	759	**mm.**	51	**mm. above summer**
Tropical	764	**mm.**	46	**mm. above summer**
Summer	810	**mm.**		**Upper edge of line through centre**
Winter	Not	**mm.**	Assigned	**mm. below summer**
Winter North Atlantic	Not	**mm.**	Assigned	**mm. below summer**
Timber Tropical fresh water	—	**mm.**	—	**mm. above timber summer**
Timber fresh water	—	**mm.**	—	**mm. above timber summer**
Timber Tropical	—	**m m.**	—	**mm. above timber summer**
Timber Summer	—	**mm.**	—	**mm. above centre of ring**
Timber Winter	—	**mm.**	—	**mm. below timber summer**
Timber Winter North Atlantic	—	**mm.**	—	**mm. below timber summer**

Valid only for unmanned operation while under tow along Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.



At Fr.52

Form No. 4111 (07-83)

DRAFT SURVEY REPORT
OF
KB-25
(YARD NO. CHISTI/012)

Rev.	Description	Date	Prepared	Review

Issued for : Approval

Client:	Project:
K.B. SHIPPING & CO. (JAMNAGAR)	**PONTOON (YARD NO. CHISTI/012)**
Client: **CHISTI MARINE ENGINEERING WORKS**	

Doc #				Document Title:	Prepared	JD
CM20-V-105-0202				**DRAFT SURVEY REPORT**	Checked	ND
Scale	Size	Sheet	Date		Review 1	
NTS	A4	1 of 9	18-03-20		Review 2	

Cybermarine
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Plot No. X-5/3, Mahape,
Maharashtra- 400710, India.

22503, Katy Freeway #7,
Katy, Texas,
USA 77450.

GENERAL PARTICULARS

Name of the Vessel	KB – 25
Client	K.B.SHIPPING & CO., JAMNAGAR
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No	CHISTI/012
Type of Vessel	PONTOON
Class Notation	卐SU-PONTOON, "INDIAN COASTAL SERVICE" DESCRIPTIVE NOTE "FOR CRANE OPERATION" "DECK STRENGTHENED FOR {10T/M^2}

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	15.000 M
Depth (Mld.)	3.000 M

References

AP at Fr.0

FP at Fr.100

Midship at Fr.50

LCB, LCF and LCG are measured from AP, Fwd of AP being Positive & Aft of AP being Negative.

VCB and VCG are measured from Baseline, with above Base Line being Positive.

TCG is measured from Centreline, with Port being Negative and Stbd Positive.

Frame spacing

Through out 500 mm

DRAFT SURVEY REPORT : KB-25 (Yard No. CHISTI/012)

Location of Test	Sikka,Jamnagar
Date	17-Mar-20
Weather	Calm
Wind	Calm
Specific Gravity	1.025
Restraining Line	Loose
Witnessed by	Mr.Chiranjeev Malviya (IRS Surveyor)

Drafts :

(w.r.t keel line)

Location		Draft Port (m)	Draft Stbd (m)	Mean Draft (m)
Fwd	(Fr.92)	0.570	0.570	0.570
Mid	(Fr.50)	0.570	0.570	0.570
Aft	(Fr.8)	0.570	0.570	0.570

Hydrostatics

Correction for Drafts

To correct for observed drafts at marks and obtain Drafts at perps.

LBP	-	50.000 m

Observed Drafts (wrt keel line)

Fwd	-	0.570 m
Aft	-	0.570 m

Correction to Drafts (wrt Keel line)

Location	At Fwd (Fr.92)	At Aft (Fr.8)	
Draft at Marks	0.570	0.570	m
Trim at Marks		0.000	m
Distance of Marks from PP	4.000	4.000	m
Distance between Marks		42.000	m
Draft correction factor		0.000	
Correction at PP	0.000	0.000	
Corrected Draft at PP	0.570	0.570	
Trim		0.000	m
Mean Draft		0.570	m
From Hydrostatics against above Mean Draft and Trim			
Displacement at density(T/M^3)=1.025 T/m^3		368.27	T
LCB		25.000	m
KMt		35.027	m
LCG		25.000	m
TCG		0.000	m

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	68.53	25.000f	0.044	6.21	25.000f	18.72	1365.62	152.943
0.212	131.38	25.000f	0.095	6.36	25.000f	19.61	746.26	84.110
0.312	195.72	25.000f	0.146	6.51	25.000f	20.41	521.46	59.400
0.412	261.52	25.000f	0.197	6.65	25.000f	21.24	406.00	46.750
0.512	328.77	25.000f	0.249	6.79	25.000f	22.05	335.40	38.975
0.570	368.27	25.000f	0.279	6.83	25.000f	22.40	304.09	35.027
0.612	396.99	25.000f	0.301	6.85	25.000f	22.65	285.27	32.654
0.712	465.83	25.000f	0.352	6.91	25.000f	23.27	249.73	28.175
0.812	535.26	25.000f	0.404	6.97	25.000f	23.88	223.09	24.822
0.912	605.29	25.000f	0.456	7.03	25.000f	24.52	202.53	22.237
1.012	675.92	25.000f	0.507	7.09	25.000f	25.16	186.15	20.181
1.112	747.15	25.000f	0.559	7.15	25.000f	25.82	172.79	18.506
1.212	818.99	25.000f	0.611	7.21	25.000f	26.50	161.77	17.125
1.312	891.43	25.000f	0.663	7.27	25.000f	27.18	152.43	15.959
1.412	964.46	25.000f	0.715	7.33	25.000f	27.87	144.50	14.970
1.512	1,038.11	25.000f	0.767	7.39	25.000f	28.59	137.68	14.122
1.612	1,112.35	25.000f	0.819	7.45	25.000f	29.30	131.72	13.382
1.712	1,187.19	25.000f	0.872	7.51	25.000f	30.04	126.53	12.740
1.812	1,262.64	25.000f	0.924	7.57	25.000f	30.79	121.92	12.171
1.912	1,338.68	25.000f	0.977	7.63	25.000f	31.55	117.84	11.669
2.012	1,415.33	25.000f	1.029	7.69	25.000f	32.24	113.91	11.217
2.112	1,492.22	25.000f	1.082	7.69	25.000f	32.31	108.27	10.755
2.212	1,569.15	25.000f	1.134	7.69	25.000f	32.38	103.19	10.344

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

DRAFT SURVEY REPORT : KB-25 (Yard No. CHISTI/012)

Weights to go off Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.00	0.000	0.00	0.000	0.00	0.000	0.00
	TOTAL	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Weights to go on Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.000	0.000	0.00	0.000	0.00	0.000	0.00
	TOTAL	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Lightship Particulars

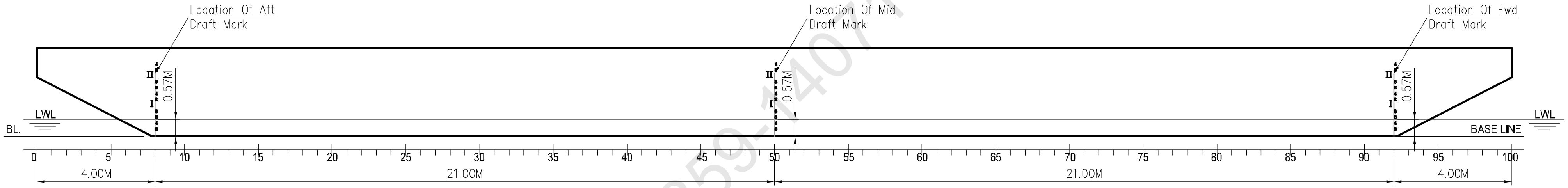
Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
Particulars as Draft Survey	368.27	25.000	9206.750	3.000	1104.810	0.000	0.000
Weights to go Off board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Weights to go on board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Lightship Particulars	368.27	25.000	9206.75	3.000	1104.81	0.000	0.00

Lightship Particulars

Displacement	**368.27**	**T**
LCG	**25.000**	**m From AP**
TCG	**0.000**	**m Off-Centreline**
VCG	**3.000**	**m above BL**

NOTE
1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

DRAFT READING



PROFILE

LIGHTSHIP SURVEY PROCEDURE AND REPORTING FORMAT
Part II

REPORTING FORM FOR LIGHTSHIP SURVEY

Attending Surveyor should witness Light-Ship Survey only if the Part-I submitted by Owner/Builder/Operator/Consultant is found satisfactory

General Particulars

Name of Vessel		K-B-25
Builder/ Owner		Chisti Marine Eng. Works / K-B. Shipping
Yard Number/ IR Number		Chisti-012
Type of Vessel	Passenger Non Passenger	☐ ☐ Pontoon.
Date(dd/mm/yyyy);Local Time of inspection/test		17/03/2020 19:00-20:00 hrs.
Wind / Weather condition		1 knots, Calm
Location of Test		Sikka Jamnagar.
Depth of water where vessel is moored, m		4.5 meter
Method of reading Draft(By Boat-Y/N)		Yes.

Location of Draft Marks	Frame Number	Draft Reading (Port side),m	Draft Reading (STBD Side),m
Aft	8	0.570 m	0.570 m
Midship	50	0.570 m	0.570 m
Forward	92	0.570 m.	0.570 m.

Note : *Recommended to attach Photographs of draft; as record*

Density of Water

Results of 3 samples from Forward, Aft and Mid of the vessel		
Sample 1: (ford)	1.025	t/m3
Sample 2: (mid)	1.025	t/m3
Sample 3: (aft)	1.025	t/m3

Verification of weights to go on / go off /re-location	***SEE PART 1***
Note: *See Part 1 (the sheet for recording the above data) , verify and endorse.*	

Special Comments on Light Ship Survey, if any.
Draft Survey carried out satisfactarily.

Attending surveyor to forward both forms Part 1 and Part 2 together with completed attachments 1 and 2 to HO.

Name & designation of Attending Surveyor: --

(CHIRANJEEV MALVIYA)

17/03/2020

INDIAN REGISTER OF SHIPPING

03-04-19 10:01:31   Cybermarine Technologies PTE. Ltd.
GHS 16.50

LATERAL PLANE STATUS
USK draft: 0.500 @ 50.00f, 0.500 @ 25.00f, 0.500 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	21.6	25.000f	-0.248	121.7	25.000f	1.289
CRANE.C				35.2	7.500f	4.712
DECKCARGO.C				40.0	25.000f	3.512
Total Lateral Plane->	21.6	25.000f	-0.248	196.9	21.871f	2.352

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.600 @ 50.00f, 0.600 @ 25.00f, 0.600 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	26.0	25.000f	-0.297	117.2	25.000f	1.236
CRANE.C				35.2	7.500f	4.612
DECKCARGO.C				40.0	25.000f	3.412
Total Lateral Plane->	26.0	25.000f	-0.297	192.4	21.799f	2.306

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.700 @ 50.00f, 0.700 @ 25.00f, 0.700 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	30.5	25.000f	-0.346	112.8	25.000f	1.183
CRANE.C				35.2	7.500f	4.512
DECKCARGO.C				40.0	25.000f	3.312
Total Lateral Plane->	30.5	25.000f	-0.346	188.0	21.723f	2.259

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.800 @ 50.00f, 0.800 @ 25.00f, 0.800 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	35.0	25.000f	-0.395	108.3	25.000f	1.130
CRANE.C				35.2	7.500f	4.412
DECKCARGO.C				40.0	25.000f	3.212
Total Lateral Plane->	35.0	25.000f	-0.395	183.5	21.642f	2.214

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.900 @ 50.00f, 0.900 @ 25.00f, 0.900 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	39.6	25.000f	-0.443	103.7	25.000f	1.077
CRANE.C				35.2	7.500f	4.312
DECKCARGO.C				40.0	25.000f	3.112
Total Lateral Plane->	39.6	25.000f	-0.443	178.9	21.557f	2.169

Distances in METERS.

03-04-19 10:01:31 Cybermarine Technologies PTE. Ltd.
GHS 16.50

LATERAL PLANE STATUS
USK draft: 1.000 @ 50.00f, 1.000 @ 25.00f, 1.000 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	44.1	25.000f	-0.492	99.1	25.000f	1.025
CRANE.C				35.2	7.500f	4.212
DECKCARGO.C				40.0	25.000f	3.012
Total Lateral Plane->	44.1	25.000f	-0.492	174.3	21.467f	2.124

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.100 @ 50.00f, 1.100 @ 25.00f, 1.100 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	48.8	25.000f	-0.540	94.5	25.000f	0.972
CRANE.C				35.2	7.500f	4.112
DECKCARGO.C				40.0	25.000f	2.912
Total Lateral Plane->	48.8	25.000f	-0.540	169.7	21.370f	2.081

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.200 @ 50.00f, 1.200 @ 25.00f, 1.200 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	53.4	25.000f	-0.589	89.8	25.000f	0.920
CRANE.C				35.2	7.500f	4.012
DECKCARGO.C				40.0	25.000f	2.812
Total Lateral Plane->	53.4	25.000f	-0.589	165.0	21.267f	2.038

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.300 @ 50.00f, 1.300 @ 25.00f, 1.300 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	58.2	25.000f	-0.637	85.1	25.000f	0.868
CRANE.C				35.2	7.500f	3.912
DECKCARGO.C				40.0	25.000f	2.712
Total Lateral Plane->	58.2	25.000f	-0.637	160.3	21.158f	1.997

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.400 @ 50.00f, 1.400 @ 25.00f, 1.400 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	62.9	25.000f	-0.685	80.4	25.000f	0.817
CRANE.C				35.2	7.500f	3.812
DECKCARGO.C				40.0	25.000f	2.612
Total Lateral Plane->	62.9	25.000f	-0.685	155.6	21.040f	1.956

Distances in METERS.

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03-04-19 10:01:31    Cybermarine Technologies PTE. Ltd.
GHS 16.50

                            LATERAL PLANE STATUS
           USK draft: 1.500 @ 50.00f, 1.500 @ 25.00f, 1.500 @ 0.00
                          Trim: zero,  Heel: zero
Part---------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                          67.7  25.000f  -0.733      75.6  25.000f   0.765
CRANE.C                                                  35.2   7.500f   3.712
DECKCARGO.C                                              40.0  25.000f   2.512
  Total Lateral Plane->       67.7  25.000f  -0.733     150.8  20.915f   1.916
Distances in METERS.----------------------------------------------------------

                            LATERAL PLANE STATUS
           USK draft: 1.600 @ 50.00f, 1.600 @ 25.00f, 1.600 @ 0.00
                          Trim: zero,  Heel: zero
Part---------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                          72.5  25.000f  -0.781      70.8  25.000f   0.714
CRANE.C                                                  35.2   7.500f   3.612
DECKCARGO.C                                              40.0  25.000f   2.412
  Total Lateral Plane->       72.5  25.000f  -0.781     146.0  20.780f   1.878
Distances in METERS.----------------------------------------------------------

                            LATERAL PLANE STATUS
           USK draft: 1.700 @ 50.00f, 1.700 @ 25.00f, 1.700 @ 0.00
                          Trim: zero,  Heel: zero
Part---------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                          77.4  25.000f  -0.829      65.9  25.000f   0.663
CRANE.C                                                  35.2   7.500f   3.512
DECKCARGO.C                                              40.0  25.000f   2.312
  Total Lateral Plane->       77.4  25.000f  -0.829     141.1  20.634f   1.841
Distances in METERS.----------------------------------------------------------

                            LATERAL PLANE STATUS
           USK draft: 1.800 @ 50.00f, 1.800 @ 25.00f, 1.800 @ 0.00
                          Trim: zero,  Heel: zero
Part---------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                          82.3  25.000f  -0.876      61.0  25.000f   0.612
CRANE.C                                                  35.2   7.500f   3.412
DECKCARGO.C                                              40.0  25.000f   2.212
  Total Lateral Plane->       82.3  25.000f  -0.876     136.2  20.477f   1.806
Distances in METERS.----------------------------------------------------------

                            LATERAL PLANE STATUS
           USK draft: 1.900 @ 50.00f, 1.900 @ 25.00f, 1.900 @ 0.00
                          Trim: zero,  Heel: zero
Part---------------------------LPA------LCP-----HCP-------LPA------LCP-----HCP
HULL                          87.2  25.000f  -0.924      56.0  25.000f   0.561
CRANE.C                                                  35.2   7.500f   3.312
DECKCARGO.C                                              40.0  25.000f   2.112
  Total Lateral Plane->       87.2  25.000f  -0.924     131.2  20.307f   1.772
Distances in METERS.----------------------------------------------------------
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03-04-19 10:01:31 Cybermarine Technologies PTE. Ltd.
GHS 16.50

LATERAL PLANE STATUS
USK draft: 2.000 @ 50.00f, 2.000 @ 25.00f, 2.000 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	92.2	25.000f	-0.971	51.1	25.000f	0.511
CRANE.C				35.2	7.500f	3.212
DECKCARGO.C				40.0	25.000f	2.012
Total Lateral Plane->	92.2	25.000f	-0.971	126.3	20.123f	1.739

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.100 @ 50.00f, 2.100 @ 25.00f, 2.100 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	97.2	25.000f	-1.019	46.1	25.000f	0.461
CRANE.C				35.2	7.500f	3.112
DECKCARGO.C				40.0	25.000f	1.912
Total Lateral Plane->	97.2	25.000f	-1.019	121.3	19.922f	1.709

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	102.8	25.000f	-1.072	40.5	25.000f	0.405
CRANE.C				35.2	7.500f	3.000
DECKCARGO.C				40.0	25.000f	1.800
Total Lateral Plane->	102.8	25.000f	-1.072	115.7	19.676f	1.677

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.300 @ 50.00f, 2.300 @ 25.00f, 2.300 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	107.2	25.000f	-1.114	36.1	25.000f	0.361
CRANE.C				35.2	7.500f	2.912
DECKCARGO.C				40.0	25.000f	1.712
Total Lateral Plane->	107.2	25.000f	-1.114	111.3	19.465f	1.653

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.400 @ 50.00f, 2.400 @ 25.00f, 2.400 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	112.2	25.000f	-1.162	31.1	25.000f	0.311
CRANE.C				35.2	7.500f	2.812
DECKCARGO.C				40.0	25.000f	1.612
Total Lateral Plane->	112.2	25.000f	-1.162	106.3	19.205f	1.629

Distances in METERS.